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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 423 — Past Questions & Materials
by Renance

Course: CSC 423
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 10
Levels: 400

This study pack was compiled and restructured by Renance for CSC 423 students at Federal University of Petroleum Resources, Effurun. It brings together 10 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

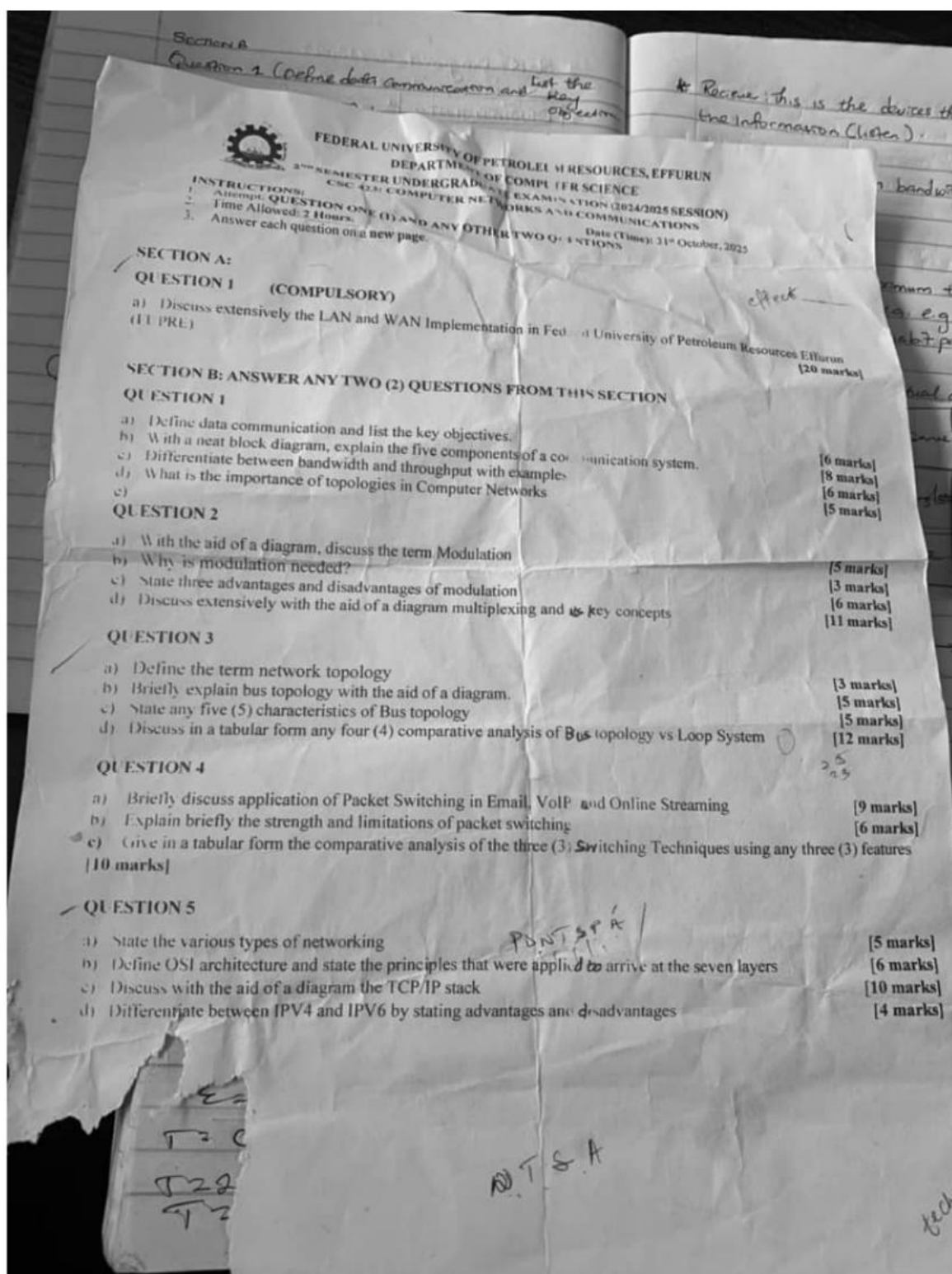
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Paper 1 of 10 — CSC 423 COMPUTER NETWORK AND COMMUNICATION 2024/2025

Level 400 • Second Semester • PastQuestion

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Paper 2 of 10 — CSC 423 FIRST SEMESTER 2020/2021

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2020/2021

Course Code/Title: CSC 423 - Computer Networks/Communication

Instructions: Attempt question one (1) and any other three (3) questions

: Allowed: 2 1/2 Hour

Question One (25 Marks)

Today's communication system is a complex system of numerous components working in a comprehensive fashion together round the clock. However, one of the key components of the Communication system is the MODEM. With the aid of a labelled diagram explain the working concept of a modem.

(10 Marks)

List one advantage and one disadvantage of peer-to-peer model for networked

application when compared to the

Client-server model. (3 Marks)

Web caching. (3 Marks)

Layering. (3 Marks)

Frequency modulation and demodulation (6 Marks).

Question Two (15 Marks)

Explain periodic function. (5 Marks)

Dirichlet's condition for a function to be expanded as a Fourier

series. (5 Marks)

Marks)

State the convergence condition on Fourier series. (5 Marks)

Question Three (15 Marks)

a. Why is modulation used in telecommunication? (5 Marks)

(5 Marks)

b. Explain the mechanism to recover data from modulated carrier.

c. Though modulation can occur in a number of different primary

modulation mechanisms. (5 Marks)

Paper 3 of 10 — CSC 423 2024 PQ

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
DEPARTMENT OF COMPUTER SCIENCE
2ND SEMESTER BSC EXAMINATION (2023/2024 SESSION)
CSC 423 – COMPUTER NETWORKS & COMMUNICATIONS

INSTRUCTIONS:

1. Attempt Question 1 And Any Other 3 Questions.
2. Time allowed: 2½ hours.
3. Answer each question on a new page.

Date: 10th Oct 2024

Question 1. [Compulsory]

- a) Define the Internet, computer network, and data communications. (3mks).
- b) Discuss the classifications of computer networks based on:
 - i) Transmission mode. (3½mks).
 - ii) Transmission time. (3½mks).
- c) Give any three (3) functions of a Bridge. (3mks).
- d) Briefly describe the functions of each layer of the OSI model. (5mks).
- e) Discuss the following network topologies (with diagrams):
 - i) Bus. (2mks).
 - ii) Ring. (2mks).
 - iii) Star. (2mks).
 - iv) Mesh. (2mks).
 - v) Hybrid. (2mks).

Question 2.

- a) What is the TCP/IP model? Give three (3) differences between the OSI and TCP/IP model. (8½mks).
- b) What are the three (3) key elements of a network protocol? (4½mks).
- c) Give any two (2) commonly used network security protocol. (3mks).

Question 3.

- a) With the aid of a diagram, define a switched network. (6mks).
- b) List any three (3) differences between packet switching and circuit switching. (5mks).
- c) List any three (3) application protocol. (3mks).

Question 4.

- a) What is Network Address Translation? (4mks).
- b) What are the two (2) primary responsibilities of an IP address? (4mks).
- c) Give five (5) descriptions captured in an IP Packet field. (6mks).

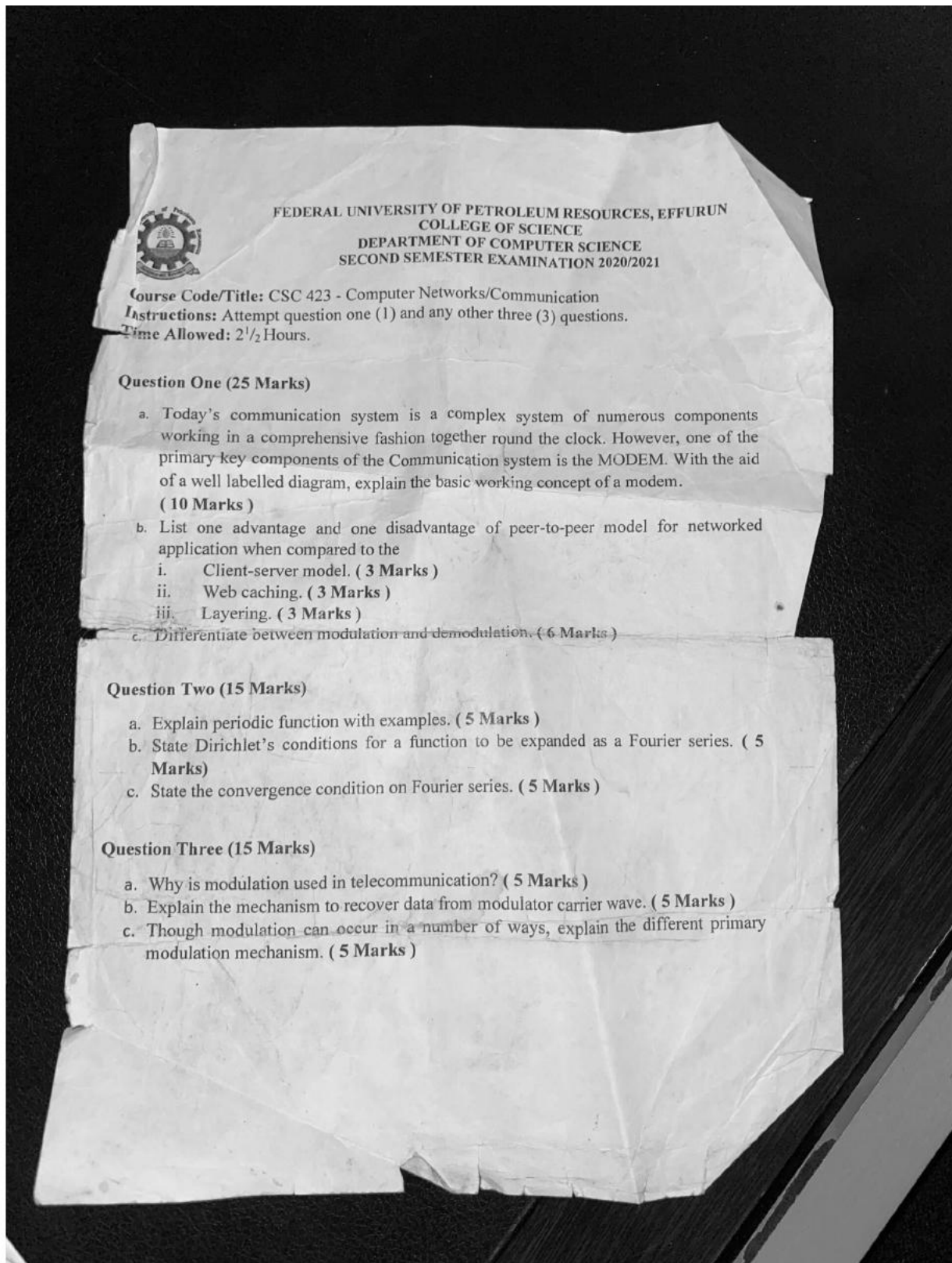
Question 5.

- a) Describe two (2) types of network security policies. (5mks).
- b) List three (3) ways in which the vulnerabilities TCP/IP can be exploited. (4½mks).
- c) Recommend three (3) solutions to the security problems listed in question 5(b). (4½mks).

Paper 4 of 10 — CSC423

Level 400 • Second Semester • PastQuestion

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Question Four (15 Marks)

- a. Explain the operation of the traditional telephone network using the concept of switching (5 Marks)
- b. Explain in details the term packet switching (4 Marks)
- c. Compare circuit-switching and packet-switching network based on the following criteria
 - i. Reserving network resources ahead of data been sent. (2 Marks)
 - ii. Utility network resources efficiently. (2 Marks)
 - iii. Guaranteeing uniform network quality of service during the communication. (2 Marks)

Question Five (15 Marks)

- a. To understand the complexities of deploying WLANs, the concept of radio wave is fundamental. Discuss. (5 Marks)
- b. With the aid of a diagram, explain the basic Radio Frequency system components. (4 Marks)
- c. Define the following terms:
 - i. Amplitude-shift keying. (2 Marks)
 - ii. Frequency-shift keying. (2 Marks)
 - iii. Phase-shift keying. (2 Marks)

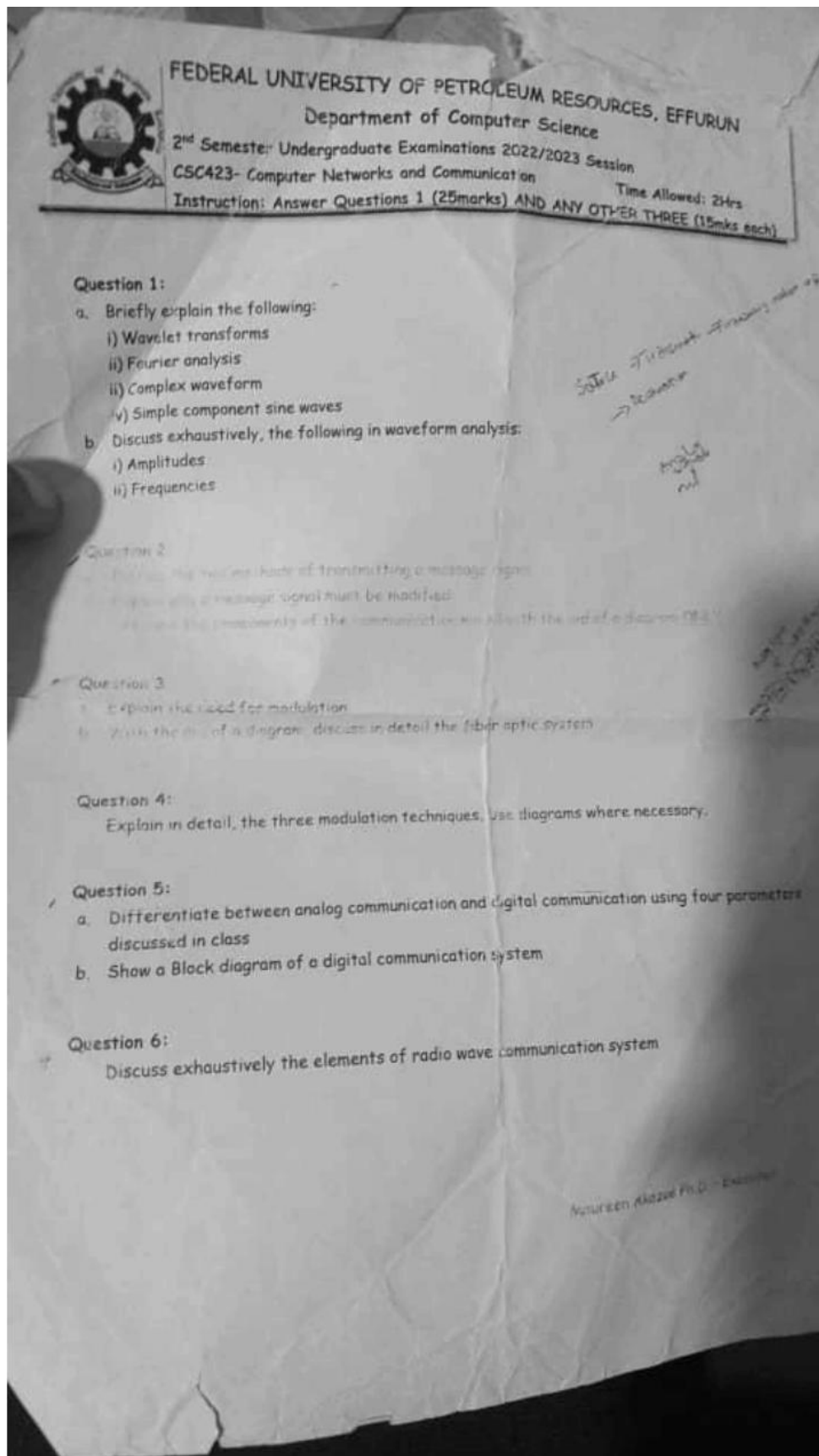
Question Six (15 Marks)

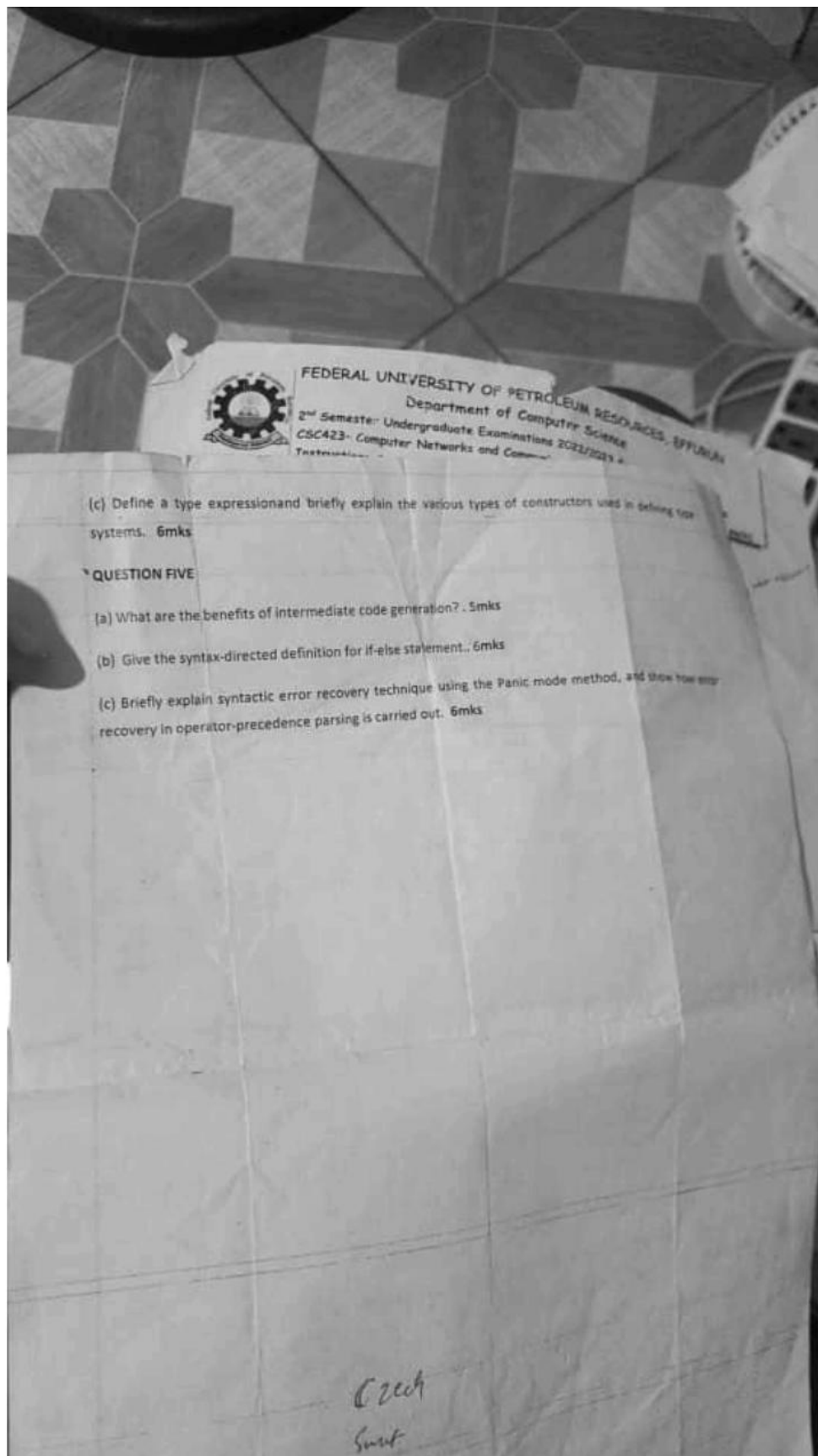
- a. A company with many offices have a computer at each location to keep track of customer orders, monitor sales and do local payroll. Previously, each of these computers worked in isolation. But at some point, management decided to connect them together. In your opinion, why did management take this decision? (3 Marks)
- b. The selection of topology for a network cannot be done in isolation as it affects the choice of media and the access method used. Explain 3 factors to be considered in making this choice. (3 Marks)
- c. Explain 3 network topologies with the aid of a diagram. State their advantages and disadvantages. (9 Marks)

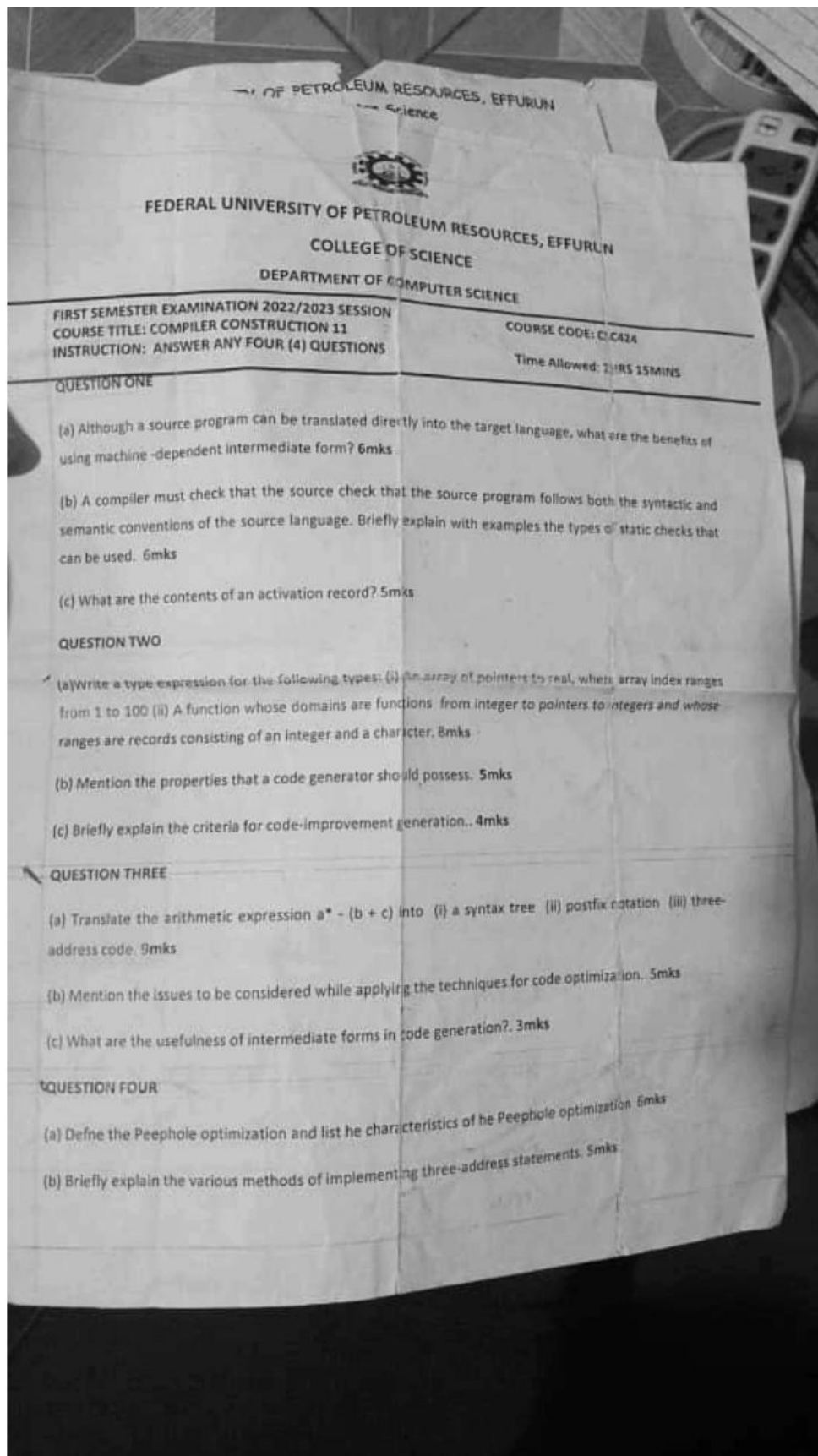
Paper 5 of 10 — CSC 423

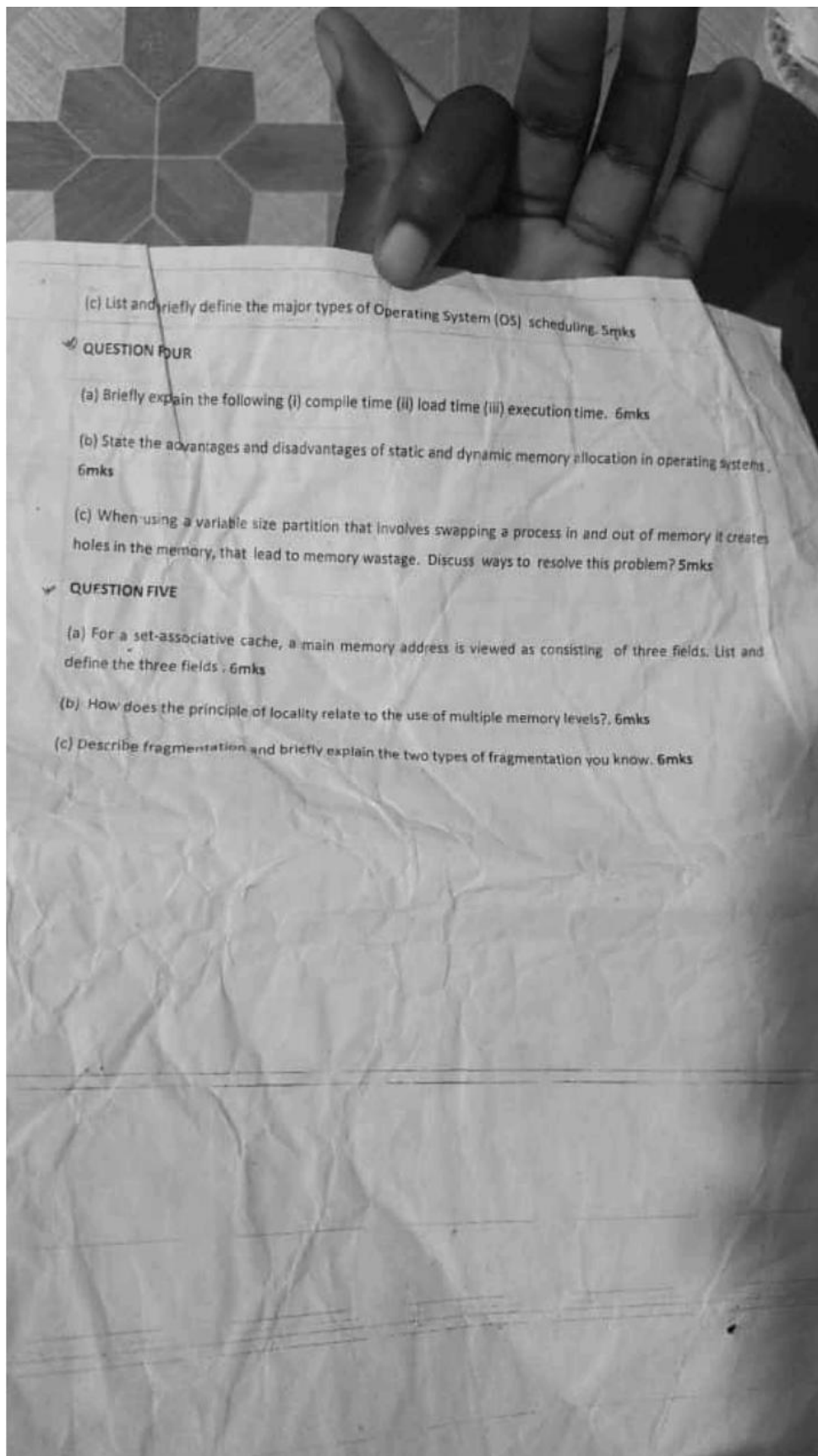
Level 400 • Second Semester • PastQuestion

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(c) List and briefly define the major types of Operating System (OS) scheduling. 5mks

QUESTION FOUR

(a) Briefly explain the following (i) compile time (ii) load time (iii) execution time. 6mks

(b) State the advantages and disadvantages of static and dynamic memory allocation in operating systems. 6mks

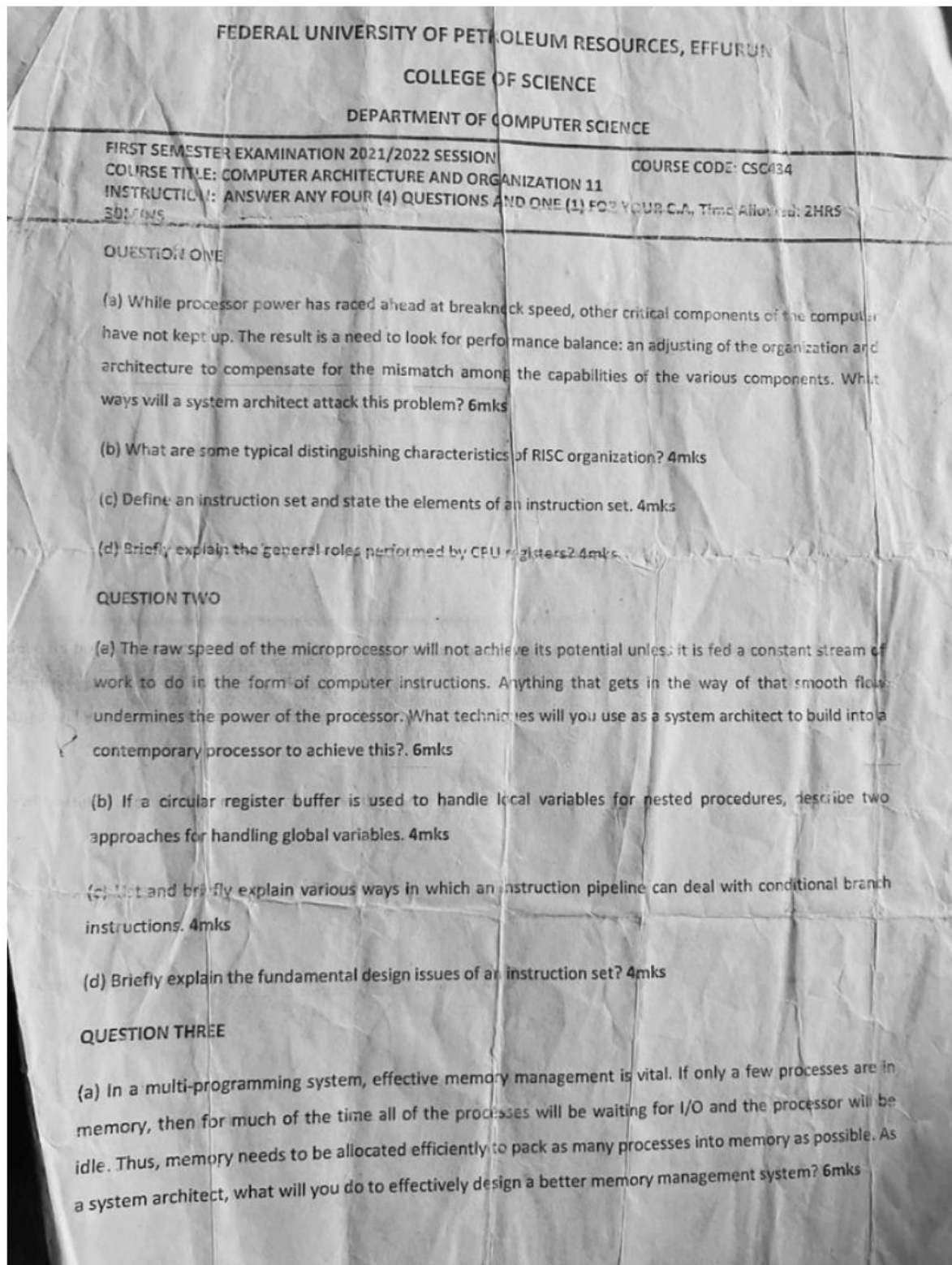
(c) When using a variable size partition that involves swapping a process in and out of memory it creates holes in the memory, that lead to memory wastage. Discuss ways to resolve this problem? 5mks

QUESTION FIVE

(a) For a set-associative cache, a main memory address is viewed as consisting of three fields. List and define the three fields. 6mks

(b) How does the principle of locality relate to the use of multiple memory levels? 6mks

(c) Describe fragmentation and briefly explain the two types of fragmentation you know. 6mks



(b) If a process is dynamically assigned to different locations in main memory, what is the implication for the addressing mechanism? 4mks

(c) List and briefly define the major types of Operating System (OS) scheduling. 4mks

(d) Define a cache memory and give reasons why we cannot have a big volume of cache memory. 4mks

QUESTION FOUR

(a) A computer is a complex system; contemporary computer contains millions of elementary electronic components. How, then, can one clearly describe them? 5mks

(b) Justify the assertion that a 32-bit instruction is probably less than twice as useful as a 16-bit instruction. 3mks

(c) Outline a three-step process that leads to a characterization of the control unit. 5mks

(d) When using a variable-size partition that involves swapping a process in and out of memory it creates holes in the memory, that lead to memory wastage. Discuss ways to resolve this problem? 5mks

QUESTION FIVE

(a) For a direct-mapped cache, a main memory address is viewed as consisting of three fields. List and define the three fields. 4mks

(b) Distinguish between spatial locality and temporal locality? 4mks

(c) Briefly explain the differences between sequential access of data and random access. 6mks

(d) Is it necessary for all of the pages of a process to be in main memory while the process is executing? Discuss. 4mks

QUESTION SIX

(a) Briefly explain the concept of stack and its implementation. 4mks

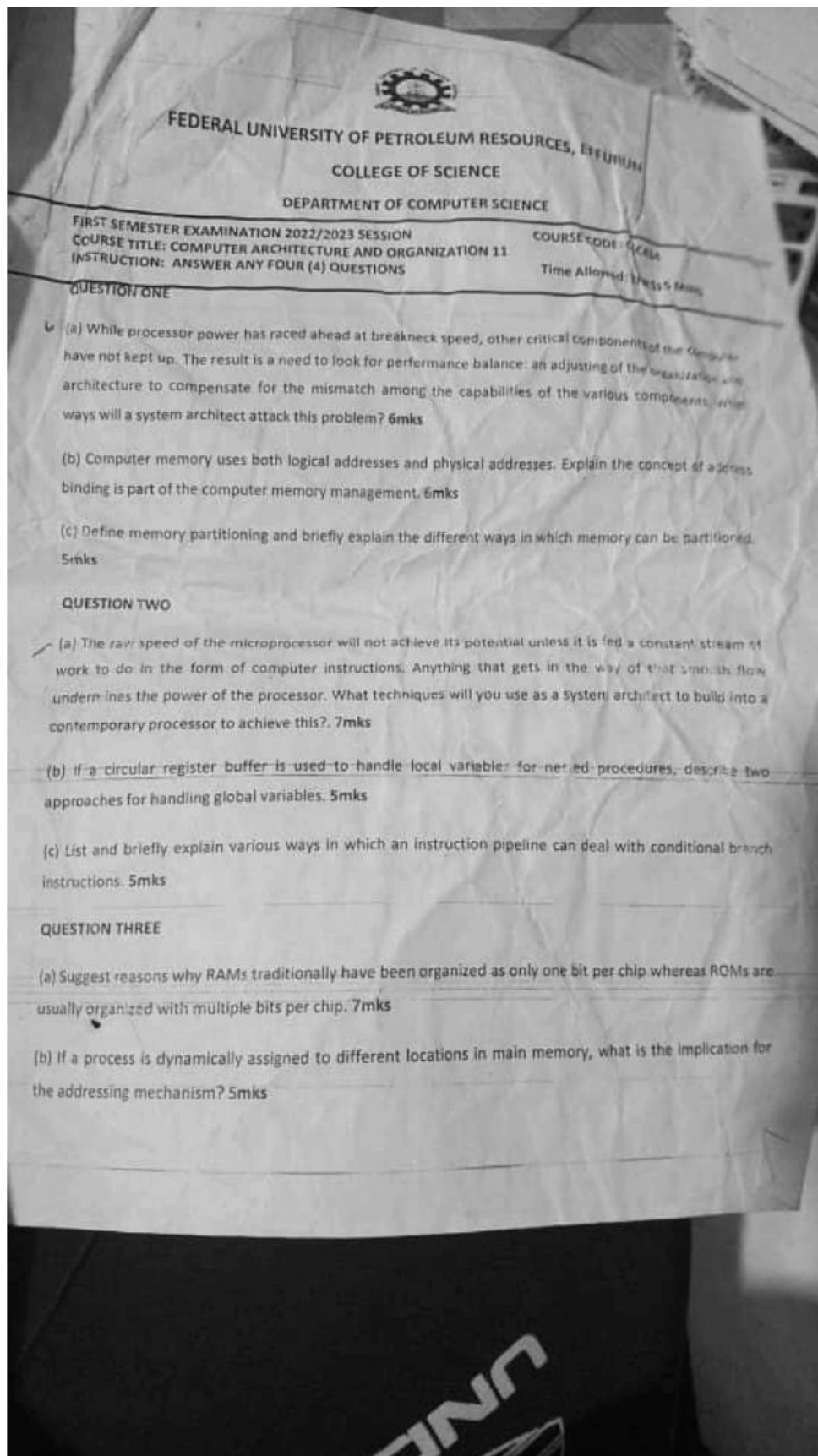
(b) Explain the concept of Page Replacement Algorithm in memory management. 5mks

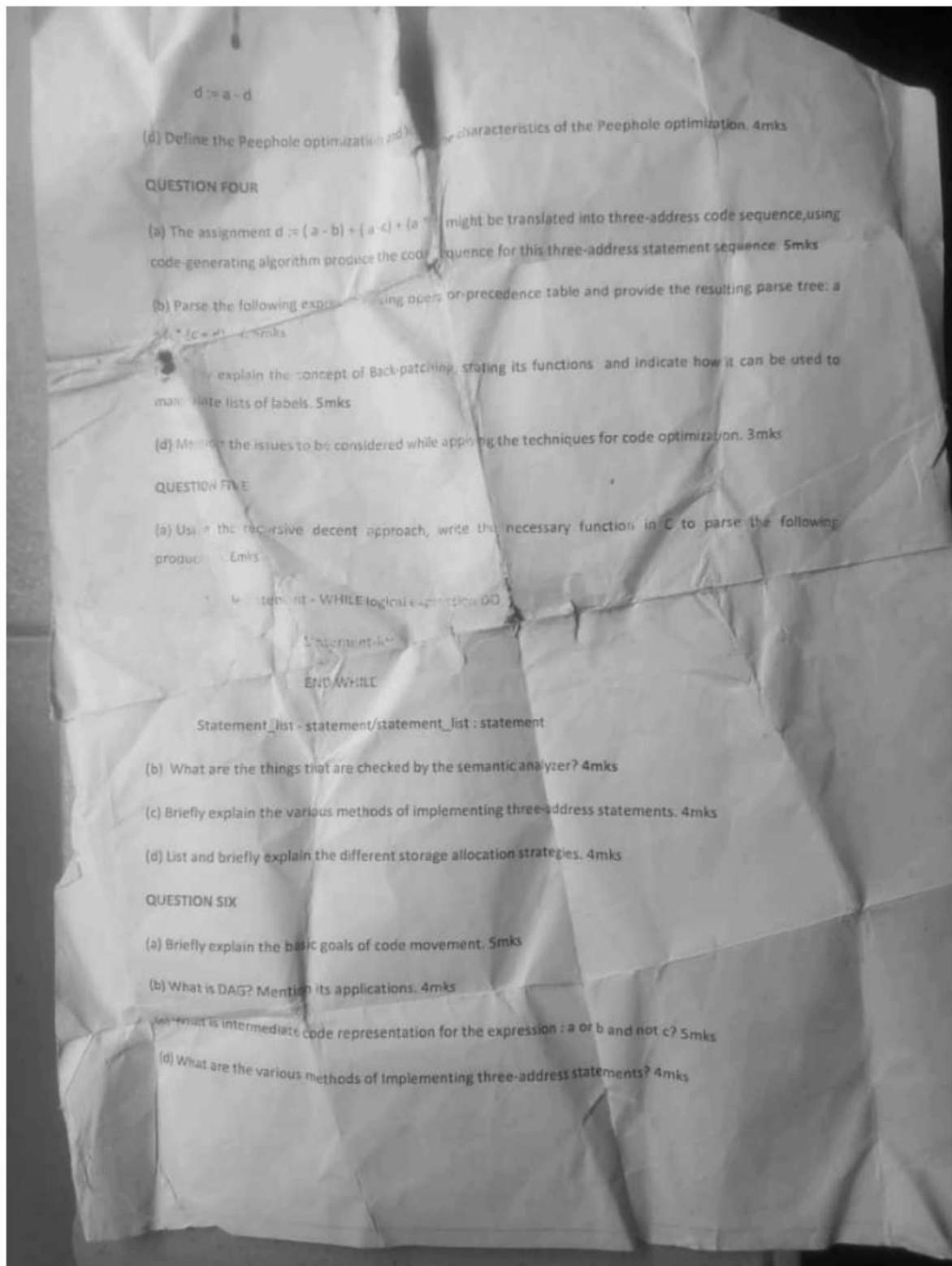
(c) How does the principle of locality of reference relate to the use of multiple memory levels? 4mks

(d) A set of 4 associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses. 5mks

Mode: med

Page 1 of 1 17/11/2022






FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUMI
DEPARTMENT OF COMPUTER SCIENCE
 2nd Semester: BSc (Computer Science Examination) 2022/2023 Session
CSC423 - MANAGEMENT SCIENCE
 Instruction: Attempt Questions 1, 2 AND Any Other

Question 1:
 A small plant assembles personal computers via 9-interlinked stages according to the following precedence and process: (1-2 = 4), (1-3 = 12), (1-4 = 10), (2-5 = 8), (2-6 = 6), (3-6 = 8), (4-6 = 10), (5-7 = 10), (6-7 = 9), (6-8 = 8), (7-8 = 10) and (8-9 = 6) respectively. Evaluate the following:

- Draw the network arrow diagram representing the above assembly line. 4mks
- Show workings of the Forward-Passes and Backward-Passes for the Assembly line. 8mks
- Tabulate the Start, Finish and Float/Slack time for the Critical Path Method. 4mks
- Compute the critical path and the assembly duration. 6mks
- Evaluate the total slack time for the project. 3mks

Question 2:
 Given the matrix diagram below, resolve the transportation problem using:
 The Vogel's Approximation Method 25mks

From \ To	Destinations				Total
	1	2	3	4	
1	8	4	6	10	300
2	12	9	10	8	250
3	7	8	8	9	400
4	6	7	10	12	250
	300	400	200	300	1200

Question 3:

- Show the schematics sources of changes in a Project. 2mks
- Describe: (a) Work-to-be-done, (b) Work-Done, (c) Fraction Complete, and (d) Rework. 4mks
- Evaluate the Capacity Network in Question 1 using the North-West Corner Rule. 14mks

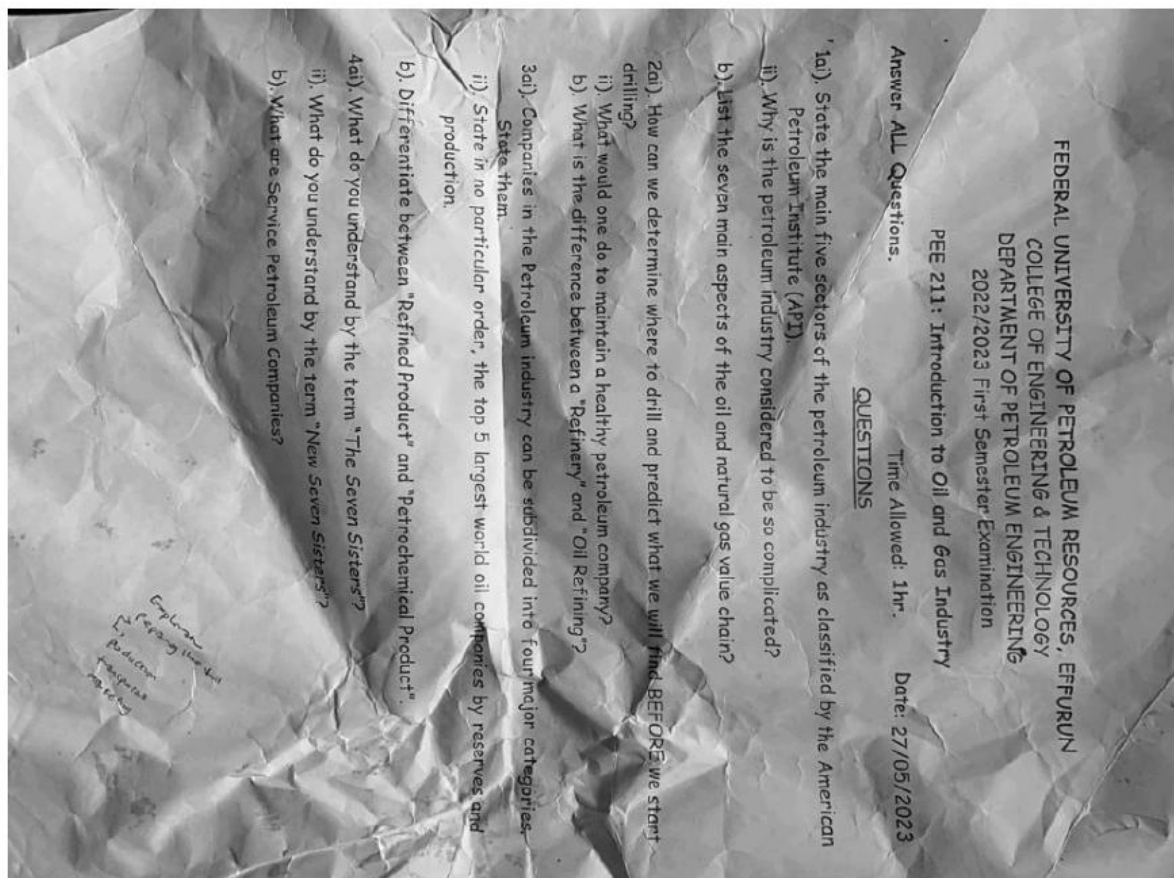
Question 4:

- Describe Project Management with a well-labelled Diagram. 4mks
- Evaluate the Capacity Network in Question 1 using the Least-Cost Method. 16mks

Question 5:

- What are the responsibilities of a project manager? 5mks
- State 7-major reasons why do Projects fail? 7mks
- Show the Schematics of an Integrated Project Management. 8mks

OJUGO, Arnold Adimabus (PhD) - Examiner



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING & TECHNOLOGY
2019/2020 First Semester Examination

PEE 211: Introduction to Oil and Gas Industry
Time Allowed: 1hr.30mins Date: 16/03/2021

Answer ALL Questions.

QUESTIONS

1a). State the main sectors of the petroleum industry.
 ii). Why is there need for alternate hydrocarbon sources?

2a). What is the first period in the producing life of a reservoir?
 ii). State the three natural forces that move the fluids in a reservoir.

3a). List the six components of a rotary drilling rig, stating one equipment that each of the components is made of.
 ii). List the personnel on a rotary drilling rig.

4a). What is the difference between a Servicing company and an Operating company? State two examples each.
 ii). Differentiate between a Kick and a Blowout.

5a). What are the 3 ways in which unconventional supplies differ from conventional supplies of oil and natural gas?
 ii). State the physical characteristics of natural gas.

6a). Why is natural gas considered as one of the cleanest forms of gas?
 ii). Explain the need for treating crude oil.

7a). What is Revenue?
 ii). What is the difference between Direct expenditures and Indirect expenditures?

8a). The table below shows a company's investment of \$10,000.00, which is estimated to generate the \$20,000.00 annually over a 5 years period, with an operating expenses of \$14,000.00 per annum and a Federal Income Tax (FIT) of \$1000. Compute the Net Cash Flow and the Cumulative Net Cash Flow over the 5 year period as shown in the table.

Year	Investment (\$)	Expenses (\$)	Revenue (\$)	FIT (\$)
0	10,000.00			
1		14,000.00	20,000.00	1,000.00
2		14,000.00	20,000.00	1,000.00
3		14,000.00	20,000.00	1,000.00
4		14,000.00	20,000.00	1,000.00
5		14,000.00	20,000.00	1,000.00

Paper 6 of 10 — CSC 423 PAST QUESTION

Level 400 • Second Semester • PastQuestion

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Ea, CSCA23- Computer Networks and Communication *023 Session

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a. Briefly explain the following:

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ii) Feurier onalysis My

i) Complex waveform ah AX

f iy) Simple component sine waves a

piscuss exhaustively, the following in waveform analysts: A

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i) Frequencies 3 i

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Question 4:

Explain in deta'l, the three modulation techniques, Use tliagrams where necessary,

Question 5:

Ditterentiate between analog communication and iugitel communication using for

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discussed in closs

b. Show a Block diagram of o digital communication system

Question 6:

communication system

Discuss exhaustively the clements of radio wave

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(b} Give the syntax-directed definition for f-elve

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FIRST SEMESTER EXAMINATION 2022/2023 Session

COURSE TITLE: COMPILER CONSTRUCTION 11 a

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS ote.

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Hurston ONE

(4) Although @ source program Can be translated d rertly into the

tary en

Bet language, what ere the benefits of

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sing machine -deper dent intermediate form? Gmks

} wilee must check that the source check that th

{b) A compiler must © ec at the source program fi

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of the source language. Briefly

emant mventions 2

semantic conven *plain with examples the types of static checks that

can be used, Smks

(ce) What are the contents of an activation record? Smks

QUESTION TWO

* (aywrite e type expression for the following types (0) Pe array of poittiersts real, whetn array index ninges

! + to 100 (li) A function whose domains are functions from integer ta pointers to /ntegers and whose

ranges are records consisting ¢ fan integer and a chartcter, Bmks

{b) Mention the properties that a code generator should possess. Smks

{c} Briefly explain the cntera for code-improvement generation. amks

F

". QUESTION THREE

- (b+ c) Into {i} @ syntax tree {il} postin nafation (iii) three

a} Transtate the arithmetic expression a
address code. Smks

b) Mention the issves to be considered while applying the techniques for code optimization. Smks

{c} What are the usefulness of intermediate forms in fode generation? 3mks

QUESTION FOUR

' fimks

fe optimisation

{a} Defne the Peephole optimization and list he charactertstics of he Peephole oP
smks

: ; < statements

(b) Briefly explain the various methods of implementing three-addres

[c} List andhriefly define the major types Of Oberating system (05) seh
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ry ilocation in Operating systems \

(b) State the advantages and disadvantages of static and dynamic memai

mks \

(c) When-using 3 variate size Partition that Involves swapping @ process in and out of memory It
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holes in the memory, that lead to memory wastage. Discuss ways to resolve this problem? Smks |

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x EXAMINATION 2021/2022 SESSION —o

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COLLEGE OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

First

) FIRST SEMESTER EXAMINATION 2022/2023 SESSION COURSE ROD Ace

yaoi TITLE: COMPUTER ARCHITECTURE AND ORGANIZATION 11 \ale =

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS Time Allard Ah

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SUESTION ONE eee

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\(a) While processor power has raced ahead at tireaknect speed, other critical ee

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have not kept pg. The result is a need to look for Performance balance: at adjusting of thee =

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architecture to compensate for the mismatch among the capabilities of the varlous

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ways will a system architect attack this problem? 6mks

{b) Computer memory uses both logical addresses and physical addresses Explain the concent 6f atçee.

binding is part of the computer memory management, 6mks

(ç) Define memory partitioning and briefly explain the different ways in which memory can be partitionsa

Smks

QUESTION TWO

(a) The rave speed of the microprocessor will Hot achieve its potential uniess it ls “ed & Conatent stream af

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work to do in the form of campiter instructions, Anything that gets in the wey of thet sen in flow undere ines the power of the processor. What techniques will you use as a System) arctytect to build into a

contemporary processor to achieve this?. 7mks

{b) if-a-circular-register buffer-is-used-to-handie local variables for-nesed- procedures, des 2 lwo approaches for handling global variabies. 55mks

d briefly explain various ways In which an Instruction pipeline can deal with conditional branch nd Ori e ve

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{c) List a

uctions. 55mks

QUESTION THREE

s why RAMs traditionally have been organized as only one Sit per-chip whereas ROMs 3

(a) Sugeest reas

with multiple bits per chip. 7mks

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(b) If 2 process is dynamically assigned to different locations in main memory, what is the Imp the addres

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(d) @ Issues to be considered while appy ret

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(b) What are the things thot are checked by the'

(c) Briefly explain thi us methods of .

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QUESTION Six

(a) Briefly explain th

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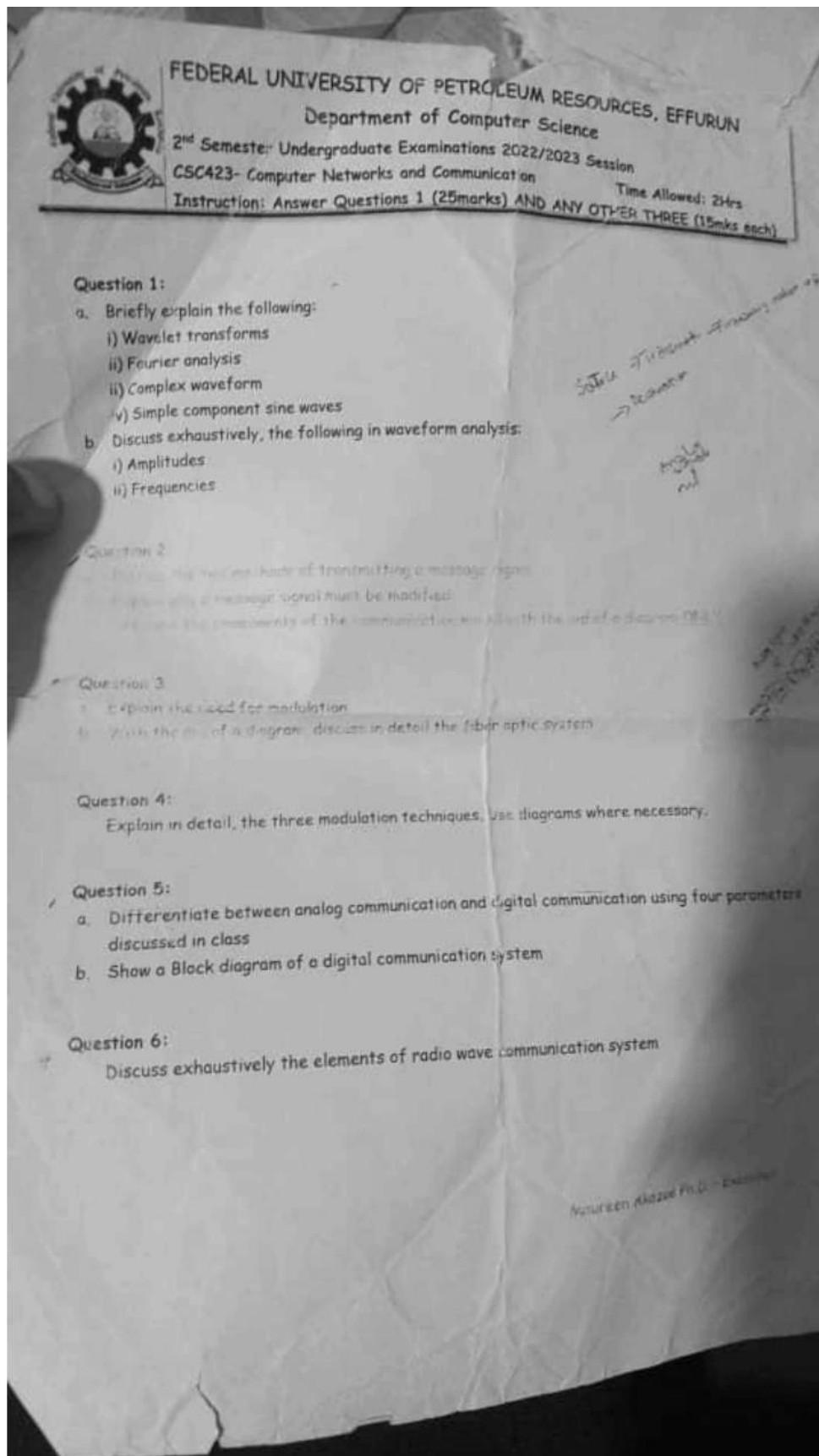
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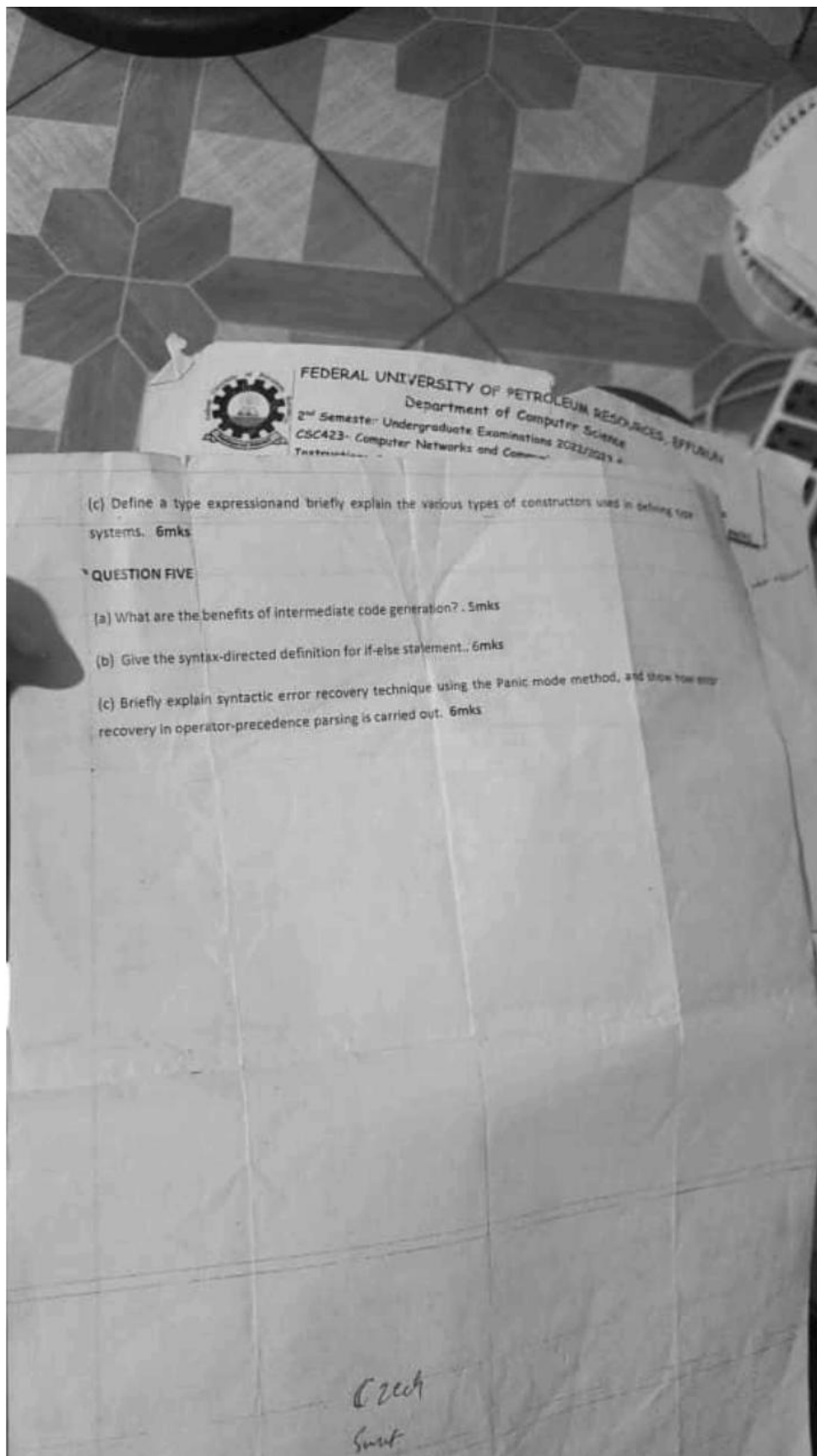
OJUGO, Arnold Adimabus (PAD) ~ Examiner

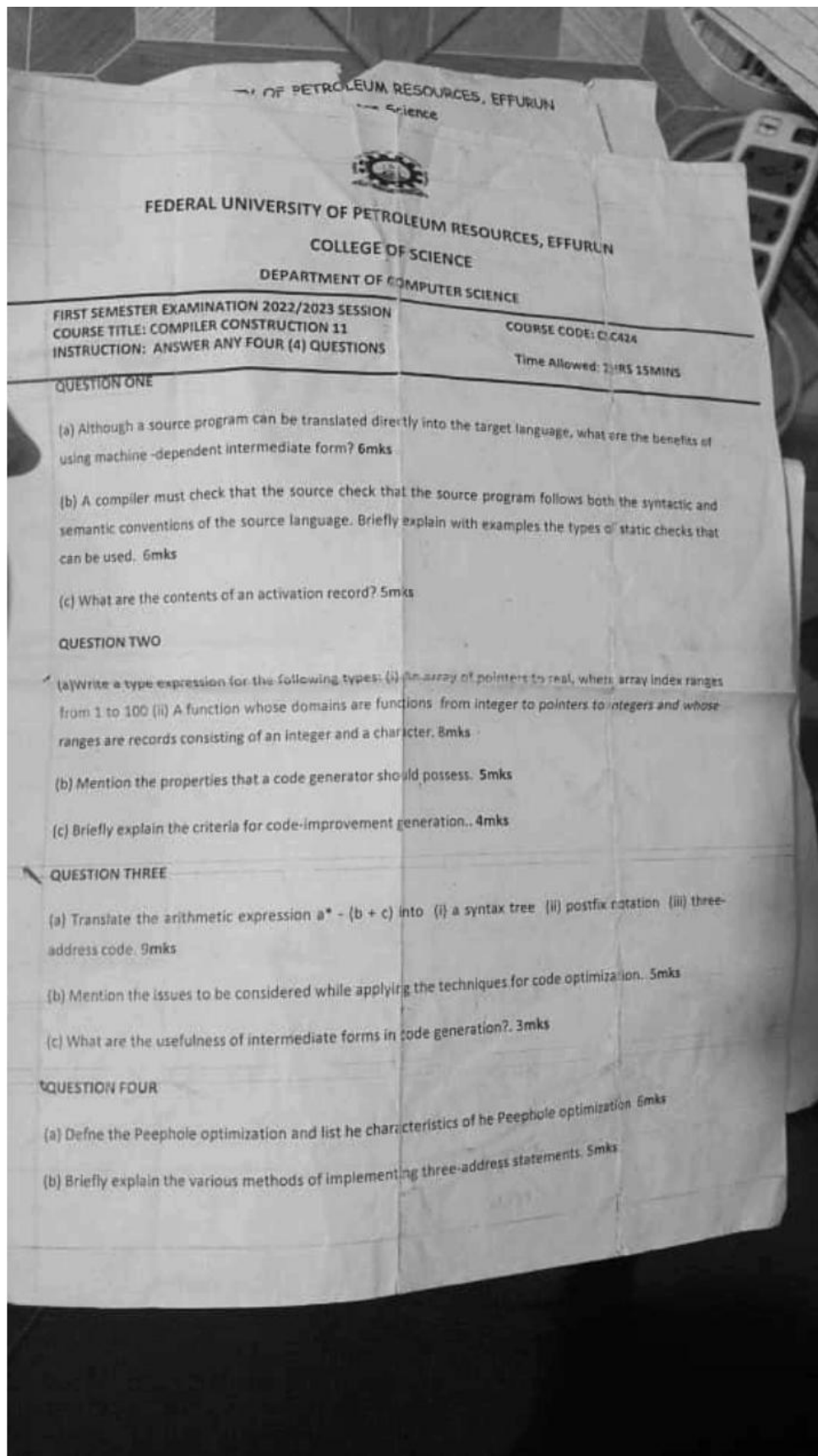
Paper 7 of 10 — CSC 423 PAST QUESTION

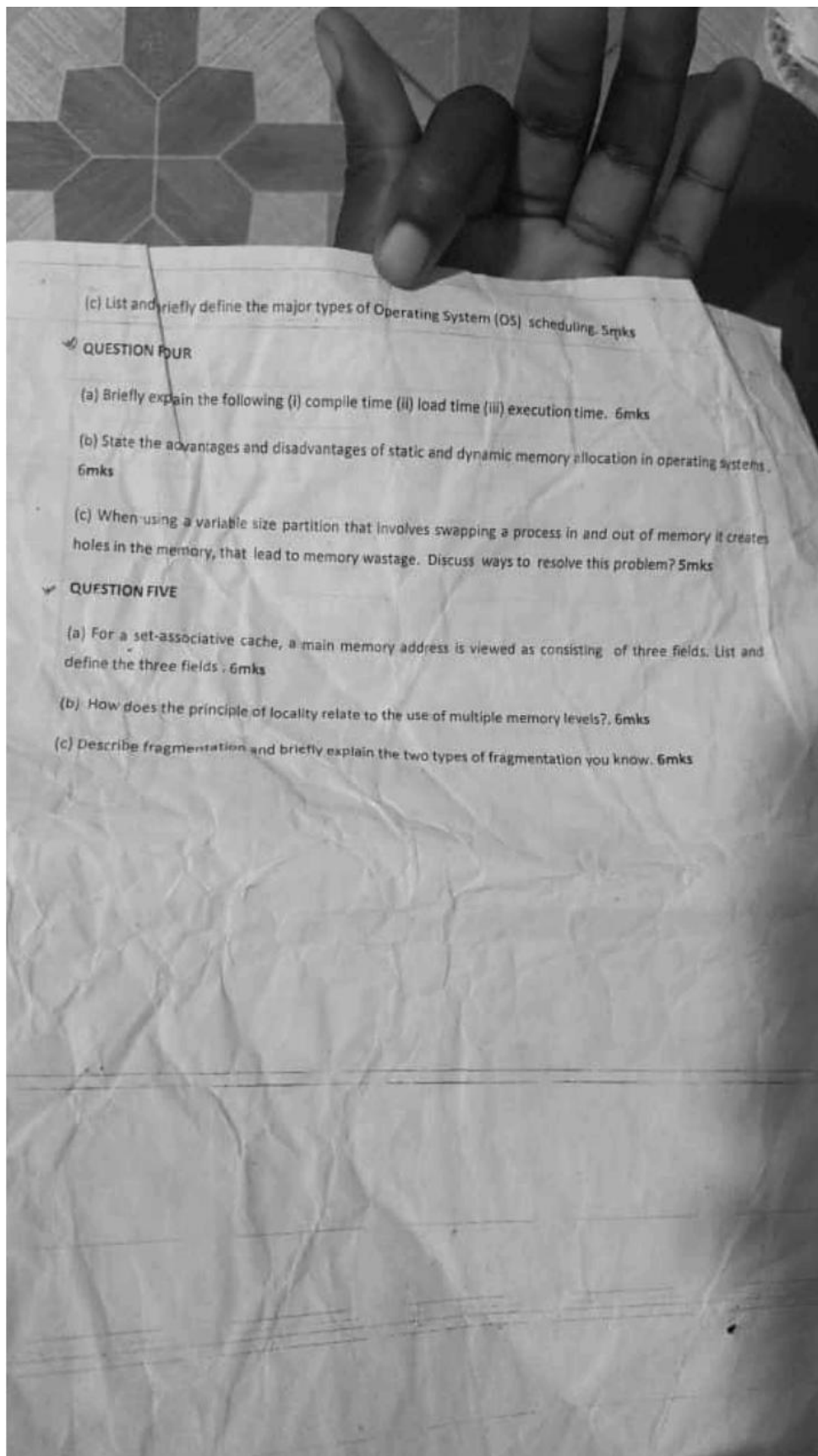
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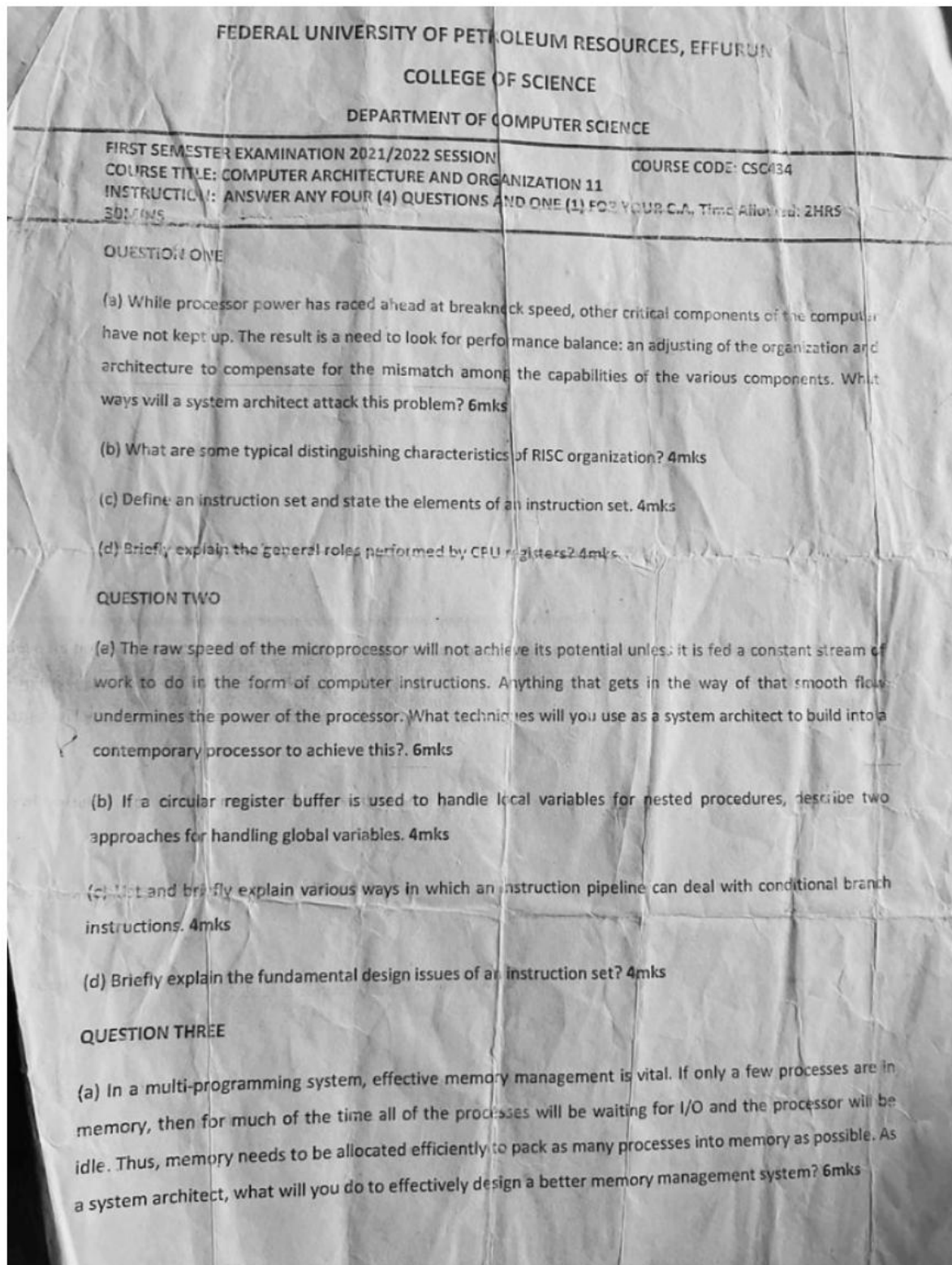
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QUESTION FIVE

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(d) Is it necessary for all of the pages of a process to be in main memory while the process is executing? Discuss. 4mks

QUESTION SIX

(a) Briefly explain the concept of stack and its implementation. 4mks

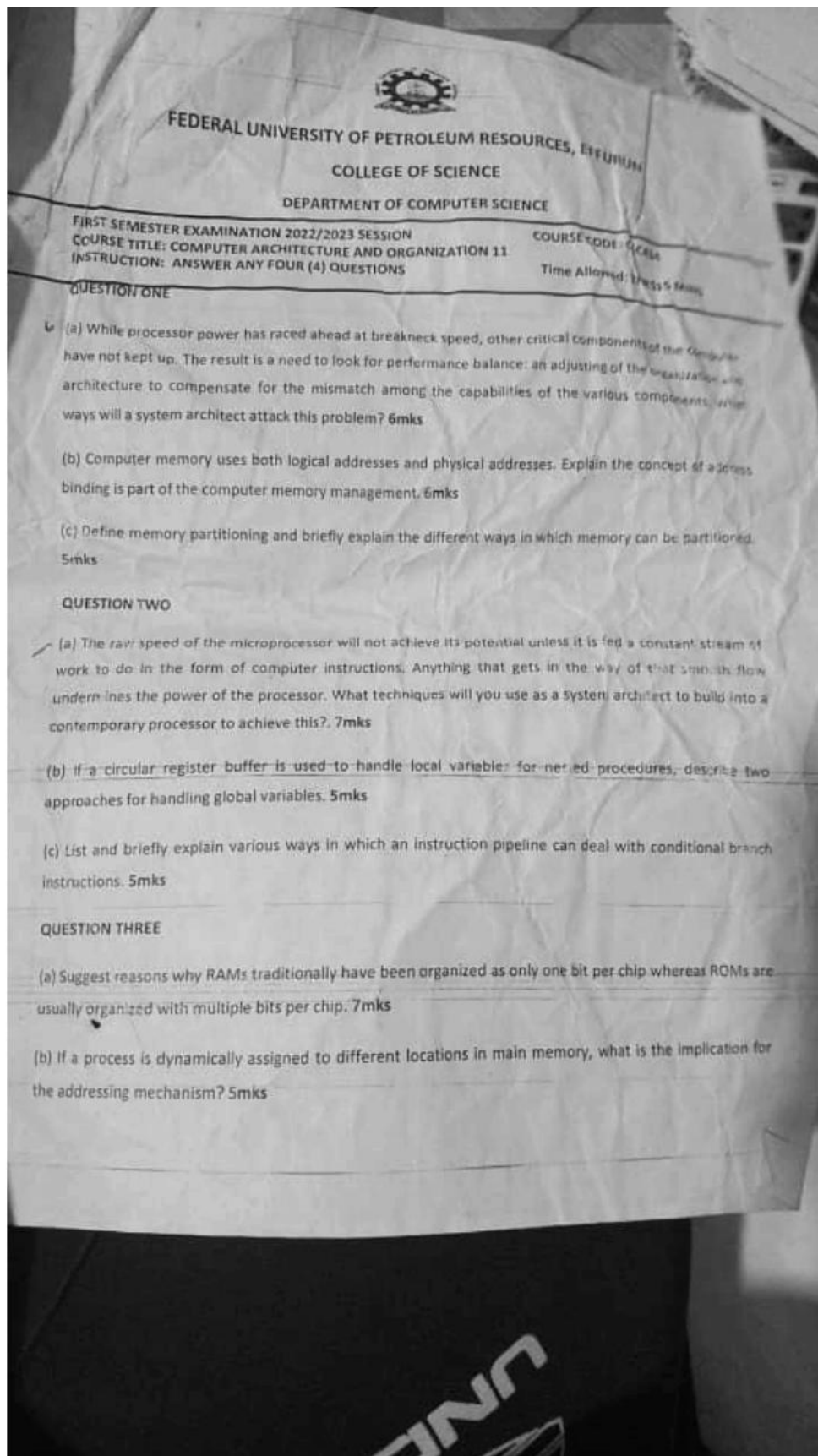
(b) Explain the concept of Page Replacement Algorithm in memory management. 5mks

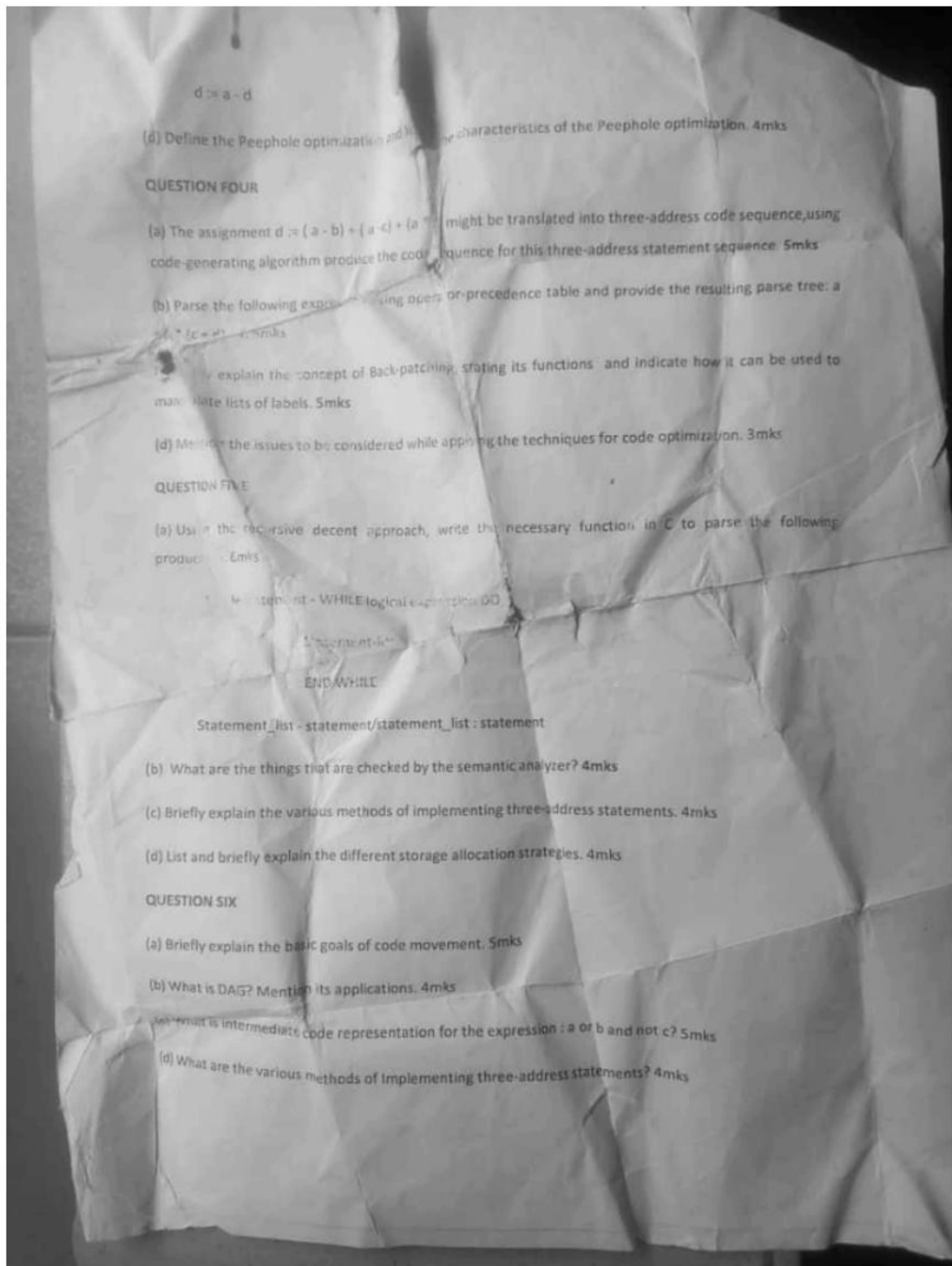
(c) How does the principle of locality of reference relate to the use of multiple memory levels? 4mks

(d) A set of 4-way associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses. 5mks

Mode: med

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUMI
DEPARTMENT OF COMPUTER SCIENCE
 2nd Semester: BSc (Computer Science Examination) 2022/2023 Session
CSC423 - MANAGEMENT SCIENCE
 Instruction: Attempt Questions 1, 2 AND Any Other

Question 1:
 A small plant assembles personal computers via 9-interlinked stages according to the following precedence and process: (1-2 = 4), (1-3 = 12), (1-4 = 10), (2-5 = 8), (2-6 = 6), (3-6 = 8), (4-6 = 10), (5-7 = 10), (6-7 = 9), (6-8 = 8), (7-8 = 10) and (8-9 = 6) respectively. Evaluate the following:

- Draw the network arrow diagram representing the above assembly line. 4mks
- Show workings of the Forward-Passes and Backward-Passes for the Assembly line. 8mks
- Tabulate the Start, Finish and Float/Slack time for the Critical Path Method. 4mks
- Compute the critical path and the assembly duration. 6mks
- Evaluate the total slack time for the project. 3mks

Question 2:
 Given the matrix diagram below, resolve the transportation problem using:
 The Vogel's Approximation Method 25mks

From \ To	Destinations				Total
	1	2	3	4	
1	8	4	6	10	300
2	12	9	10	8	250
3	7	8	8	9	400
4	6	7	10	12	250
	300	400	200	300	1200

Question 3:

- Show the schematics sources of changes in a Project. 2mks
- Describe: (a) Work-to-be-done, (b) Work-Done, (c) Fraction Complete, and (d) Rework. 4mks
- Evaluate the Capacity Network in Question 1 using the North-West Corner Rule. 14mks

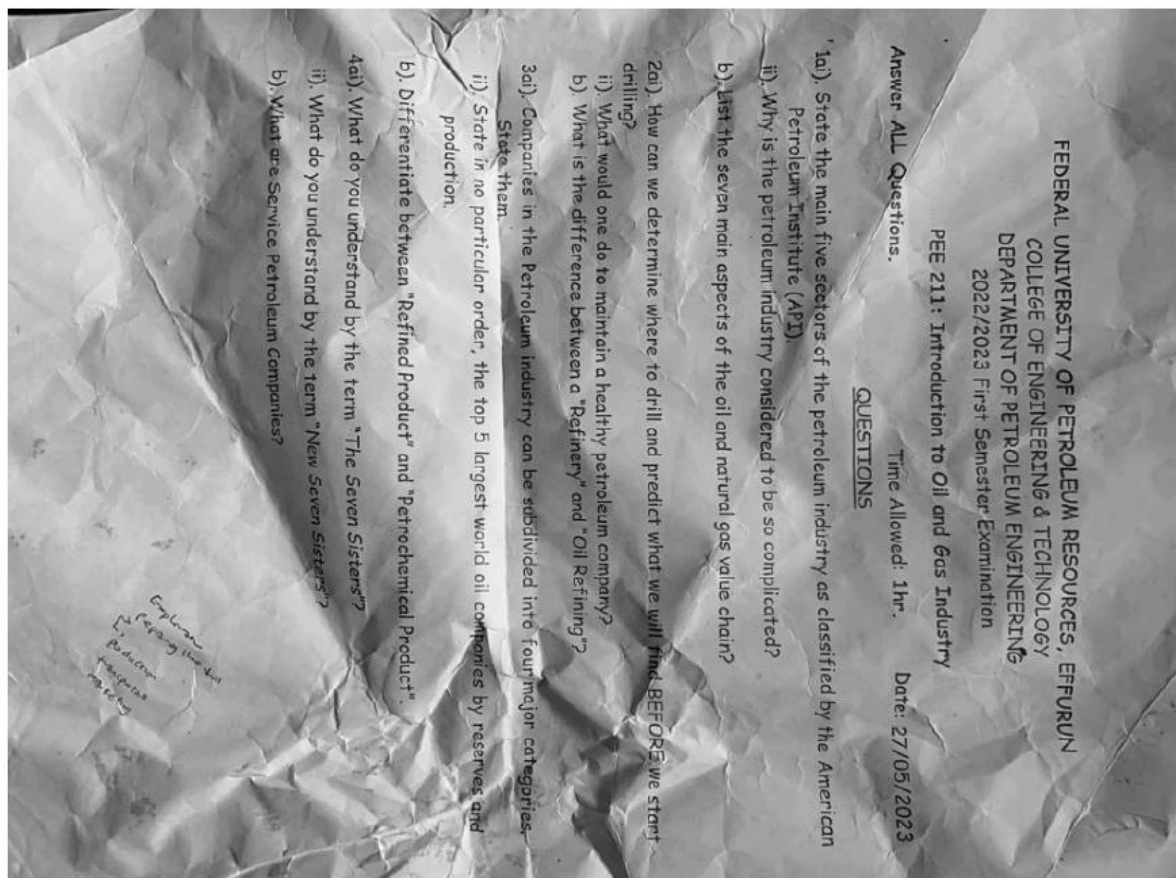
Question 4:

- Describe Project Management with a well-labelled Diagram. 4mks
- Evaluate the Capacity Network in Question 1 using the Least-Cost Method. 16mks

Question 5:

- What are the responsibilities of a project manager? 5mks
- State 7-major reasons why do Projects fail? 7mks
- Show the Schematics of an Integrated Project Management. 8mks

OJUGO, Arnold Adimabus (PhD) - Examiner



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING & TECHNOLOGY
2019/2020 First Semester Examination

PEE 211: Introduction to Oil and Gas Industry
Time Allowed: 1hr.30mins Date: 16/03/2021

Answer ALL Questions.

QUESTIONS

1a). State the main sectors of the petroleum industry
 ii). Why is there need for alternate hydrocarbon sources?

2a). What is the first period in the producing life of a reservoir?
 ii). State the three natural forces that move the fluids in a reservoir.

3a). List the six components of a rotary drilling rig, stating one equipment that each of the components is made of.
 ii). List the personnel on a rotary drilling rig.

4a). What is the difference between a Servicing company and an Operating company? State two examples each.
 ii). Differentiate between a Kick and a Blowout.

5a). What are the 3 ways in which unconventional supplies differ from conventional supplies of oil and natural gas?
 ii). State the physical characteristics of natural gas.

6a). Why is natural gas considered as one of the cleanest forms of gas?
 ii). Explain the need for treating crude oil.

7a). What is Revenue?
 ii). What is the difference between Direct expenditures and Indirect expenditures?

8a). The table below shows a company's investment of \$10,000.00, which is estimated to generate the \$20,000.00 annually over a 5 years period, with an operating expenses of \$14,000.00 per annum and a Federal Income Tax (FIT) of \$1000. Compute the Net Cash Flow and the Cumulative Net Cash Flow over the 5 year period as shown in the table.

Year	Investment (\$)	Expenses (\$)	Revenue (\$)	FIT (\$)
0	10,000.00			
1		14,000.00	20,000.00	1,000.00
2		14,000.00	20,000.00	1,000.00
3		14,000.00	20,000.00	1,000.00
4		14,000.00	20,000.00	1,000.00
5		14,000.00	20,000.00	1,000.00

Paper 8 of 10 — CSC 423 PRESENTATIONS 2025

Level 400 • Second Semester • Handout

GROUP PRESENTATION

ON MODULATION, DEMODULATION, MULTIPLEXING,
DATA TRANSMISSION

BY

COMPUTER SCIENCE DEPARTMENT STUDENTS
(GROUP 1)

SUBMITTED TO

DR. AKO RITA

COMPUTER SCIENCE DEPARTMENT

COLLEGE OF SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN, DELTA STATE, NIGERIA

AUGUST, 2025

Modulation

What is Modulation?

Modulation is the process of encoding a message signal onto a carrier wave by systematically changing one of its properties: amplitude, frequency, or phase. The message signal (also known as the baseband signal) is the original low-frequency information you want to send, such as voice, music, or data. The carrier wave is a high-frequency sinusoidal wave that acts as a vessel to carry the message signal over long distances.

Think of it like this: if you want to send a letter (your message signal) across the ocean, you don't just throw it in the water. Instead, you put it in a bottle (the carrier wave) that is designed to withstand the journey. Modulation is the process of putting the letter inside the bottle.

Fig 1: Modulation

Signal Modulator

A modulator is an electronic circuit that superimposes a low-frequency (information) signal onto a high-frequency (carrier) signal for the purpose of wireless transmission.

Modulation Process

Its primary function is to combine a low-frequency information-bearing signal (like audio or data) with a high-frequency carrier wave. It changes one of the carrier wave's properties—its amplitude, frequency, or phase—

to encode the information.

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Fig 2: Signal Modulator Device

Why do we need modulation?

1. **Reduced Antenna Size:** Low-frequency signals require very large, impractical antennas. By modulating the signal onto a high-frequency carrier, we can use much smaller, more practical antennas.
2. **Long-Distance Transmission:** High-frequency waves suffer less from signal degradation over long distances compared to low-frequency waves.
3. **Avoiding Interference:** If every radio station broadcasted at the same frequency, all the signals would mix together. Modulation allows each station to use a different carrier frequency, enabling multiple signals to be transmitted simultaneously without interfering with each other.

Advantages and Disadvantages of Modulation

Advantages of Modulation

1. **Antenna Size Reduction:** By modulating a signal onto a high-frequency carrier wave, the antenna size required is dramatically reduced to a practical length.
2. **Long-Distance Transmission:** High-frequency carrier waves give signals the energy needed to travel across vast distances without significant loss.
3. **Interference Avoidance:** Modulation allows each signal to be assigned a unique carrier frequency, preventing them from overlapping and enabling simultaneous transmission.
4. **Better Reception Quality:** Modulation makes signals more robust against noise and interference by carrying information on a powerful, high-frequency signal.

Disadvantages of Modulation

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1. **Increased Complexity:** Modulation and demodulation require extra circuitry and components, which raises the cost and complexity of the system.
2. **Bandwidth Requirement:** The process of modulation often demands a wider frequency range than the original signal, which is a significant issue since the frequency spectrum is limited.

3. Non-Linearity: Some modulation techniques can introduce signal distortion due to non-linearity if not managed correctly.

Types of Modulation

1. Amplitude Modulation (AM): The amplitude (strength) of the carrier wave is varied in proportion to the message signal's amplitude. The carrier's frequency and phase remain constant. AM is simple to implement but is susceptible to noise. It's widely used in AM radio broadcasting.

Fig 3: Amplitude Modulation

Advantages and Disadvantages of Analog Modulation

Advantages:

- a. AM receivers are relatively simple to design and manufacture, making them cheap.
- b. AM signals can be transmitted over very long distances, especially at night when atmospheric conditions change.

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Disadvantages:

- a. AM has a narrow bandwidth, which limits the quality of audio it can transmit.
 - b. AM is very sensitive to static and electrical interference, which can cause significant distortion in the received signal.
 - c. The majority of the power in an AM signal is in the carrier component, which carries no information, making it power-inefficient.
2. Frequency Modulation (FM): The frequency of the carrier wave is varied in accordance with the message signal's amplitude. The carrier's amplitude remains constant. FM is more resistant to noise than AM and provides higher quality audio, making it suitable for FM radio and television audio.

Fig 4: Frequency Modulation Signal

Advantages:

- a. Because the information is in the frequency and not the amplitude, FM is highly immune to common noise sources like static.

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- b. The wider bandwidth of FM allows for the transmission of a larger range of audio frequencies, resulting in superior sound quality, which is why it is the standard for music broadcasting.

c. The amplitude of the FM signal is constant, leading to more efficient power usage in the transmitter.

Disadvantages:

- a. FM signals typically require a much wider bandwidth compared to AM, which can limit the number of stations that can operate in a given frequency range.
- b. The circuitry for FM transmitters and receivers is more complex and, therefore, more expensive than for AM.
- c. FM signals operate in the VHF/UHF bands, which do not propagate over long distances by reflecting off the ionosphere like AM signals. This limits the reception range to a "line-of-sight" path from the transmitter.

3. Phase Modulation (PM): Phase Modulation (PM) is an analog modulation technique where the phase of a high-frequency carrier wave is varied in direct proportion to the instantaneous amplitude of the message signal. During this process, both the carrier's amplitude and frequency remain constant. PM is a cousin to FM, as a change in phase is directly related to a change in frequency, which is why they are often grouped together as angle modulation techniques.

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Fig 5: Phase Modulation

Demodulation

What is Demodulation?

After a signal is transmitted, the modulated signal must be converted back into the original signal to retrieve the information. This back-conversion is known as demodulation.

Demodulation is the process of extracting the original information from a modulated carrier wave. It's a crucial function in communication systems that allows for the recovery of transmitted data at the receiver.

Fig 7: Demodulation

The Demodulator

This process is performed by a demodulator, a device that converts variations in a carrier's amplitude, frequency, or phase back into the original message signal.

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How it works

A demodulator works by reversing the process of a modulator. While a modulator encodes an information signal onto a high-frequency carrier

wave, a demodulator takes that modulated signal and strips away the carrier wave to retrieve the original, low-frequency information. Think of it like unwrapping a gift—the demodulator is the person who carefully removes the wrapping paper to reveal the present inside.

Fig 8: Carrier Demodulator

Demodulation Techniques

The specific demodulation technique used depends on how the baseband signal's parameters (amplitude, frequency, or phase) were transmitted in the carrier signal.

1. For Amplitude Modulation (AM): A synchronous detector can be used.
2. For Angular Modulation (e.g., FM or PM): A frequency modulation (FM) demodulator or a phase modulation (PM) demodulator is required.

Demodulators can perform many other functions beyond simple signal recovery, including:

1. Carrier Recovery: This is a process at the receiver that reconstructs the carrier signal to allow for accurate demodulation.
2. Clock Recovery: This function extracts a synchronized timing clock from the received data, which is vital for correctly sampling the incoming digital bits.
3. Bit Slip and Frame Synchronization: These techniques are used to identify and fix the boundaries of individual bits and data frames within a continuous stream.

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4. Pulse Compression: A signal processing method that turns a long, low-power pulse into a short, high-power one to improve a signal's clarity and range.
5. Error Detection and Correction: These are methods that add redundant data to a signal so the receiver can find and, in some cases, fix errors from transmission.

Advantages and disadvantages of demodulation

Advantages of Demodulation

Demodulation is a crucial process that allows for the retrieval of the original signal, which is essential for communication systems to function.

Its key advantages include:

- a. Signal Recovery: Demodulation restores the original signal (like audio or data) from the modulated carrier wave, making it usable by the receiver.

- b. Noise Filtering: Some demodulation techniques can help filter out noise and interference, which improves the clarity and reliability of the signal.
- c. Bandwidth Efficiency: By working with modulation, demodulation allows for the efficient use of bandwidth by letting a single channel carry multiple signals.
- d. Widespread Application: Demodulation is a key process for a wide variety of technologies, including radio, television, satellite communication, and radar systems.

Disadvantages of Demodulation

Despite its importance, demodulation can have certain drawbacks, often related to the complexity and conditions of the communication channel:

- a. Complexity: Some advanced demodulation techniques require complex circuitry and precise synchronization, which makes them difficult and costly to implement.
- b. Signal Degradation: The demodulation process itself can sometimes cause signal degradation, especially if the communication channel is of poor quality.
- c. Power Consumption: Certain demodulation techniques, particularly those in digital systems, can have high power consumption, which is a key consideration for portable devices.
- d. Crosstalk: In crowded frequency spectra, there is a risk of crosstalk or interference from other signals, which presents a significant challenge for demodulators.

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Amplitude Modulation(AM) Radio: An AM signal encodes the information into the carrier wave by varying its amplitude in direct sympathy with the analogue signal to be sent. There are two methods used to demodulate AM signals namely,

Frequency Modulation (FM): Frequency modulation (FM) has numerous advantages over AM such as better fidelity and noise immunity. However, it is much more complex to both modulate and demodulate a carrier wave with FM, and AM predates it by several decades.

There are several common types of FM demodulators:

1. The quadrature detector, which phase shifts the signal by 90 degrees and multiplies it with the unshifted version. One of the terms that drops out from this operation is the original information signal, which is selected and amplified.
2. The signal is fed into a PLL and the error signal is used as the

demodulated signal.

3. The most common is a Foster-Seeley discriminator. This is composed of an electronic filter which decreases the amplitude of some frequencies relative to others, followed by an AM demodulator. If the filter response changes linearly with frequency, the final analog output will be proportional to the input frequency, as desired.

4. A variant of the Foster-Seeley discriminator called the ratio detector

5. Another method uses two AM demodulators, one tuned to the high end of the band and the other to the low end, and feed the outputs into a difference amplifier.

Phase Modulation (PM): Phase modulation (PM) decoding involves retrieving the original message signal from a carrier wave whose phase is modulated according to the message. Key techniques include:

1. **Coherent Detection:** This method requires a synchronized reference signal at the receiver, with the same frequency as the transmitter's carrier. The received signal is multiplied by this reference, and the product is integrated over a bit period. This process effectively extracts the phase information.
2. **Phase Estimation Algorithms:** In more advanced systems, especially with higher-order PSK or QAM, phase estimation algorithms like the Viterbi algorithm are employed to recover the transmitted phase.

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3. **Differential Encoding:** Since phase estimation can be ambiguous, differential encoding is often used. This means the phase shift between consecutive symbols, rather than the absolute phase, carries the information.

Multiplexing

What is multiplexing

Multiplexing (often called muxing) is a process in telecommunications and computer networking that combines multiple signals or data streams into a single signal for transmission over a shared physical medium. The primary purpose is to efficiently use a scarce resource—the transmission medium (like a single wire or cable)—by transmitting multiple channels of information simultaneously.

Fig 9: Multiplexing

Origin: The concept originated in telegraphy in the 1870s and was later applied to telephony by George Owen Squier in 1910.

Purpose: To share a limited resource and increase transmission efficiency. For example, carrying several phone calls over a single wire.

Key Concepts

1. Multiplexer (MUX): This is the device that performs the multiplexing process. It takes several individual signals and combines them into one multiplexed signal.
2. Demultiplexer (DEMUX or DMX): At the receiving end, the demultiplexer performs the reverse process. It receives the multiplexed signal and separates it back into the original individual channels or data streams.
3. Inverse Multiplexing (IMUX): This is the opposite of multiplexing. It involves breaking a single data stream into several

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smaller streams, transmitting them simultaneously over multiple channels, and then reassembling the original stream at the other end.

Types of Multiplexing

Multiplexing divides the capacity of a communication channel into several logical channels. Here are a few examples:

Frequency-Division Multiplexing (FDM)

This is an analog technology that allows multiple signals to be transmitted simultaneously over a single medium. FDM works by dividing the total available bandwidth of a communication medium into a series of non-overlapping frequency bands. Each individual signal is then modulated onto a different frequency band, allowing them all to travel over the same medium without interfering with one another.

Early attempts at FDM, known as acoustic or harmonic telegraphy, were made in the 1860s and 1870s. While these early efforts were unsuccessful, they led to a basic understanding of electroacoustic technology, which ultimately resulted in the invention of the telephone.

Fig : Frequency-Division Multiplexing (FDM)

Signals: In FDM, the signals being combined are typically electrical signals.

Receivers: To access a specific signal, a receiver must be tuned to the correct frequency band (often called a "channel").

Application of the FDM Principle

FDM separates signals by assigning each a distinct frequency range. This prevents them from overlapping and causing interference.

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1. Subcarriers: FDM can also be used to combine signals before a final modulation step. In this case, the individual carrier signals are referred to as subcarriers. A classic example is stereo FM radio, where a 38 kHz subcarrier is used to carry the left-right difference signal, separating it from the main left-right sum signal.

2. Frequency Duplex: A variation of FDM is frequency duplex, which uses different frequencies for upstream and downstream data transmission. A common example is broadband DSL, which uses separate frequencies for data going to and from your computer on the same phone line.

Application Examples

FDM is a foundational technology with many practical applications:

1. Radio and Television Broadcasting: This is the most common example of FDM, where multiple radio and TV signals are broadcast at different frequencies at the same time.
2. Cable Television: A single cable uses FDM to carry many television channels simultaneously, with each channel assigned its own distinct frequency band.
3. Telephony: FDM is used in telephone systems to transmit multiple calls over high-capacity trunk lines.
4. Communications Satellites: Satellites use FDM to transmit multiple channels of data on their uplink and downlink radio beams.

Frequency-Division Multiple Access (FDMA)

When FDM is used to allow multiple users to share a single physical communication channel, it is called Frequency-Division Multiple Access (FDMA). FDMA is the traditional method for separating radio signals from different transmitters.

Fig: Frequency Division Multiple Access (FDMA)

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Advantages and Disadvantages of FDM

Advantages of FDM

- a. Efficient Use of Bandwidth: FDM efficiently uses bandwidth by transmitting multiple signals at the same time over a single channel.
- b. No Time Synchronization Required: Unlike other methods, FDM doesn't need precise timing synchronization between the transmitter and receiver, which simplifies implementation.

c. Low Implementation Cost: Because of its analog nature and simpler hardware needs, FDM is a cost-effective technology to implement.

Disadvantages of FDM

- a. Limited Capacity: FDM is limited in the number of signals it can carry because each requires its own unique frequency band, which can be a disadvantage in applications that need many channels.
- b. Interference: FDM is susceptible to interference from signals on nearby or overlapping frequencies, which can reduce signal quality.
- c. Difficulty in Assigning Frequencies: Assigning frequencies to signals is a complex process that requires careful management to ensure "guard bands" are in place to prevent interference.

Time-Division Multiplexing (TDM)

Time-Division Multiplexing (TDM) is a method for transmitting and receiving multiple, independent signals over a common signal path. It works by dividing the transmission time on the line into a series of fixed-length, recurring time slots. Each signal gets its own dedicated time slot, taking turns on the channel. This technique is most effective when the bit rate of the transmission medium is higher than the individual signals being transmitted.

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Fig: Time Division Multiplexing

Primary Use: TDM is used primarily for digital signals, although it can be applied to analog signals as well.

Synchronization: TDM relies on synchronized switches at both the transmitting and receiving ends to ensure each signal is correctly placed into its time slot and then reassembled at the destination.

How TDM Works

The time domain is divided into a series of recurrent time slots of fixed length. A TDM frame is a complete cycle that contains one time slot for each sub-channel, along with a synchronization channel and sometimes an error correction channel.

Process:

1. A data sample from the first channel is transmitted during the first time slot.
2. A data sample from the second channel is transmitted during the

second time slot.

3. This continues for all sub-channels.

4. After all sub-channels have had their turn, the cycle repeats with a new frame.

This process of "interleaving" data allows multiple signals to share the same physical channel, giving the illusion that they are being transmitted simultaneously.

Application Examples

TDM is a fundamental technology in modern telecommunications.

Examples of its application include:

a. Digital Telephony:

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i. Plesiochronous Digital Hierarchy (PDH) systems, like the T-carrier or E-carrier, used for digital transmission of multiple telephone calls over copper cables or fiber.

ii. The Synchronous Digital Hierarchy (SDH) and Synchronous Optical Networking (SONET) standards, which have replaced PDH.

b. ISDN: The Basic Rate Interface and Primary Rate Interface for the Integrated Services Digital Network (ISDN) use TDM.

c. Audio Standards: The RIFF (WAV) audio standard uses TDM to interleave left and right stereo signals on a per-sample basis.

Time-Division Multiple Access (TDMA)

TDM can be extended into a more advanced scheme called Time-Division Multiple Access (TDMA). In TDMA, multiple stations connected to the same physical medium (e.g., sharing the same frequency channel) can communicate by taking turns to access the medium.

Application Examples:

a. The GSM mobile telephone system.

b. Tactical Data Links such as Link 16 and Link 22.

Fig : Time-Division Multiple Access (TDMA)

Advantages and Disadvantages of TDM

Advantages of TDM

a. High Capacity: TDM is highly efficient for digital transmission and can support a large number of signals over a single channel.

b. Simple Implementation for Digital Signals: This technique is a

straightforward and cost-effective solution to implement, particularly for digital data, as it doesn't require complex filtering for each channel.

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c. Precise Time Synchronization: The precise timing between the transmitter and receiver, a core requirement of TDM, ensures the accurate and reliable transmission of signals by preventing them from getting mixed up.

Disadvantages of TDM

- a. Inefficient Use of Bandwidth: TDM can be inefficient with bandwidth because time slots remain empty and unused if a channel has no data to transmit.
- b. High Cost of Implementation: The precise synchronization required for TDM can necessitate sophisticated and expensive hardware or software, making it more costly to implement.
- c. Vulnerable to Timing Jitter: The reliance on precise timing makes TDM vulnerable to "timing jitter," where a slight drift in clocks can cause the receiver to misinterpret data, leading to errors.

Wavelength-Division Multiplexing

What is Wavelength-Division Multiplexing?

In the world of fiber-optic communications, Wavelength-Division Multiplexing (WDM) is a technology that takes a number of optical carrier signals and combines (multiplexes) them onto a single optical fiber. It achieves this by using different wavelengths (i.e., different colors) of laser light.

□ Purpose: This technique is a key enabler for modern high-speed data networks. It allows for the multiplication of a fiber's capacity and even enables bidirectional communications over a single strand of fiber, a process also known as wavelength-division duplexing.

Wavelength vs. Frequency

The term WDM is typically applied to an optical carrier, which is described by its wavelength, while Frequency-Division Multiplexing (FDM) typically applies to a radio carrier, described by its frequency. This is purely a matter of convention, as wavelength and frequency are two sides of the same coin and convey the same information.

□ The Relationship: The relationship is defined by the formula:
 $\text{Frequency} \times \text{Wavelength} = \text{Velocity of the wave}.$

o In a vacuum, this velocity is the speed of light (c).

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o In a glass fiber, the velocity is a bit slower—around 0.7 times the speed of light.

Essentially, different wavelengths of light correspond to different frequencies. By using a different wavelength for each signal, WDM separates them in a way that is analogous to how FDM separates radio signals by frequency.

Fig : Wavelength-Division Multiplexing (WDM)

How Wavelength-Division Multiplexing (WDM) Works

Wavelength-Division Multiplexing (WDM) is a core technology in fiber-optic communications that dramatically increases the capacity of a single optical fiber. The fundamental principle is to transmit multiple optical carrier signals, each at a different wavelength (i.e., a different color of laser light), over the same fiber simultaneously. Think of it like a single highway with multiple lanes, where each lane is a different color of light carrying a separate stream of data.

Process of Wavelength-Division Multiplexing (WDM)

The process involves two main components:

- a. At the Transmitter: A multiplexer combines multiple optical signals, each at a different wavelength, into a single, multi-wavelength signal.
- b. During Transmission: The different wavelengths travel together down a single optical fiber, and optical amplifiers can be used to boost the signal's strength over long distances.
- c. At the Receiver: A demultiplexer separates the single multi-wavelength signal back into its individual wavelengths, sending each to its respective receiver.

Applications of Wavelength-Division Multiplexing (WDM)

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WDM is indispensable for modern high-speed networks because it allows for significant capacity expansion without the need to lay new fiber optic cables. Its key applications include:

1. Increasing Network Capacity: The primary application is to maximize the use of existing fiber infrastructure by allowing a single fiber to carry terabits of data per second.
2. Long-Haul Networks: WDM is widely used in long-distance networks, such as intercontinental undersea cables and long-haul terrestrial links, where laying new fiber is extremely costly and impractical.

3. Data Center Interconnects (DCI): It provides high-capacity links between data centers, supporting the massive data transfer needs of cloud computing and online services.

4. Telecommunications: WDM is a key enabler for modern telecom backbones, including technologies like 5G and Fiber-to-the-Home (FTTH), by providing the necessary bandwidth to handle increasing traffic.

5. Bidirectional Communication: WDM also enables bidirectional communication over a single fiber strand by using different wavelengths for transmitting and receiving signals, which is highly efficient.

Advantages and Disadvantages of Wavelength-Division

Multiplexing (WDM)

Advantages of WDM

- a. Increased Capacity: WDM dramatically increases the data-carrying capacity of a single optical fiber by allowing multiple signals to be transmitted at the same time.
- b. Cost-Effectiveness: WDM is a cost-effective way to increase bandwidth, as it avoids the significant expense and disruption of laying new fiber optic cables.
- c. Full-Duplex Transmission: This technique can enable simultaneous two-way communication on a single fiber by using a different wavelength for each direction.
- d. Flexibility and Scalability: WDM systems are highly flexible and scalable, allowing new wavelengths to be added as needed to meet growing bandwidth demands without interrupting existing services.
- e. Good Compatibility: WDM systems are transparent to the type of data, so each wavelength can carry a different signal format independently.

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Disadvantages of WDM

- a. Complexity and Cost: WDM systems, especially Dense WDM (DWDM), are complex and costly to implement due to the need for high-precision components like lasers and multiplexers.
- b. Limited Number of Wavelengths: There is a physical limit to how many wavelengths can be effectively packed onto a single fiber without causing interference, which limits the total capacity.
- c. Component Sensitivity: WDM components are very sensitive and can be vulnerable to environmental factors like temperature fluctuations.

- d. Distance Limitations: Simpler WDM technology (CWDM) has a more limited reach because it lacks amplification and uses wider wavelength spacing.
- e. Potential for Interference: Despite using guard bands, there is still a risk of crosstalk between signals if the wavelengths are not managed properly.

Code-Division Multiplexing

What is Code-Division Multiplexing?

Code-division multiplexing (CDM), also known as Code-division multiple access (CDMA) or spread spectrum, is a class of techniques where several channels simultaneously share the same frequency spectrum. Unlike other multiplexing methods that separate signals by time or frequency, CDM separates them using unique digital codes. This means multiple users can transmit at the same time and on the same frequency without interfering with one another. The bandwidth used for transmission is intentionally much higher than the actual bit rate or symbol rate.

3.2.4.2. How Code-Division Multiplexing Works

The core of CDM lies in its use of unique, channel-specific codes to spread the signal over a wide frequency band. This is done through one of two primary methods:

1. Direct Sequence Spread Spectrum (DSSS): In this method, each bit of data is represented by a unique, time-dependent sequence of short pulses called chips. The number of chips per bit is known as the spreading factor. For example, a "1" might be represented by a code like 1011011, while a "0" is represented by the inverse 0100100. All channels, each with a different code, can transmit their data on the same medium (e.g., a radio channel or

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fiber). A receiver, knowing the specific code for the channel it wants to listen to, can use it to "despread" the signal and recover the original data, while the signals from other channels appear as random noise and are rejected.

2. Frequency Hopping Spread Spectrum (FHSS): This technique involves a rapid change of the carrier frequency according to a pseudo-random sequence. Both the transmitter and receiver must know this sequence in order to stay synchronized and follow the "hopping" pattern.

Process of Code-Division Multiplexing

The process is divided into two main stages:

1. Transmission (Encoding)

1. Data Signal Generation: The original digital data stream is created.
2. Code Assignment: Each data bit is assigned a unique, channel-specific spreading code (a sequence of chips).
3. Chip Sequence Modulation: The data bits are multiplied by their spreading code, creating a much faster chip sequence with a wider bandwidth.
4. Carrier Modulation: The high-rate chip sequence modulates a carrier wave for transmission.

2. Reception (Decoding)

1. Signal Reception: The receiver captures the combined signal from all channels.
2. Local Code Generation: The receiver generates a local replica of the desired channel's spreading code.
3. Signal Despreading: The received signal is multiplied by the local code, which despreads the desired signal while the other signals remain spread and are rejected as noise.
4. Data Recovery: A low-pass filter removes the remaining high-frequency noise, and the original data is recovered.

3.2.4.4. Applications Code-Division Multiplexing

CDM is a foundational technology in modern wireless communication and has several key applications:

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1. Cellular Networks: CDM is a key technology for CDMA (Code-Division Multiple Access) cellular standards, such as IS-95 and CDMA2000. It allows multiple users to share a single radio frequency, which greatly increases the network's capacity.
2. Satellite Communications: CDM is used in satellite systems to allow multiple ground stations to communicate with a single satellite, making efficient use of the limited satellite bandwidth.
3. GPS (Global Positioning System): The GPS system uses a form of CDM to allow multiple satellites to broadcast signals on the same frequency. A GPS receiver can distinguish between the signals from different satellites based on their unique spreading codes.
4. Wi-Fi and Wireless LANs: Spread spectrum techniques, a

broader class of which CDM is a part, are also used in Wi-Fi networks to provide robust and reliable wireless communication.

Advantages and Disadvantages of Code-Division Multiplexing

Advantages of CDM

- a. Improved Signal Quality: It has better signal quality.
- b. Prevents Interference: Since the sender and receiver only know the spreading code, it prevents eavesdropping and disturbance.
- c. Increased Protection: Enhance security, avoid crashes and collisions, and use bandwidth effectively. CDMA's spread spectrum technology makes it difficult for eavesdroppers to catch signals, and the special spread spectrum code makes it possible to avoid interference and block transmission.
- d. Easy and Increased User Capacity: Adding users is easy and there is no limit to the number of users.
- e. Large signal bandwidth reduce multipath fading.
- f. Effective use of private spectrum.
- g. Resource allocation is flexible.
- h. It is very efficient process.
- i. No synchronization required.
- j. In this multiplexing, many users can share the same bandwidth.
- k. CDM is scalable.
- l. Compatible with other types of cellular technology.
- m. Uses a fixed frequency spectrum.

Disadvantages of CDM

1. As the number of users increases, the overall service quality will decrease.

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2. It needs time synchronization.
3. In CDM, the transmitted bandwidth of each user is increased than the digital data speed of the source.
4. The data transfer rate is low.
5. CDM is complex mechanism.

Data Transmission

What is Data Transmission?

Data transmission is the process of transferring digital information from a source to a destination. This occurs over a communication channel and involves both the physical movement of data and the protocols that ensure its accurate and reliable transfer. Data is represented as an electromagnetic signal, which can be an electrical voltage, a radiowave,

or an infrared signal.

Types of Data Transmission

There are two main types of data transmission:

1. **Analog Transmission:** This method conveys information using a continuous signal that varies in amplitude, phase, or frequency. Analog signals can represent both analog data (like a human voice) and digital data (like a sequence of bits).
2. **Digital Transmission:** This method transfers a discrete stream of bits. It can either be a digitized analog signal (like a scanned image) or a "born-digital" bitstream (like a computer file).

Components of a Communication System

A communication system is built from several essential components that work together to enable the successful transfer of a message.

1. **Message:** This is the information to be transmitted. It can be in various forms, such as text, audio, video, or a file.
2. **Sender:** This is the device that sends the message. Examples include a computer, a mobile phone, a video camera, or a workstation.
3. **Receiver:** This is the device that receives the message. It can be a computer, a phone, or any other device capable of receiving the data.
4. **Transmission Medium (Communication Channel):** This is the physical or wireless medium that connects the sender and receiver.

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Examples include copper wires, optical fibers, or the radio spectrum.

5. **Set of Rules (Protocol):** This is a critical component that ensures the transmitted data is understandable to the receiver. Protocols are the agreed-upon rules that govern how data is formatted, transmitted, and received. For example, without a common protocol, the message would be meaningless to the receiver.

Fig: Data Transmission

Serial Transmission

Serial transmission is a method of data transfer where bits are sent one after another, sequentially, over a single communication channel. This channel can be a single wire, a frequency, or an optical path.

Fig: Serial Communication in Microprocessor

How it Works:

Data is sent bit by bit in a specific order. To ensure the receiver knows when a new byte of data is starting and ending, additional bits (like a start bit and stop bit) may be added. This method is highly reliable over 23

long distances because it requires less complex signal processing and is less prone to timing issues.

Key Characteristics:

1. Sequential: Bits are transmitted one after the other.
2. Long-Distance: Ideal for transmitting data over long distances.
3. Fewer Wires: Requires only a single communication channel, such as a pair of wires.
4. Synchronization: Requires a common clock or timing signal to ensure the receiver can correctly interpret each bit.

Examples:

1. USB (Universal Serial Bus): Used for connecting peripherals like keyboards, mice, and flash drives.
2. SATA (Serial Advanced Technology Attachment): Connects hard drives and solid-state drives.
3. Ethernet: Modern network cables transmit data serially.

Parallel Transmission

Parallel transmission is a method of data transfer where multiple bits are sent simultaneously over several separate communication channels.

Fig: Parallel Communication in Microprocessor

How it Works:

Data is sent in groups of bits, such as 8 or 16 bits at a time, using multiple wires. This allows for a much higher data transfer rate over short distances. However, it is prone to a problem called timing skew,

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where bits on different wires arrive at slightly different times, which can corrupt the message. This issue becomes worse with distance.

Key Characteristics:

1. Simultaneous: Multiple bits are transmitted at the same time.
2. Short-Distance: Best suited for internal communication within a computer or over short distances.
3. Multiple Wires: Requires more wires than serial transmission (e.g., a 25-pin cable).
4. Timing Skew: A significant disadvantage, as it limits the effective

distance over which data can be transmitted reliably.

Examples:

1. Internal Computer Buses: Connections between the CPU and RAM often use parallel transmission.
2. Printer Parallel Ports (Centronics port): Used in older PCs to connect to printers.
3. Parallel ATA (PATA/IDE): An older standard for connecting hard drives and optical drives before the widespread adoption of SATA.

Differences Between Serial and Parallel Transmission

SERIAL TRANSMISSION	PARALLEL TRANSMISSION
In this type, a single communication link is used to transfer data from one end to another	In this type, multiple parallel links are used to transmit the data
In serial transmission, data flows in bi-direction.	In Parallel Transmission, data flows in multiple lines.
Serial Transmission is cost-efficient.	Parallel Transmission is not cost-efficient.
In serial transmission, one bit is transferred at one clock pulse.	In Parallel Transmission, eight bits are transferred at one clock pulse.
Serial Transmission is slow in comparison of Parallel Transmission.	Parallel Transmission is fast in comparison of Serial Transmission.
Generally, Serial Transmission is used for long-distance.	Generally, Parallel Transmission is used for short distance.
The circuit used in Serial Transmission is simple.	The circuit used in Parallel Transmission is relatively complex.
Serial Transmission is full duplex	Parallel Transmission is half-duplex
Converters are required in a serial transmission to convert the data between internal and parallel form	No converters are required in Parallel Transmission
Serial transmission is reliable and straightforward.	Parallel transmission is unreliable and complicated in comparison to serial transmission.
Synchronous Transmission	
Synchronous transmission is a method of data transfer where data is sent	

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in blocks or frames at a constant rate. Both the sender and receiver are synchronized by a common clock signal. This means they operate at the same speed and know exactly when to send and receive data bits, eliminating any time gaps between data.

Fig: Synchronous Transmission

How it Works

Instead of using start and stop bits for each individual byte, a continuous stream of data is sent. The receiver relies on the shared clock signal to correctly interpret the incoming data stream, identifying where each bit begins and ends. This makes synchronous transmission highly efficient for transferring large amounts of data.

Key Characteristics

1. Blocked Transmission: Data is sent in blocks or frames.
2. Common Clock: Both devices are synchronized by a shared clock.
3. Efficient: More efficient for large data transfers as there are no start/stop bits overhead.
4. Full-Duplex: The communication is a two-way, full-duplex process.

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Examples:

1. Ethernet (LAN communication): Data frames are sent in sync with a network clock.
2. SONET/SDH: Used in fiber-optic networks for telecommunications.
3. USB (Universal Serial Bus): Uses a clock-based protocol.
4. SPI (Serial Peripheral Interface): A synchronous communication protocol commonly used in microcontrollers and embedded systems.

Asynchronous Transmission

Asynchronous transmission is a method of data transfer where data is sent one byte or character at a time. The sender and receiver do not share a common clock signal. Instead, each byte is sent independently with its own timing and control bits.

How it Works

To ensure the receiver can correctly interpret each byte, a start bit is added to signal the beginning of a new byte, and a stop bit is added to mark its end. This allows for flexible communication where data is sent intermittently, as there can be variable gaps between the transmission of each byte. It's a simpler, more flexible method for transmitting data that isn't in a continuous stream.

Key Characteristics

1. Byte-based: Data is sent a byte or character at a time.
2. No Common Clock: Synchronization is not required between the sender and receiver.
3. Start/Stop Bits: Each byte includes these extra bits, which adds overhead but provides flexibility.
4. Half-Duplex: This type of transmission is often half-duplex, meaning data is sent in one direction at a time.

Examples

1. Emails and Forums: Data is not transmitted in a continuous stream.
2. RS-232: A standard for serial communication in older computers and modems.

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3. UART (Universal Asynchronous Receiver/Transmitter): A common hardware component in microcontrollers that handles asynchronous serial communication.
4. SMS Text Messages: Sent in individual bursts rather than a continuous stream.

Fig: Asynchronous Transmission

Difference Between Synchronous and Asynchronous Transmission

SYNCHRONOUS TRANSMISSION	ASYNCHRONOUS TRANSMISSION
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In Synchronous transmission, data is sent in form of blocks or frames	In Asynchronous transmission, data is sent in form of bytes or characters
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Synchronous transmission is fast.	Asynchronous transmission is slow.
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Synchronous transmission is costly.	Asynchronous transmission is economical.
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In Synchronous transmission, the time interval of transmission is constant.	In Asynchronous transmission, the time interval of transmission is not constant, it is random.
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In this transmission, users have to wait till the transmission is complete before getting a response back from the server.	Here, users do not have to wait for the completion of transmission in order to get a response from the server.
--	--

In Synchronous transmission, there is no gap present between data.	In Asynchronous transmission, there is a gap present between data.
--	--

Efficient use of transmission lines While in Asynchronous is done in synchronous transmission, the transmission line transmission. remains empty during a gap in character transmission.

The start and stop bits are not The start and stop bits are used in used in transmitting data. transmitting data that imposes

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extra overhead.

Synchronous transmission needs Asynchronous transmission does precisely synchronized clocks for not need synchronized clocks as the information of new bytes. parity bit is used in this transmission for information of new bytes.

Low latency due to real-time High latency due to processing communication. time and waiting for data to become available.

Errors are detected and corrected Errors are detected and corrected in real time. when the data is received.

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GROUP PRESENTATION

ON

NETWORKING MODELS, PROTOCOLS AND TOPOLOGIES

COURSE CODE: CSC 423

COURSE TITLE: COMPUTER NETWORKS/COMMUNICATIONS

BY

400 LEVEL COMPUTER SCIENCE DEPARTMENT STUDENTS

(GROUP 3)

SUBMITTED TO

DR. AKO RITA

COMPUTER SCIENCE DEPARTMENT

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA

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AUGUST, 2025

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INTRODUCTION

1.1 Overview of the Topic

Networking is the backbone of modern communication. Every time you send a WhatsApp message, stream a YouTube video, or log into an online class, there is a computer network working silently in the background to make that possible. Networking is basically the process of connecting two or more devices (computers, phones, servers, etc.) so they can share information and resources like files, the internet, or even printers.

In today's world, networking is not just about cables and wires; it also includes wireless technologies, cloud computing, and advanced protocols that make the internet fast, secure, and reliable. Without networking, we wouldn't have the internet, email, social media, or even online banking.

1.2 Objectives and Purpose

The main goal of this document (and our group presentation) is to help us understand the core concepts of networking in a structured way. By the end of this, you should be able to:

- Understand basic networking concepts (like how data travels from one computer to another).
- Learn about different networking models (OSI and TCP/IP).
- Explore key network protocols (like HTTP, FTP, and DNS).
- Identify network devices (like routers, switches, and servers) and their roles.
- Compare network topologies (star, mesh, ring) and know their advantages/disadvantages.
- Apply these concepts to real-world scenarios, such as how a company sets up its office network or how your phone connects to the internet.

1.2 Scope of the Document

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Agenda / Content Overview

2.1 List of Main Sections

In this presentation, our discussion will be structured into six major sections to ensure clarity and logical flow. These include:

1. Introduction & Overview
2. Networking Models
3. Network Protocols
4. Network Devices
5. Network Topologies
6. Implementation & Conclusion

2.2 Presentation Flow and Structure

Our presentation will follow a logical sequence to make sure the audience clearly understands how each topic connects to the next:

- We will begin with basic networking concepts (Introduction & Overview).
- Then, we will move into networking models, to understand the foundation.
- After that, we will explore network protocols, which define communication standards.

- This will be followed by network devices, showing the physical tools behind networks.
- Next, we will analyze different network topologies, their pros, and their applications.
- Finally, we will conclude with real-world implementations, exercises, and a summary of key findings.

Detailed Content Sections

2.1 Introduction & Overview

Each of the past three centuries has been dominated by a single technology. People were doing lot of paper work in organizations because, lack of advance systems which will help them in their day today work. The 18th century was the time of the great mechanical systems accompanying the Industrial revolution. Computer industry has made spectacular progress in short time.

During the first two decades of their existence. Computer systems were highly centralized, usually within the single large room. A medium size company or university might have had one or two computers, while large institutions had at most few dozen. The idea that within 20 years equally powerful computers smaller than postage stamps would be mass produced by the millions was pure science fiction. The merging of computers and communications has had a profound influence on the way computer systems are organized. The old model of single computer serving all of the organization computational need has been replaced by one which the-large no of separate but interconnected computers do the fob. These systems are called has computer network.

A network is a group of two of more computer systems sharing services and interacting in some manner. This interaction is, accomplished through a shared communication link, with the shared components being data. Put simply a network is a collection of machines linked both physically and through software components to facilitate communication and the sharing of information. A physical pathway known as transmission medium, connects the systems and a set of rules determines how they communicate.

These rules are known as protocols. A network protocol is software installed on a machine that determines the agreed upon set of rules for two or more machine to communicate with each other. One common metaphor used to describe different protocols is to compare them to human languages. Think of a group of people in the same room who know nothing about each other. In order for them to communicate, this group must determine what language to speak, how to handle identifying each other, whether to make general announcements or have private conversations and so on. Machines using different protocols installed can't communicate with each other. Networks are widely used by companies or on personal level also. Network for companies should provide high reliability, cost efficient, and recourse sharing

3.1.2 Types of Networking

The network can be divided into geographical areas and fall into one of two major

categories

- Local Area Network (LANs)
- Metropolitan Area Network (MANs)
- Wide Area Network (WANs)
- Wireless Networks

1. Local Area Network

A LAN is a network that covers a small geographical area, such as a single building, floor, or small office. It is typically confined to a specific location and uses a single type of transmission medium, such as cabling.

LANs are generally inexpensive to set up and can achieve high speeds (10 to 100 Mbps, or even higher in newer networks) with low delay and very few errors. They are commonly used to connect personal computers and workstations to share resources like printers and files.

2. Metropolitan Area Network (MAN)

MAN is a larger version of a LAN that spans a broader area, like a group of corporate offices in a city or an entire city itself. It can be a private network for a company or a public network. A MAN typically uses similar technology to a LAN but covers a broader area, often serving as a backbone for services like phone carriers. Unlike more complex networks, a MAN usually consists of one or two cables and does not contain many switching elements.

3. Wide Area Network (WAN)

A WAN covers a large geographical area, such as a country or continent. It is formed by connecting multiple LANs that are separated by geographical distance. For example, connecting a corporate office's LAN in one city to a branch office's LAN in another city creates a WAN. WANs contain numerous cables or telephone lines, and routers are used to connect different networks.

If two routers are not directly connected, they communicate indirectly, often using devices like modems.

4. Wireless Networks

Wireless networks are used when a wired connection is not feasible. They are essential for mobile devices like laptops and smartphones, allowing users to connect to a network even when they are away from a physical wired connection. A wireless network can be a part of a traditional LAN, using a radio link instead of a hardwired line to connect devices.

3.1.3 Importance in Modern Technology

- **Foundation of the Digital World:** Networking is the core system that enables all modern digital interactions, from social media and streaming services to online banking and e-commerce. It's the "pipes" that carry all our data.
- **Enables Global Connectivity:** It allows people and devices to communicate seamlessly across the globe, breaking down geographical barriers for work,

education, and personal connections.

- **Powers Key Innovations:** Without networking, major technologies like cloud computing, the Internet of Things (IoT), and artificial intelligence (AI) would not be possible. These rely on the ability to share and process vast amounts of data in real-time.
- **Supports Remote Work and Learning:** Networking provides the essential infrastructure for remote work, online classes, and virtual meetings, which have become a critical part of our daily lives.
- **Drives Economic Growth:** The entire digital economy, including online retail, digital marketing, and app development, is built on the efficiency and speed of interconnected networks.

Networking Models

In networking, models are like a set of blueprints that define how data is transmitted from one point to another. They break down the complex process of communication into smaller, more manageable parts, or layers. This layered approach is essential because it allows different hardware and software from various manufacturers to work together seamlessly. The two most common models are the OSI (Open Systems Interconnection) Model and the TCP/IP stack.

3.2.1 OSI Architecture

ISO defines a common way to connect computer by the architecture called Open System Interconnection (OSI) architecture. Network functionality is divided into seven layers. The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

Organization of the layers

The 7 layers can be grouped into 3 subgroups

1. Network Support Layers

Layers 1,2,3 - Physical, Data link and Network are the network support layers. They deal with the physical aspects of moving data from one device to another such as electrical specifications, physical addressing, transport timing and reliability.

2. Transport Layer

Layer4, transport layer, ensures end-to-end reliable data transmission on a single

link.

3. User Support Layers

Layers 5,6,7 – Session, presentation and application are the user support layers.

They allow interoperability among unrelated software systems

Functions of the layers

1. Physical Layer

The Physical Layer is the foundational layer of the OSI model. It coordinates the functions needed to send a raw bitstream over a physical medium. Think of it as the mechanical and electrical system for a courier service: it's not concerned with the content of the package, but rather how it's physically transported.

The Physical Layer is concerned with the following functions:

- **Physical Characteristics:** It defines the technical specifications of the interface between a device and the transmission medium. This includes aspects like pin layouts, voltage levels, and cable specifications.
- **Representation of Bits:** This layer determines how a stream of bits (1s and 0s) is encoded into signals for transmission. This process allows the digital information to be sent over physical media like copper wire, fiber optics, or radio waves.
- **Data Rate (or Transmission Rate):** The physical layer sets the number of bits that are sent per second. This is the speed of the transmission.
- **Synchronization of Bits:** It ensures that the clocks of the sending and receiving devices are synchronized at the bit level. This is crucial for the receiver to correctly interpret the bitstream.
- **Line Configuration:** This defines how two or more devices are connected. A point-to-point configuration involves a dedicated link between two devices, while a multipoint configuration allows a single link to be shared among several devices.
- **Physical Topology:** It describes the geometric layout of the network. Common topologies include mesh, bus, star, and ring, which define how devices are interconnected.
- **Transmission Mode:** The layer specifies the direction of data flow between two devices.
 - o Simplex: Data flows in one direction only.
 - o Half-Duplex: Data can flow in both directions, but not simultaneously.
 - o Full-Duplex: Data can flow in both directions at the same time.

2. Data Link Layer

The Data Link Layer is responsible for transmitting data frames from one node (device) to the next on the same network. It ensures that the data is prepared correctly for physical transmission and that it can be passed reliably from one point to another. Think of it as the local delivery service that handles packages within a single city, ensuring each one gets to the correct street address.

- **Framing:** It divides the stream of bits it receives into manageable data units called

frames.

- **Physical Addressing:** It adds a header to each frame with the physical addresses of the sender and receiver, like a street address on a package.
- **Flow Control:** This mechanism ensures that the sender doesn't send data faster than the receiver can process it, preventing data overload.
- **Error Control:** It detects and corrects errors, retransmits lost or damaged frames, and prevents duplicate frames. This is achieved by adding a trailer at the end of each frame.
- **Access Control:** It determines which device has the right to use the shared network link at any given time.
- **Hop-to-Hop Delivery:** It ensures the frame successfully moves from one device to the very next device on the network.

3. Network Layer

The Network Layer handles the delivery of packets from a source to a final destination, even if they are on different networks. This layer is crucial for communicating across network boundaries, like sending a letter from one city to another using a mail sorting system.

- **Logical Addressing:** It uses a logical addressing system, such as IP addresses, to identify the source and destination devices across different networks.
- **Routing:** It uses devices called routers to determine the best path for a packet to travel from its source to its final destination.
- **Host-to-Host Delivery:** It is responsible for delivering a packet from the initial sending device (the source host) to the final receiving device (the destination host).

4. Transport Layer

The Transport Layer is responsible for process-to-process delivery, ensuring that the data reaches the correct application on the destination computer. It is also concerned with the order and integrity of the data. Think of it as the receptionist at an office building who ensures a package is delivered to the correct person inside, not just the building itself.

- **Port Addressing:** It uses a port address in its header to direct the data to the correct application process on the destination computer.
- **Segmentation and Reassembly:** It divides the message into smaller segments, assigns a sequence number to each, and then reassembles them in the correct order at the destination.
- **Connection Control:** This layer can be connection-oriented (like a phone call, establishing a link before sending data) or connectionless (like a text message, sending data without a prior connection).
- **Flow and Error Control:** Similar to the Data Link Layer, but it operates at the process-to-process level to ensure all data arrives correctly.

5. Session Layer

The Session Layer is responsible for establishing, managing, and terminating sessions between two different applications. Its main job is to control the dialogue and synchronize

the communication.

- Dialogue Control: This function determines whether the communication is half-duplex (one-way at a time) or full-duplex (both ways simultaneously).
- Synchronization: The session layer adds checkpoints or synchronization points to a stream of data. If the network fails, the transmission can be resumed from the last checkpoint rather than starting all over again.

6. Presentation Layer

The Presentation Layer acts as a translator, ensuring that the information exchanged between two systems has a common syntax. It's like a universal language translator for the network.

- Translation: It handles the interoperability between different encoding methods used by various computers, converting the message into a common format.
- Encryption and Decryption: This layer is responsible for securing data. It transforms original data into an unreadable format (encryption) before transmission and then converts it back into its original form at the destination (decryption).
- Compression and Expansion: To improve efficiency and speed, this layer reduces the number of bits in the data (compression) and then restores the data to its original size (expansion) at the other end.

7. Application Layer

The Application Layer is the layer that directly interacts with the user. It provides the services that allow users to access the network.

- FTAM (File Transfer, Access, Management): This service enables a user to access, transfer, and manage files on a remote host.
- Mail Services: This provides email functions such as forwarding and storage.
- Directory Services: It gives users access to a database of information about various network resources and objects.
- Network Virtual Terminal (Remote Log-in): This allows a user to remotely log in to another computer, as if they were physically using it.
- Accessing the World Wide Web: This is the service that makes it possible for you to browse websites using your web browser.

3.2.2 TCP/IP stack.

The TCP/IP protocol suite operates on a four-layer conceptual model known as the DARPA model, named after the agency that created it. This model simplifies network communication into four distinct layers: Application, Transport, Internet, and Network Interface. Each of these layers corresponds to one or more layers of the more detailed, seven-layer Open Systems Interconnection (OSI) model, providing a practical framework for how data is transmitted across the internet.

1. Network Interface Layer

The Network Interface Layer, also known as the Host-to-Network Layer, is the lowest layer of the TCP/IP model. It combines the functions of the OSI model's Data Link and

Physical layers.

This layer is responsible for how a computer connects to a network and transmits data over the physical medium. It handles the details of the specific network technology being used, such as Ethernet or Wi-Fi, including the cables, network interface cards, and physical addressing.

In short, it's the layer that defines how a device physically gets an IP packet onto the network and how it receives one from the network.

2. Internet Layer

The Internet Layer is a connectionless network layer responsible for the host-to-host delivery of packets. Its primary job is to get data packets from their source to their destination, even if they have to pass through multiple networks. This layer is very similar in function to the OSI model's Network layer.

Key characteristics of the Internet Layer and its main protocol, IP (Internet Protocol), are:

- **Best-Effort Delivery:** The Internet Layer does its best to deliver packets but provides no guarantee that they will arrive. It's like sending a letter through the mail—you trust the system, but there's no promise it will arrive safely or on time.
- **No Error Checking:** This layer does not check for errors within the data. This task is left to the higher layers (specifically the Transport Layer).
- **No Tracking:** It does not keep track of the packets it sends. Packets can arrive at the destination out of the original order in which they were sent. It is up to the higher layers to reassemble them correctly.
- **Packet Routing:** A crucial function of this layer is routing, which ensures that packets find the most efficient path to their destination, helping to avoid network congestion.
- **Host-to-Host Protocol:** IP is a host-to-host protocol, meaning it is responsible for the overall journey of a packet from the source device to the destination device.

3. Transport Layer

The Transport Layer sits above the Internet Layer in the TCP/IP model. Its main purpose is to enable two hosts—a source and a destination—to have a direct conversation. This layer ensures that data is reliably transferred from one application process to another.

The Transport Layer specifies two primary end-to-end protocols:

- **TCP (Transmission Control Protocol):** TCP is a reliable, connection-oriented protocol. It ensures that a stream of bytes sent from one machine is received on the destination machine without any errors. It achieves this by breaking the data into smaller messages and passing them to the Internet Layer. At the destination, TCP reassembles the messages into the original data stream. It also has built-in flow control to prevent a fast sender from overwhelming a slower receiver with too much data. Think of TCP like a certified mail service that requires a signature and confirmation of delivery.

- **UDP (User Datagram Protocol):** UDP is an unreliable, connectionless protocol. It is used for applications where speed is more important than perfect accuracy. Unlike TCP, UDP does not perform sequencing or flow control. This makes it ideal for real-time applications such as video streaming, online gaming, and voice calls, where a few dropped packets are acceptable and re-transmitting them would cause noticeable delays. Think of UDP like a standard postcard—you send it, but you don't get a confirmation that it arrived.

4. Application Layer

The Application Layer is the top layer in the TCP/IP model, sitting directly above the Transport Layer. It is where all the high-level protocols that users interact with reside. Unlike the OSI model, which separates the Session and Presentation layers, the TCP/IP Application Layer includes their functionalities. It enables services like remote login, file transfer, and email.

The Application Layer defines several key protocols:

- **File Transfer Protocol (FTP):** Designed for the reliable transfer of files between different machines. It uses TCP at the Transport Layer to ensure reliability. FTP works on a client-server model, where an FTP client interacts with an FTP server (often called FTPd). It provides a way to move data efficiently while making differences in storage methods transparent to the user.
- **Hypertext Transfer Protocol (HTTP):** This protocol allows applications like web browsers to upload and download web pages. HTTP is a connectionless protocol that sends a request and receives a response before disconnecting. It uses TCP for reliable delivery of HTML documents, JavaScript, and other web components.
- **Simple Mail Transfer Protocol (SMTP):** SMTP is used to send emails from a local mail host to a remote one. It uses TCP to establish a connection, send the mail, and then disconnect. While it supervises the sending of email, it relies on the local mail system to distribute received mail to recipients.
- **Simple Network Management Protocol (SNMP):** This protocol is used for transporting network management information. It allows a central computer to monitor managed network devices, check their status, and receive alerts if performance drops. Unlike the other protocols listed, SNMP uses UDP for transport, which reduces network traffic overhead because it doesn't require a connection to be established and maintained.

Comparison of OSI and TCP/IP Models

Both the OSI and TCP/IP models are based on the concept of a stack of independent protocols, and they both provide end-to-end, network-independent transport services. However, there are some key differences:

- **Number of Layers:** The OSI model has seven layers, while the TCP/IP model has only four. The TCP/IP model combines the functions of the OSI's Data Link and Physical layers into the Network Interface layer, and the Session, Presentation, and Application layers into a single Application layer.

- **Connection-Oriented vs. Connectionless:** The OSI model supports both connection types at the Network layer but only connection-oriented communication at the Transport layer. The TCP/IP model, however, has only a connectionless mode at its Internet layer but supports both connection-oriented (TCP) and connectionless (UDP) modes at the Transport layer, giving users a choice.
- **Development Order:** A major difference is how they were developed. The OSI model was a theoretical framework created before the protocols were invented. This made it very general and not biased toward any specific technology. Conversely, the TCP/IP model was created after the protocols were already in use. As a result, the model fits the existing TCP/IP protocols perfectly but is less suitable for other protocol stacks.
- **Central Concepts:** The OSI model makes a clear and explicit distinction between services (what a layer does for the layer above it), interfaces (how to access those services), and protocols (how the layer performs its job). The TCP/IP model originally did not make these distinctions, making its protocols less independent and harder to replace as technology evolves.

3.3 Network Protocols

Network protocols are the official rules and standards that govern how data is formatted and transmitted between devices in a network. Without them, devices wouldn't be able to understand each other, much like people from different countries can't communicate without a shared language or a translator.

Application Layer Network Protocols

The Application Layer is the top layer, providing services that enable user-facing applications to access the network.

- **DHCP (Dynamic Host Configuration Protocol):** Automates the assignment of unique IP addresses to devices on a network from a central point.
 - o **Advantages:** Centralized IP management, easy addition of new clients, and efficient IP address reuse.
 - o **Disadvantages:** Difficult to track internet activity for a single device over time and unsuitable for devices that need to act as servers.
- **DNS (Domain Name System):** Translates human-friendly hostnames (like www.google.com) into numerical IP addresses. It uses a distributed database to resolve these names.
 - o **Advantages:** Makes the internet easy to use by eliminating the need to memorize complex IP addresses.
 - o **Disadvantages:** Susceptible to security risks like phishing attacks if root servers are compromised.
- **FTP (File Transfer Protocol):** Facilitates file sharing between hosts by creating two separate TCP connections: a control connection for commands and a data connection for the actual file transfer.

- o Advantages: Can share large files and directories, and allows for resuming interrupted transfers.
- o Disadvantages: Lacks security as data, usernames, and passwords are sent in plain text.
- HTTP (Hypertext Transfer Protocol): A request-and-response protocol for transferring information systems like web pages. It works on a client-server model and is stateless, meaning it doesn't retain information between requests.
- o Advantages: Low memory and CPU usage due to fewer concurrent connections.
- o Disadvantages: Lacks encryption and security, and requires more power to establish connections.
- IMAP (Internet Message Access Protocol): An email protocol that lets users manage emails directly on a mail server. This means emails are not downloaded and can be accessed by multiple clients concurrently.
- o Advantages: Saves local storage and makes it easy to retrieve emails since they are stored on the server.
- o Disadvantages: Requires an active internet connection and can increase storage costs for the mail server.
- POP (Post Office Protocol): An email protocol that downloads emails from the mail server to the user's local device. Once downloaded, emails are often deleted from the server.
- o Advantages: Allows users to read emails without an internet connection and frees up space on the mail server.
- o Disadvantages: Emails can be lost if the local device crashes or is stolen.
- SMTP (Simple Mail Transfer Protocol): A push protocol used to send emails from one mail system to another.
- o Advantages: Easy to install and can connect to any system without restrictions.
- o Disadvantages: Can be slow due to back-and-forth conversations between servers, and some firewalls can block its ports.
- Telnet (Terminal Emulation Protocol): Enables a user to communicate with and manage a remote device's command line interface.
- o Advantages: Compatible with multiple operating systems and provides swift connectivity for remote management.
- o Disadvantages: Lacks encryption, sending critical information in plain text.
- SNMP (Simple Network Management Protocol): Used to monitor and manage nodes on an IP network. It has three components: a managed device, an SNMP agent (on the device), and an SNMP manager (for monitoring).
- o Advantages: Used to diagnose network issues.
- o Disadvantages: Version 1 and 2 lack strong security.

Presentation Layer Network Protocols

- LPP (Lightweight Presentation Protocol): Provides support for OSI application services in specific, constrained environments that use TCP/IP. It is a streamlined protocol for applications with a limited context.

Session Layer Network Protocols

- RPC (Remote Procedure Call): A protocol that allows a program to request a service from a program on a remote computer without needing to understand the underlying network. It can use either TCP or UDP.
 - o Advantages: Omits many protocol layers to improve performance and minimizes code rewriting.
 - o Disadvantages: Not proven to work effectively over wide-area networks and has limited support for transport protocols beyond TCP/IP.

Transport Layer Network Protocols

- TCP (Transmission Control Protocol): A connection-oriented protocol that provides a reliable, ordered, and error-checked stream of data. It ensures data reaches the destination completely and without duplication.
 - o Advantages: Guarantees data delivery and automatically breaks data into packets.
 - o Disadvantages: Cannot be used for broadcast or multicast connections.
- UDP (User Datagram Protocol): A connectionless, unreliable protocol that provides a simple message service. It lacks flow control and error recovery.
 - o Advantages: Faster than TCP and can be used for broadcast and multicast connections.
 - o Disadvantages: Packets may not be delivered or may arrive out of order, and manual data packet disintegration is required.

Network Layer Protocols

- IP (Internet Protocol) - IPv4 & IPv6: Responsible for routing data packets from a source to a destination.
 - o IPv4: Uses a 32-bit address.
 - Advantages: Data encryption for security, scalable, and economical routing.
 - Disadvantages: Labor-intensive, complex, and prone to errors.
 - o IPv6: The latest version, using a 128-bit address to handle the exhaustion of IPv4 addresses.
 - Advantages: More efficient routing and better security.
 - Disadvantages: Not compatible with IPv4 machines, and upgrading devices is a challenge.
- ICMP (Internet Control Message Protocol): A supporting protocol used by network devices to send error messages and operational information.
 - o Advantages: Used to diagnose and troubleshoot network issues.
 - o Disadvantages: Too many ICMP messages can increase network traffic and

can be exploited by malicious users for denial-of-service attacks.

Data Link Layer Network Protocols

- ARP (Address Resolution Protocol): Maps IP addresses to physical MAC addresses on a local network using an ARP cache.
 - o Advantages: No need to memorize MAC addresses.
 - o Disadvantages: Susceptible to security attacks like ARP spoofing.
- SLIP (Serial Line IP): A simple packet framing protocol for point-to-point serial connections. It is used on dedicated serial links and for some dial-up purposes.
 - o Advantages: Small overhead, suitable for microcontrollers, and easy to deploy.
 - o Disadvantages: Does not support automatic network setup and is not suitable for synchronous connections.

3.3.4 Real-World Examples

- Shopping Online: When you buy a product on Amazon, your web browser uses HTTP to communicate with Amazon's servers. The payment information you enter is encrypted and sent using TCP to ensure it arrives correctly and securely.
- Sending an Email: When you compose an email, your email client uses SMTP to send it to your email provider's server. The DNS system is used to find the IP address of the recipient's mail server.
- Uploading a Photo: If you're a web developer and need to upload a new version of a website, you would likely use FTP to transfer the files from your computer to the web server.

Network Devices

3.4.1 Definitions and Roles

1. Hubs

Hubs are basic networking devices used in twisted-pair Ethernet networks. They act as a central connection point, but they are unintelligent. A hub simply receives a data packet and forwards it to every other device connected to it, regardless of the packet's intended destination. This makes them inefficient and can create performance bottlenecks, especially on busy networks.

- Passive Hub: A very basic type that only provides a pathway for electrical signals to travel.
- Active Hub: A more common type that also regenerates the signal to maintain its strength before forwarding it.

2. Switches

Switches are more intelligent than hubs. They also serve as central connection points but they only forward data to the specific device that is the intended recipient. They achieve this by learning the MAC address of each device attached to them. By creating a direct path between two devices, switches significantly reduce network collisions and improve performance.

- Full-Duplex Communication: Switches can communicate with devices in full-duplex mode, allowing devices to send and receive data at the same time, effectively doubling the data transmission speed.

- Switching Methods:

- o Cut-through: Forwards the packet as soon as it's received. It's fast but doesn't check for errors.

- o Store-and-forward: Receives and error-checks the entire packet before forwarding. It's slower but ensures no errors are propagated.

- o FragmentFree: A compromise that reads just enough of the packet to check for collisions before forwarding.

3. Bridges

Bridges are used to divide a large network into smaller segments. They operate by managing the flow of data between these segments based on the MAC addresses of the devices. A bridge will either forward data to another segment if the destination is on that side, or block it if the destination is on the same side as the sender.

- Learning Bridges: Modern bridges automatically build a list of MAC addresses on each interface by observing network traffic.

- Bridge Placement: Follows the 80/20 rule, where 80% of data should remain local to a segment, and only 20% should be destined for another segment.

- Types of Bridges:

- o Transparent Bridge: Operates invisibly to network devices, simply forwarding or blocking data based on MAC addresses.

- o Source Route Bridge: Used in Token Ring networks, where the packet itself contains information about the entire path it should take.

- o Translational Bridge: Converts one network data format to another (e.g., Token Ring to Ethernet).

4. Routers

Routers connect different networks together. They read the destination IP address in a data packet's header and use a routing table to determine the best path (or "route") for the packet to reach its destination.

- Static Routing: Routes and network information are manually entered into the routing table. Suitable for small networks.

- Dynamic Routing: Routers automatically exchange routing information with other routers using routing protocols, which allows them to build and maintain up-to-date routing tables.

- o Distance Vector Protocols (e.g., RIP): Routers share information about routes with directly connected neighbors, but they only know about the next "hop" in the journey.

- o Link State Protocols (e.g., OSPF): Routers send out special packets containing information about themselves to all other routers on the network, enabling them to build a complete map of the network topology.

5. Gateways

A device that translates one data format to another. This translation can occur between different network protocols, such as a router converting data from one network to another, or a software application converting between two file formats. It's important not to confuse a gateway with a default gateway, which is a router to which all network traffic not destined for the local network is sent.

6. CSU/DSU (Channel Service Unit/Data Service Unit)

A device that converts the digital signal format used on a Local Area Network (LAN) to a format used on a Wide Area Network (WAN). This is necessary because the underlying technologies for LANs and WANs are different.

7. Network Interface Cards (NICs)

A Network Interface Card (NIC) is a physical device that enables a computer to connect to a network. When installing a NIC, you must consider:

- **System Bus Compatibility:** The NIC must fit the computer's system bus, such as a PCI slot.
- **System Resources:** NICs require system resources like an IRQ and memory I/O addresses.
- **Media Compatibility:** The card must match the network's cabling, such as twisted-pair, coaxial, or fiber-optic.

8. Wireless Access Points (WAPs) and Modems

- **WAPs (Wireless Access Points):** A device that creates a wireless LAN (WLAN). It acts as a transceiver, allowing wireless devices to connect to a wired network. WAPs are used in an infrastructure wireless topology and can often combine the functions of a switch, DHCP server, router, and firewall.
- **Modems:** A device that converts digital signals from a computer into analog signals that can travel over a telephone line. A modem at the receiving end performs the reverse function. Modems can be internal or external, and their speed is affected by the modem itself and the UART chip in the computer.

9. Transceivers and Firewalls

- **Transceivers (Media Converters):** A device that is both a transmitter and a receiver of signals. They can be built into other devices like network cards or exist as standalone units. Media converters are a type of transceiver that allows different media types, such as twisted-pair and fiber, to be interconnected.
- **Firewalls:** A security device or software that controls network access to protect an organization's resources from external threats. Firewalls can be either software-based (like on a server's operating system) or hardware-based (a dedicated device). They are typically placed at network entry/exit points to control traffic flow.

3.1.2 Interaction Within Networks

The interaction between network devices follows a logical flow. Within a local network, a

hub or a switch acts as the central connection point for devices. If a network uses a hub, every data packet is broadcast to all devices. In contrast, a switch intelligently forwards the packet only to the correct device by using its MAC address.

To connect different local networks or a local network to the internet, a router is used. The router reads the destination IP address of a data packet and uses its routing table to determine the best path to the final destination, which may be on another network.

A bridge is used to segment a single network to reduce traffic, acting as a filter between two network segments. A gateway can also be used to connect different networks, but its primary function is to translate data formats between them.

Devices like network cards (NICs), modems, and ISDN adapters are the physical interfaces that allow individual computers to connect to these networks.

3.4.2 Use Cases

Office Network: A large office uses multiple switches to connect all computers and printers on different floors or departments. These switches are then connected to one or more central routers that manage all external traffic. A firewall is placed at the network's edge to protect against outside threats.

Internet Communication: The internet itself is a massive collection of interconnected networks. Billions of routers globally use dynamic routing protocols to continuously find the best paths for data packets to travel from any source to any destination in the world. This ensures that a request for a website in Europe, for example, can be routed efficiently from a computer in Asia.

Network Topologies

Network Topology refers to the layout of a network and describes how links and nodes are set up to relate to each other. There are five types of topology – Bus, Ring, Tree, Star, and Mesh.

Bus Topology

A bus topology connects all the devices on a network along a single cable running in a single direction from one end of the network to the other. It is sometimes called a line topology or backbone topology. Data flow on the network follows the route of the cable, moving in one direction.

Advantages of Bus Topology

Bus topologies are a good, cost-effective choice for smaller networks because the layout is simple, allowing all devices to be connected via a single coaxial cable. If needed, more nodes can be easily added to the network by joining additional cables.

Disadvantages of Bus Topology

However, because bus topologies use a single cable to transmit data, they are somewhat vulnerable. If the cable experience a failure, the whole network goes down, which can be time consuming and expensive to restore, which can be less of an issue with smaller network

Ring Topology

In ring topology the nodes are arranged in a circle (or ring). The data can travel through the ring network in either one direction or both directions, with each device having exactly two neighbours.

Advantages of Ring Topology

Since each device is only connected on either side, when data is transmitted, the packets also travel along the circle, moving through each of the intermediate nodes until they arrive at their destination. If a large network is arranged in a ring topology, repeaters can be used to ensure packets arrive correctly and without data loss.

Disadvantages of Ring Topology

Even though it's popular, a ring topology is still vulnerable to failure without proper network management. Since the flow of data transmission moves unidirectionally between nodes along each ring, if one node goes down, it can take the entire network with it. Moreover, the entire network must be taken offline to reconfigure, add, or remove nodes.

Tree Topology

The tree topology structure gets its name from how the central node functions as a sort of trunk for the network, with nodes extending outward in a branch-like fashion. A tree topology has a parent-child hierarchy to connect the nodes. Those connected to the central hub are connected linearly to other nodes, so two connected nodes only share one mutual connection. Because the tree topology structure is both extremely flexible and scalable, it's often used for wide area networks to support many spread-out devices.

Advantages of Tree Topology

Combining elements of the star and bus topologies allows for the easy addition of nodes and network expansion. Troubleshooting errors on the network is also a straightforward process, as each of the branches can be individually assessed for performance issues.

Disadvantages of Tree Topology

If the central hub fails, the various node branches will become disconnected, though connectivity within, but not between, branch systems will remain. Because of the hierarchical complexity and linear structure of the network layout, adding more nodes to a tree topology can quickly make proper management an unwieldy, not to mention costly, experience. Tree topologies are expensive because of the sheer amount of cabling required to connect each device to the next within the hierarchical layout.

Star Topology

A star topology, the most common network topology, is laid out so every node in the network is directly connected to one central hub via coaxial, twisted-pair, or fiber-optic cable. Acting as a server, this central node manages data transmission as information sent from any node on the network has to pass through the central one to reach its destination and functions as a repeater, which helps prevent data loss.

Advantages of Star Topology

Star topologies are common since they allow to conveniently managing the entire network from a single location. Because each of the nodes is independently connected to the central

hub, should one go down, the rest of the network will continue functioning unaffected, making the star topology a stable and secure layout. The devices can be added, removed, and modified without taking the entire network offline. The simplicity of the network design makes life easier for administrators, too, because it's easy to identify where errors or performance issues are occurring.

Disadvantages of Star Topology

On the other hand, if the central hub goes down, the rest of the network can't function. But if the central hub is properly managed and kept in good health, administrators shouldn't have too many issues. The overall bandwidth and performance of the network are also limited by the central node's configurations and technical specifications, making star topologies expensive to set up and operate.

Mesh Topology

A mesh topology is a structure of point-to-point connections where the nodes are interconnected. Mesh networks can be full or partial mesh. Partial mesh topologies are mostly interconnected, with a few nodes with only two or three connections, while full-mesh topologies are fully interconnected. The web-like structure of mesh topologies offers two different methods of data transmission: routing and flooding.

Advantages of Mesh Topology

Mesh topologies are reliable and stable, and the complex degree of interconnectivity between nodes makes the network resistant to failure. For instance, no single device going down can bring the network offline.

Disadvantages of Mesh Topology

Mesh topologies are incredibly labour intensive. Each interconnection between nodes requires a cable and configuration once deployed, so it can also be time consuming to set up. As with other topology structures, the cost of cabling adds up fast, and mesh networks require a lot of cabling.

3.5.3 Industry Applications

- **Bus Topology:** Largely obsolete in modern LANs but was historically used due to its simplicity.
- **Star Topology:** Most common in home, office, and campus networks because of its reliability and ease of management.
- **Ring Topology:** Historically used in Token Ring networks but is now uncommon. It is still used in some specific applications where controlled data flow is essential.
- **Tree Topology:** Used in large, hierarchical networks like a corporation's network where different departments have their own star networks connected by a central backbone.
- **Mesh Topology:** Found in critical systems where reliability is paramount, such as military command and control systems and parts of the internet's core infrastructure. It's also increasingly used in wireless sensor networks.

Conclusion

4.1 Recap of Key Points

We covered the foundational concepts of networking. We started with the basic building blocks of a network, including the various network devices like hubs, switches, routers, and gateways, each with a distinct role in directing data traffic. We then explored network protocols, the rules that govern communication, such as TCP/IP, HTTP, and DNS, which ensure devices can understand each other. Finally, we examined network topologies, which describe the physical layout of a network, including the common star, mesh, and ring configurations, each with its own advantages and disadvantages.

4.2 Significance and Relevance

Understanding these networking concepts is crucial because they are the foundation of all modern digital communication. From a simple home Wi-Fi setup to the global internet, these principles of hardware, software, and physical layout are what allow information to be shared seamlessly. A strong grasp of these topics is essential for anyone entering the IT field, as it provides the core knowledge needed to design, troubleshoot, and secure networks effectively.

4.3 Future Considerations

The field of networking is constantly evolving. As technology advances, we can expect to see new developments that will build on these core principles. The rise of 5G networks will demand more efficient routing protocols, while the Internet of Things (IoT) will require new methods for managing vast numbers of connected devices. Cloud computing and network virtualization will continue to shift the focus from physical hardware to software-defined networks, which will change how we design and manage network infrastructure.

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1. Bus Structures and Loop System

1.1 Introduction

Network topology refers to the arrangement or structure of the various elements in a computer network, including nodes, like computers, routers, and switches, and the connections between them.

It defines how devices are interconnected and how data flows within the network, and is a critical factor in determining both the performance and reliability of the network.

1.1.1 Overview of Network Architectures

Network architecture refers to the overall design and structure of a computer network, describing how devices are connected, how communication occurs, and how resources are shared. It serves as a blueprint that guides the development of reliable, efficient, and scalable networks. This architecture can be examined from two perspectives: the physical layout and the logical layout.

□ **Physical Layout:** Represents the tangible arrangement of devices, cables, switches, routers, and other hardware components in the network. It shows how nodes are physically interconnected.

□ **Logical Layout:** Represents the flow of data, addressing schemes, and communication protocols. It defines how information travels across the network regardless of the actual physical connections.

1.1.2 Importance of Topologies in Computer Networks

Network topology impacts various aspects of a network, including data transmission speeds, security, and efficiency. There are several types of network topology, each with its own advantages and disadvantages.

Choosing the right topology helps you to:

- Reduce network operational and maintenance costs,
- Increase network performance,
- Ensure optimal network health by the effective allocation of resources,
- Locate and troubleshoot errors faster.

1.1.3 Historical Context

In the 1970s and 1980s, bus topology was the foundation of early Ethernet standards. Thicknet (10Base-5) relied on thick coaxial cables with vampire taps to connect computers, while Thinnet (10Base-2) simplified installation by using thinner coaxial cables and BNC connectors. These designs were inexpensive and well-suited for small offices, but they suffered from frequent collisions and were vulnerable to single points of failure along the backbone cable.

At the same time, ring topology gained traction through IBM's Token Ring LANs, which introduced a token-passing method that allowed only one device to transmit at a time. This eliminated collisions

and provided more predictable performance than bus-based Ethernet. Later, the Fiber Distributed Data Interface (FDDI) advanced the concept by implementing dual counter-rotating rings, offering both redundancy and higher speeds of up to 100 Mbps for use in backbone networks.

1.1.4 Modern Relevance

Despite these innovations, the rapid improvements in switched Ethernet during the 1990s—driven by lower costs, greater scalability, and higher speeds—eventually displaced both bus and ring systems in mainstream networking. Today, their influence remains in specialized fields, and their design principles continue to shape modern approaches to reliability, access control, and data transmission.

Although bus and ring topologies are rarely used in modern LANs, their underlying concepts remain highly relevant. Within computers, bus-style connections still exist in the form of PCI buses, memory buses, and USB, which operates as a shared-line system through hubs. In the automotive industry, the Controller Area Network (CAN bus) applies the same principle by linking critical subsystems such as airbags, ABS, and engine control units, simplifying wiring while ensuring cost-effective and reliable communication.

Ring-based systems also continue to play an important role in telecommunications. Technologies such as SONET/SDH and metro fibre dual-ring networks employ counter-rotating loops to provide redundancy; if one path is cut, traffic is rerouted almost instantly in the opposite direction, minimizing downtime. Similarly, in industrial automation, fieldbus networks employ token-like deterministic access to guarantee that every device receives fair and predictable communication time, echoing the principles of classic token ring networks.

1.2 Bus Structure

1.2.1 Definition

A bus topology is a network design where all devices, or nodes, are connected to a single main cable.

Known as the backbone, this main cable runs through the office. All your data travels along this cable. When one device wants to communicate with another, it sends out a message along the wire.

The data travels the length of the backbone. Each connected device checks the message to see if it's

the intended recipient. If it is, the device processes the message. If not, it ignores it, allowing the message to move on to the next device.

1.2.2 Characteristics

1 Central Backbone Cable: A single coaxial cable forms the backbone of the network, serving as the medium through which all data is transmitted.

2 T-Connectors: Each device connects to the central cable using T-connectors, which allow nodes to send and receive data along the shared medium.

3 Terminators: Both ends of the backbone are fitted with terminators that absorb signals, preventing them from reflecting back and causing interference.

4 Shared Communication Channel: All devices use the same transmission line. When one device sends data, every connected node receives it, though only the addressed recipient processes it.

5 Collision Handling: Since multiple devices share the same medium, simultaneous transmissions can cause collisions. To manage this, bus networks often rely on CSMA/CD (Carrier Sense Multiple Access with Collision Detection), which ensures orderly data transmission.

6 Performance Dependency: The efficiency of the network decreases as more devices are added because the likelihood of collisions increases.

7 Simple Setup for Small Networks: Bus topology is straightforward and cost-effective for small-scale networks but becomes less reliable and harder to manage as the number of connected devices grows.

8 Passive Topology: Bus is considered a passive topology because it has no active electronics to regenerate signals along the backbone. Without terminators at both ends, signals would bounce back and forth (a phenomenon known as ringing), disrupting communication.

1.2.3 Advantages

1. Simplicity and ease of implementation: The bus topology is just a single cable—nothing too complex. This makes it really easy to set up. You don't need to juggle multiple cables or figure out the fancy configurations that come with more complicated network designs. You just grab a coaxial cable, throw in some T-connectors, and you're good to go. It's as simple as adding another seat at the table for a guest. If you know how to connect a DVD player or game console to your TV, you're basically set to connect a device in a bus topology setup.

2. Cost-effectiveness: You don't need a ton of expensive equipment to deploy a bus network. A single coaxial cable could do the job, which meant your expenses were significantly reduced. No need for pricey switches or hubs that come with other network types.

3. Minimizes clutter: Because it uses a single cable, a bus topology also minimizes clutter. This simplicity and tidiness can keep things looking professional, which might seem like a small detail, but in a work environment, it can make a big difference.

1.2.4 Disadvantages

1. Network performance issues: Each device has to wait its turn to send data, and this can slow things down considerably. Things get considerably more sluggish when you try to add a few more computers to your setup. It isn't too bad when you only have three or four devices, but once you hit six, it will feel like you are all trying to drive through a traffic jam.

2. Scalability: The more devices you add, the worse it gets. In a small office, adding a couple of extra computers might not be a big deal at first. But try doubling the number of devices, and you quickly run into snags. Soon, the cable becomes crowded with data collisions, slowing everything to a crawl.

3. Fault tolerance and single point of failure: In a bus topology, everything hinges on that one central cable. If something goes wrong with it—a cut, a kink, anything—your whole network can grind to a halt. It's like knocking out all the streetlights on a busy road. Say someone accidentally unplugs the cable while rearranging the desks. Suddenly, no one can access shared files or the printer. You all have to scramble to figure out what happened, and it can take some time to get everything back online.

4. Maintenance and troubleshooting: If your network goes down, you have to inspect the entire cable to find the problem. It can be a tedious and frustrating process.

1.2.5 Applications and Use Cases

1. Small Office and Home Networks: Bus topology was once commonly used in simple setups where a limited number of computers were connected to share files, printers, or basic resources without requiring complex infrastructure.
2. Early Ethernet Implementations: It formed the basis of older Ethernet standards such as 10Base-2 (Thin Ethernet) and 10Base-5 (Thick Ethernet), which relied on coaxial backbone cables and T-connectors.
3. Laboratory and Educational Networks: Due to its simplicity and low cost, bus topology has been used in temporary or experimental networks, especially in teaching environments to demonstrate fundamental networking concepts.
4. Industrial and Manufacturing Systems: In production facilities, bus configurations have been applied to connect sensors, controllers, and monitoring devices along a single communication line, allowing efficient data collection and process coordination.
5. Legacy System Support: Some older systems in organizations and industries may still rely on bus-based designs, making knowledge of this topology important for troubleshooting or maintaining backward compatibility.

1.3 Loop System (Ring Topology)

1.3.1 Definition

Ring topology is a network layout where every device connects to two others, forming a loop. Data travels sequentially, reducing collisions. This concept is used in older and high-speed networks.

Ring

topology connects devices in a closed loop, with each node having two neighbours. Data flows through this circle without a central hub.

While ring topology ensures balanced data distribution and steady traffic management, mesh topology provides redundant routing, allowing multiple paths for data to travel, reducing the risk of network failure if a node goes down.

Figure: Loop System

1.3.2 Characteristics

1. Closed Pathway: All devices are connected in a circular arrangement, with each node linked to exactly two others, forming a continuous loop.
2. Data Transmission Direction: Data typically travels in one direction (unidirectional ring), though some designs allow two-way flow (bidirectional rings) for fault tolerance.
3. Token Passing Mechanism: Many ring networks use a token-passing protocol, where a special control frame (the token) circulates around the ring, granting permission to transmit and preventing collisions.
4. Equal Access: Since the token ensures orderly communication, all nodes have equal opportunity to send data, avoiding the unfairness or congestion common in bus topologies.
5. Dependency on Nodes: Each node functions as a repeater, receiving and forwarding signals to the next. This maintains signal strength but also means that a single faulty node can disrupt the entire loop.

6. Predictable Performance: Because collisions are eliminated, performance is stable even under heavy traffic, making it suitable for real-time or time-sensitive applications.
7. Complexity in Setup and Maintenance: Installing and reconfiguring a ring network requires careful planning, and troubleshooting can be challenging compared to simpler topologies.

1.3.3 Advantages

1. Efficient Data Transmission: Since only one node can transmit at a time through the token-passing mechanism, packet collisions are virtually eliminated. This ensures smoother and more reliable data transfer compared to shared-medium topologies.
2. Equal Access for All Devices: Each node has an equal opportunity to transmit data when it receives the token, preventing any single device from dominating the network.
3. Cost-Effective Setup: Ring topologies can be relatively inexpensive to install, especially for small to medium-sized networks, since they require less cabling than star-based designs.
4. Predictable Performance Under Load: Even as traffic increases, the structured access method maintains consistent performance, making the topology suitable for real-time applications.
5. Simplified Fault Isolation: The sequential, node-to-node connection makes it easier to pinpoint where a failure occurs, improving troubleshooting and network management.
6. Supports Long-Distance Communication: Because each node amplifies and forwards the signal, data can travel across longer distances without significant degradation.

1.3.4 Disadvantages

1. Single Point of Failure: If one node or connection fails, it can disrupt the entire network, making it less reliable without fault-tolerant designs such as dual rings.
2. Slower Transmission in Large Networks: As the number of nodes increases, data must pass through more devices, which can reduce overall speed and efficiency since bandwidth is shared.
3. Risk of Overloading: Heavy traffic can overburden network resources, leading to delays in data transmission and reduced performance.
4. Difficult Scalability: Adding or removing nodes requires the network to be temporarily taken offline, which complicates maintenance and reduces flexibility.
5. Complex Setup in Larger Deployments: While simple for smaller installations, managing and configuring large ring networks can be more challenging compared to other topologies like star.

1.3.5 Applications and Use Cases

1. Metropolitan and Campus Networks: Ring topology has been used in city-wide and campus networks to connect multiple buildings or institutions in a loop, enabling efficient data routing across large geographical areas.
2. IBM Token Ring LANs: One of the most notable applications was in IBM's Token Ring Local Area Networks, which used the topology to manage access, eliminate collisions, and provide stable performance in office environments.
3. Telecommunication Backbones: Technologies such as FDDI (Fiber Distributed Data Interface) and SONET/SDH employed dual-ring structures to provide high-speed, reliable

communication in backbone networks.

4. Dual Ring Redundancy: In advanced implementations, a secondary ring is added to maintain data flow if one ring fails. This quick failover mechanism reduces downtime, minimizes revenue loss, and increases network reliability.

5. Critical Business and Industrial Systems: Ring topologies are suitable for environments where high reliability and predictable performance are required, such as banking systems, manufacturing plants, and mission-critical monitoring networks.

1.4 Comparative Analysis: Bus vs Loop System

Aspect Bus Structure Loop System (Ring Topology)

Data Flow A single shared backbone cable. Devices form a closed loop where connects all nodes. Transmission each packet is forwarded from node to node. Every node regenerates the signal, but only the addressed device signal, and data moves one hop at a time until it reaches its destination.

Bandwidth All nodes contend for a single Access is regulated by a circulating token. Only the token-holder transmits at a time, and collisions are prevented, giving equal access to all devices or waiting reduce efficiency.

Node/Cable A fault in the backbone cable or a break in the ring or a failed node failure brings down the entire network. Individual station bypass mechanisms are needed to maintain connectivity under failure.

Collision Collisions occur when multiple nodes transmit simultaneously. Missing tokens indicate faults, but a single break can disrupt the loop. Tolerance Protocols like CSMA/CD handle but a single break can disrupt the loop. Cable faults stop all communication.

Performance Performance degrades sharply as throughput remains predictable due to token control, but latency rises with more devices and traffic increase because of collisions and more hops since data must pass through each node.

Repeaters / Terminators Passive medium requires Active medium where each node terminates at both ends. External regenerates the signal for the next. Repeaters may be added to extend cable length (e.g., Token Ring, FDDI) for orderly access.

Add/Remove New devices can be added easily. Adding or removing devices requires tapping into the backbone, but breaking the loop and reconnecting, too many taps weaken the signal which typically takes the network

and may require re-termination. offline temporarily.

2. Data Switching Principles in Computer Networks

2.1 Introduction

2.1.1 General Overview of Data Transmission in Computer Networks

Data transmission is a fundamental process in computer networks that involves the transfer of information between devices. This is achieved through a variety of communication channels, which can be point-to-point, connecting two specific devices, or point-to-multipoint, linking one device to many others. The physical media for these channels range from traditional copper wires and optical fibres to wireless communication using radio waves. Regardless of the medium, all data is ultimately

translated into a stream of digital signals, pulses of light or electricity, referred to as "bits" that are interpreted by the receiving computer.

A cornerstone of efficient digital communication is the practice of breaking down large messages into smaller, more manageable segments known as "packets." Each packet is a self-contained unit of information, composed of two primary parts: the "payload" (the actual data) and a "header" that contains critical metadata, such as source and destination addresses. This segmentation allows

for more flexible and robust data transfer compared to sending an entire message as a single, large block. The principles of data communication, which include networking applications, communication protocols, and the key functions of routing and switching, are core topics within the fields of computer science and computer engineering.

2.1.2 The Role of Switching in Directing and Controlling Data Flow

Switching is the process of transferring these data packets or blocks from one device or network to another using specialized hardware called switches. Unlike routers, which forward packets between different networks, a switch is primarily concerned with forwarding data within a single network, such as a Local Area Network (LAN). Switches operate at the data link layer (Layer 2) of the OSI (Open Systems Interconnection) model, making them foundational to local network infrastructure.

The main function of a switch is to receive an incoming data frame or packet and examine its header

to identify the destination's unique Media Access Control (MAC) address. By maintaining an internal table that maps MAC addresses to specific physical ports, the switch can intelligently forward the packet to the correct port. This targeted delivery mechanism is a radical departure from

the behaviour of older networking devices like hubs, which would indiscriminately broadcast all incoming traffic to every connected device. This ability to direct data to a single intended recipient is what enables the sophisticated management of data flow in modern networks.

2.1.3 The Importance of Switching for Efficiency in Communication Systems

Effective switching is an indispensable element of modern IT infrastructure, playing a crucial role in enhancing network performance, optimizing efficiency, and supporting scalability. The fundamental principle behind this efficiency is intelligent, directed forwarding. The shift from a simple "fire and forget" broadcast model to a "directed and controlled" approach represents a

major architectural advancement in network design.

Older network topologies often relied on devices like hubs that broadcast all data to all connected devices, creating a "collision domain" where multiple simultaneous transmissions could collide, requiring costly retransmissions and reducing overall output. The introduction of the switch fundamentally changed this by creating dedicated, single-device collision domains for each connected port. By directing data only to its intended destination, switches significantly reduce network congestion, prevent data packet collisions, and alleviate the processing load on individual devices. This targeted delivery optimizes the utilization of network bandwidth, leading to faster and more reliable data transmission. Furthermore, the ability of switches to segment networks and intelligently manage traffic allows for easier expansion and the accommodation of a greater number

of devices and users without a corresponding decline in performance, directly enabling the massive scalability required by contemporary networks.

2.2 The Concept of Switching in Networks

2.2.1 Definition of Switching

Switching is a core process in computer networking that facilitates the movement of data between devices. This process can occur within a single local network, where switches connect devices like computers, smartphones, and printers to each other, or between different networks. Switches operate

by reading the destination address contained in a data unit's header typically the MAC address for local networks to determine the optimal port for transmission. This intelligent forwarding stands in contrast to the indiscriminate broadcasting of data by a hub, which simply sends all traffic to all ports.

2.2.2 Purpose and Necessity in Data Communication

The necessity of switching arises from the need to build scalable and efficient networks. Without this technology, network designers would have to rely on complex, inefficient technologies that require a direct, dedicated physical connection between every pair of devices. The purpose of switching is to solve this problem by introducing a managed, central point for data exchange. The key functions served by switching include:

- **Reducing Congestion:** Switches divide a network into smaller, isolated segments, which significantly reduces the volume of traffic and minimizes the risk of packet collisions. This results in a substantial improvement in network performance and output.
- **Enhancing Security:** By isolating traffic to specific ports, switches provide a higher level of data security. They can restrict unauthorized access and contain the scope of a security breach to a specific network segment.
- **Enabling Scalability:** As businesses and user bases grow, network requirements increase. Switches facilitate this growth by providing a simple way to add more devices and ports, making network expansion straightforward without the need for a complete redesign.
- **Providing Quality of Service (QoS):** Many modern switches offer advanced capabilities like Quality of Service (QoS), which allows network administrators to prioritize certain types of traffic, such as voice or video data, over less time-sensitive traffic like email. This ensures a

high-quality experience for critical applications.

2.3 General Switching Model

A simple conceptual model for switching demonstrates how it facilitates communication between devices. Imagine a central switch with multiple hosts (e.g., computers, servers) connected to its ports. When Host A wishes to send data to Host C, it transmits the data to the switch. The switch receives the data, examines its header to find the destination address, and then consults its internal

forwarding table to identify the specific port corresponding to Host C. The data is then forwarded exclusively to Host C, without being broadcast to any other connected devices.

This directed forwarding is a core component of the switching model and is what differentiates a switch from a hub. A hub would have broadcast the data to all hosts, including Host B and Host D, creating unnecessary traffic. This model highlights how a switch, by intelligently managing data flow, becomes an essential tool for creating a more efficient and secure network environment.

Figure 1: Data reaching connected devices through switch. (Source: <https://circuitglobe.com/wp-content/uploads/2020/06/switch1.jpg>)

Figure 2: Data reaching connected devices through switch. (Source: <https://circuitglobe.com/wp-content/uploads/2020/06/switch1.jpg>)

2.4 Switching Techniques

Switching is the mechanism of moving or transferring information between different networks. In large networks there might be multiple paths linking sender and receiver. Information may be switched as it travels through different channels.

In a computer network multiple devices are present and these devices share information with each other. These devices share same communication channel for sending information. Communication channel can either be wired or wireless. Thus, switching techniques is used when multiple devices share the same communication channel simultaneously. This technique improves the efficiency of the network. We have different switching techniques they are Circuit, Packet and Message Switching.

2.4.1 Circuit Switching

Circuit switching is a networking paradigm that establishes a dedicated, end-to-end communication channel between a source and a destination before any data transmission begins. This method is “connection oriented” and operates in three distinct sequential phases:

1. **Circuit Establishment:** Establishes a dedicated circuit for a single connection between the sender and recipient during the communication session once the connection is established between the two parties it will be available till the end of the conversation.
2. **Data Transfer:** Once the dedicated circuit is established, data flows a continuous, uninterrupted stream from the source to the destination usually in the form of voice.
3. **Circuit Disconnection:** After the communication session concludes, the connection is terminated. The switches on the path release the reserved resources, making the bandwidth and other network components available for new connections.

A key characteristic of circuit switching is that the circuit remains reserved for the entire duration of

the session, even during periods of silence or inactivity. This means that no other communication can use that dedicated path until it is released.

2.4.1.1 Example: Public Switched Telephone Network (PSTN)

The most well-known example of circuit switching is the traditional landline telephone system commonly known as the Public Switched Telephone Network (PSTN). When a user dials a phone number, the telephone exchanges (switching nodes) work to create a continuous electrical circuit between the two phones. For the entire duration of the call, this circuit is exclusively dedicated to that conversation, ensuring a consistent and reliable connection. This is why, even if the parties are silent, the line remains open and no one else can use that specific circuit. The use of circuit switching

in telephone communications provided the guaranteed bandwidth and low latency necessary for clear voice communication, a feature that was critical in the pre-digital era.

Figure 3: PSTN Network Connection.

2.4.1.2 Application of Circuit Switching in Emergency Call Systems and Telephones

Circuit switching remains critical for applications where a guaranteed connection and constant data stream are paramount. The most common and enduring example is traditional landline telephone (PSTN), which relies on circuit switching to provide a reliable, uninterrupted connection for voice calls. The guaranteed quality and low latency of this method are essential for ensuring a clear, continuous conversation. This same principle makes circuit switching the preferred technology for mission-critical applications like emergency services, in these situations, the ability to establish a reliable, dedicated communication path with minimal delay is a non-negotiable requirement for ensuring public safety and real time voice communication via Telephones.

Strengths:

- Dedicated communication channel circuit between two endpoints.
- Constant Transmission Rate: Data flows at a predictable and consistent speed, ideal for real time voice communication.
- No Interference During Session: Once connected, no other user can disturb or make use of the connected that circuit.
- Simple to Use: Easy to establish and understand since it behaves like a physical wire connection.
- More secure than any other switching techniques

Limitations:

- Inefficient Use of Resources: Even if no data is being transmitted (e.g. silence in a call), the reserved path stays unused but still blocked from others leading to wasted network capacity.
- Setup Delay: Time is needed to establish the circuit before data transfer can start. Take for example you dial a number and the network takes time to connect before you hear a "hello". In other words, it implies that extra time is required to establish a connection before data is sent.
- Used only for voice communication.
- Failure of Path/ Communication Loss: If any link in the dedicated circuit fails, the entire

communication breaks.

2.4.2 Packet Switching

Packet switching is a flexible method of data transmission that serves as the backbone of the modern Internet. Instead of using a fixed circuit, it divides a message into smaller units called packets. Each packet carries part of the original message, known as the payload, along with a header that contains important details such as the source address, destination address, and a sequence number.

These packets travel independently through the network. At every intermediate device, such as a router or a switch, routing algorithms decide the best path for each packet based on the current network conditions. Because of this, packets belonging to the same message may take different routes before arriving at their destination. Once received, the destination device uses the sequence numbers to arrange them back into the correct order. This process depends on the store-and-forward principle, where each packet is received, stored briefly in memory, and then forwarded to the next node.

2.4.2.1 Subtypes of Packet Switching

Packet switching is broadly divided into two main subtypes based on how the packets are handled:

- **Datagram Switching (Connectionless):** In this method, each packet, or datagram, is handled as an independent unit. No path is reserved in advance, so every datagram must carry the full addressing information. As each packet is routed separately, they may arrive in a different order than they were sent. Protocols such as Internet Protocol (IP) and User Datagram Protocol (UDP) follow this model. In this case, it is the responsibility of the receiving device to reorder packets and manage any errors.

- **Virtual Circuit Switching (Connection-Oriented):** This method combines features of both circuit and packet switching. Before communication begins, a logical path known as a virtual circuit is set up. While this is not a permanent circuit, all packets in the session follow the same route, which ensures they arrive in the correct sequence. Because of the shared path, packet headers are smaller and only need to include a connection identifier. Examples of technologies that use virtual circuits include Asynchronous Transfer Mode (ATM) and Multiprotocol Label Switching (MPLS), which is widely applied in high-performance telecommunication networks.

2.4.2.2 Example: Internet Data Communication

The Internet is the clearest example of packet switching in action. Activities such as browsing websites, sending emails, or streaming videos all rely on this method. Messages are split into thousands of packets, which may travel across different routes or nodes to avoid congestion. The receiving system then reassembles the packets to recreate the original content. This design makes the Internet highly reliable, since packets can be rerouted if one node or path fails.

Figure 4: Data transmission via data packets.

2.4.2.3 Application of Packet Switching in Email, VoIP, and Online Streaming

Packet switching is the engine behind a vast majority of modern digital communication, enabling applications that would be impractical or impossible with older switching techniques.

- **Email:** The delivery of electronic mail perfectly illustrates the principles of packet switching and its similarities to message switching. When an email is sent, the underlying network, which is packet-switched, breaks the message down into packets that are transmitted over the Internet. These packets are forwarded across various nodes until they reach the recipient's mail server, where the full message is reassembled and stored. This process operates on a "store-and-forward" principle at the application layer, which provides an asynchronous, delay-tolerant model that is highly efficient for email traffic.
- **VoIP (Voice over Internet Protocol):** VoIP technology allows for voice calls to be transmitted as IP packets over a packet-switched network, replacing the need for a dedicated, circuit-switched telephone line. To ensure high-quality, real-time audio, VoIP employs specialized protocols like the Real-time Transport Protocol (RTP) and the User Datagram Protocol (UDP). UDP is favoured for its speed and low overhead, which is critical for voice, while RTP adds timing information to the packets to help synchronize playback at the destination. To counteract the variable latency and jitter inherent in packet switching, VoIP applications use a "playout buffer," which temporarily stores incoming packets to smooth out the audio stream and deliver a consistent listening experience.
- **Online Streaming:** Modern video and audio streaming services are built on packet switching. The data is broken into packets and sent over the Internet, allowing for efficient resource sharing with other users. This is particularly important for streaming, as it allows for adaptive bitrate streaming, where the quality of the video or audio can be dynamically adjusted based on the user's current network conditions and available bandwidth. This flexibility and efficiency make packet switching ideal for handling the massive data volumes and fluctuating traffic demands of streaming services.

Strengths:

- **High Efficiency and Resource Utilization:** Packet switching allows multiple users to share the same network resources, which prevents wasted bandwidth compared to circuit switching.
- **Robustness and Scalability:** Since packets can take alternate routes, the network can handle failures and congestion effectively. It also supports the easy addition of new devices, making it suitable for large and expanding networks.
- **Flexibility:** The method supports different types of data such as text, voice, and video and adapts well to varying traffic loads.

Limitations:

- **Variable Latency and Packet Loss:** Packets may face delays, arrive out of order, or even be lost due to congestion or route changes. This can affect time-sensitive applications like voice calls and video streaming.
- **Complexity:** Managing packet switching requires advanced protocols for reordering, error control, and congestion handling, making the system more complex than circuit switching.
- **Overhead:** Each packet must carry header information for routing and reassembly. This adds extra bits that slightly reduce the effective data transfer rate.

2.4.3 Message Switching

Message switching is a technique where the entire message is treated as a single data unit and transmitted from the source to the destination via intermediate nodes (like post offices). Each intermediate node stores the complete message in memory, then forwards it to the next node (this is called store-and-forward).

How does it work?

- The sender transmits the whole message to the first switching node.
- The node stores the message in its buffer (storage).
- The message is then forwarded to the next node, and so on, until it reaches the destination.
- If the next node is busy or unavailable, the message is queued until it can be forwarded.

Characteristics of Message Switching:

- Store-and-forward Mechanism: Entire message is stored before forwarding.
- No Dedicated Path Required: Messages can take different routes depending on availability.
- Variable Message Length: Unlike packets, messages may be long or short.
- Delay Dependent on Message Size: Larger messages take longer to transmit.
- Queuing at Nodes: If a link is busy, the message is stored in a buffer until the link is free.

2.4.3.1 Early Use in Telegraph and Older Computer Networks

Message switching emerged in the 1950s as a significant improvement over earlier telegraph networks, which relied on manual relay systems. Western Union, for instance, used a message switching system called Plan 55-A for processing telegrams. The technique was also foundational to early computer networks, particularly for military and government communication, where message delivery reliability was more important than real-time performance. The original design for the ARPANET, the precursor to the Internet, was based on a message switching model using Interface Message Processors (IMPs).

Figure 5: Data transmission via intermediate nodes storing and then forwarding data to the next node. (Source:

<https://networkinterview.com/wp-content/uploads/2021/01/message-switching-1-768x437.jpg.webp>
)

2.4.3.2 Application of Message Switching in Historical Computer Networks

As a precursor to packet switching, message switching played a significant role in the early days of computer networking. It was widely used in environments where real-time communication was not a requirement, such as in early military and government communication systems. The foundational design for the ARPANET, which evolved into the Internet, was initially based on a message switching model before it was replaced by packet switching. While the technology is largely obsolete

for direct data transmission today, its "store-and-forward" principle has had a lasting impact. This concept remains a core component of many application-layer services, most notably email, where messages are sent and stored on a server until the recipient retrieves them, and in other specialized

networks with intermittent connectivity.

Strengths:

- **Efficient Bandwidth Use:** Since entire messages are stored and forwarded only when a path is available, the network can better manage bandwidth compared to circuit switching, which reserves a path even if it's idle.
- **No Dedicated Path Needed:** Unlike circuit switching, message switching doesn't require a dedicated communication line between sender and receiver. This allows more flexibility and scalability in the network.
- **Reliable Delivery:** Messages can wait in queue until the network is available.
- **Suitable for non-real time communication** such as data transfer, emails and telegrams.

Limitations:

- **Delay in Transmission:** Because entire messages are stored at each node before forwarding, it introduces significant delays, especially with large messages or during high network load.
- **Requires Large Storage Buffers:** Each node must have enough memory to store complete messages. If storage is full, incoming messages may be delayed or dropped.
- **No Real-Time Communication Support:** The delay and store-and-forward nature make it unsuitable for real-time services like voice or video calls, which require continuous and timely data flow.
- **Risk of Message Loss:** If an intermediate node fails, the stored message may be lost.

2.4.3.3 Distinction from Packet Switching

While both message switching and packet switching rely on a store-and-forward mechanism, they are fundamentally different in their operation.

- **Unit of Transmission:** The most significant difference is the unit of transmission. Message switching transmits the entire message as a single, indivisible block, which requires each intermediate node to have sufficient buffer space to store the complete message. Packet switching, conversely, divides the message into smaller packets of a limited size, which are then transmitted independently.
- **Storage and Delay:** Message switching requires a node to store the entire message, which can lead to significant delays and necessitates the use of large storage capacity, such as a hard drive. Packet switching stores small, fixed-length packets in much faster main memory, resulting in significantly lower latency.
- **Data Location:** In a message-switched network, the complete message exists in only one location within the network at a time. In a packet-switched network, different parts of the same message (the packets) can exist simultaneously at multiple locations as they traverse different paths.

The chronological development of these three techniques (Message → Circuit → Packet) reflects a progression driven by changing communication needs and technological constraints. Message switching's delay-tolerant nature was well-suited for early, unreliable networks where real-time communication was not a priority. The demand for clear, low-latency voice communication led to the development of circuit switching, which sacrifices resource efficiency for guaranteed quality. However, the rise of computer data traffic, which is inherently vast and non-continuous, exposed the

inefficiency of dedicated circuits, which were often idle. This inefficiency spurred the development of packet switching, which, by allowing multiple users to share resources dynamically, became the ideal solution for the asynchronous nature of modern data. This evolution demonstrates a consistent

trade-off between performance guarantees and resource efficiency, a core principle of network design.

Furthermore, while message and packet switching appear to be distinct, they share a fundamental architectural concept: the "store-and-forward" principle. In message switching, the entire message is buffered and stored at each intermediate node. Packet switching employs the same principle, but for each individual packet, and at a much smaller scale, using fast memory instead of slow disk storage. This illustrates a deeper concept: every network node acts as a temporary relay, a fundamental departure from the fixed-path, non-buffered nature of pure circuit switching. The refinement of the store-and-forward principle from a macro-level for entire messages to a micro-level for individual packets is what enabled the massive performance gains and scalability of modern networks.

2.5 Comparative Analysis of Switching Techniques

Table 2.5.1 Comparative Analysis of Switching Principles

This table provides a concise overview of the key differences between the three main data switching techniques, highlighting the trade-offs in their design and application.

Feature	Circuit Switching	Packet Switching	Message Switching
---------	-------------------	------------------	-------------------

Connection-oriented	Yes	No (Datagram)	No (Connectionless)
---------------------	-----	---------------	---------------------

(connectionless) or			
---------------------	--	--	--

Connection Type			
-----------------	--	--	--

Virtual Circuit			
-----------------	--	--	--

(connection-oriented)			
-----------------------	--	--	--

Continuous stream of data	Yes	No (Small, fixed- or variable-length)	No
---------------------------	-----	---------------------------------------	----

Unit of Transmission	packets		
----------------------	---------	--	--

Entire message			Yes
----------------	--	--	-----

Three phases: setup, data	Yes	No explicit	No phases
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Phases transfer, and teardown	Yes	Yes (setup/teardown in datagram; three)	No
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Feature	Circuit Switching	Packet Switching	Message Switching
---------	-------------------	------------------	-------------------

phases in virtual			
-------------------	--	--	--

circuit			
---------	--	--	--

Fixed and often wasted, as	Yes	Shared and highly	Shared and efficient,
----------------------------	-----	-------------------	-----------------------

Bandwidth Usage	bandwidth is reserved even when idle	allocated	needed
-----------------	--------------------------------------	-----------	--------

High setup delay, minimal Low setup delay, High delay, as entire transfer delay variable transfer message must be

Speed & Delay

delay due to stored

congestion

High, no packet loss High, with Low, as there is no

Reliability retransmission direct link between

protocols (e.g., TCP) source and destination

High, due to dedicated Low, due to shared Lower, as it uses

paths and resources resources network links more

Cost

efficiently than circuit

switching

Limited; complex and High; easily handles Limited; requires large

Scalability costly to scale growing traffic and storage at intermediate

devices nodes

No buffering during data Small packets Entire message stored

Memory/Buffering transfer buffered in main on disk

memory

2.5.2 Suitability for Different Communication Scenarios

The selection of a switching technique is not a simple choice between old and new technology, but a strategic decision based on the specific requirements of the communication system. This decision involves a fundamental trade-off between guaranteed performance and resource efficiency.

- Circuit Switching is best suited for applications that require long-duration, real-time, and continuous connections where consistent, predictable performance is non-negotiable. Its low latency and guaranteed bandwidth make it ideal for traditional voice telephony and mission-critical applications like emergency call systems. In such scenarios, the cost of a dropped or delayed connection is far greater than the cost of potentially wasted bandwidth.

- Packet Switching has become the dominant method for modern data networks due to its high efficiency and flexibility. It is perfectly suited for vast data traffic, such as web browsing, file transfers, and email, where data is transmitted in short, irregular bursts. Its ability to dynamically share network resources and reroute around congestion makes it the foundation of the Internet and its vast ecosystem of applications.

- Message Switching is a largely historical technique that has been superseded by packet switching for most data communication. Its use was primarily in environments where real-time delivery was not a concern, such as early telegraph networks. While no longer used as a primary network-level switching method, its "store-and-forward" principle lives on in modern application-layer services, particularly in email and some specialized networks where intermittent connectivity is expected.

This strategic choice highlights how network design balances diverse requirements. While a modern

network is typically packet-switched, some applications, like VoIP (Voice over Internet Protocol), use higher-layer protocols and buffering mechanisms to effectively emulate the performance benefits of a circuit-switched network. This approach demonstrates that network architecture can be

a hybrid of these principles, combining the resource efficiency of packet switching with the performance consistency of circuit switching to meet the demands of modern communication.

2.5.3 Benefits of Effective Switching Techniques

The widespread adoption of intelligent switching techniques has provided numerous benefits that have fundamentally reshaped computer networks and enabled the digital world as we know it. They include:

- **Improved Network Performance and Reduced Congestion:** By intelligently directing data to its specific destination, modern switches eliminate the broadcasting and collisions that characterized older networks. This targeted forwarding reduces network congestion, leading to a significant increase in data transfer speeds and overall network reliability.

- **Optimal Use of Bandwidth and Resources:** Packet switching, in particular, maximizes the efficiency of network resources. By dynamically sharing bandwidth among multiple users and transmitting data only when needed, it prevents the waste of capacity that is a hallmark of circuit switching.

- **Support for Scalability and Modern Communication Demands:** The efficiency and flexibility of packet switching allow networks to easily expand to accommodate a growing number of devices and users without a proportionate increase in infrastructure costs. This scalability, combined with its robustness against network failures, makes it a preferred method for handling the immense data volumes and dynamic traffic patterns of modern communication, from online gaming to cloud computing.

2.6 Conclusion

The principles of data switching: circuit, packet, and message switching, each represent a unique approach to managing the flow of information in a network. Circuit switching, a legacy of the telephone era, prioritizes a dedicated, continuous connection to guarantee quality and low latency. This makes it ideal for real-time, mission-critical applications where a predictable, high-quality connection is essential. In contrast, packet switching, the dominant method of the modern Internet, prioritizes efficiency and scalability by breaking data into small, independently routed packets that share network resources. While this can introduce variable delays, it provides a highly flexible and cost-effective solution for handling the immense and diverse data traffic of today's digital landscape.

Message switching, the historical precursor to packet switching, prioritized a simple store-and-forward principle, which, while no longer used as a primary network-level technique, continues to influence the architecture of many application-layer services, most notably email.

The choice between these switching techniques is not a matter of which is "best" but rather a strategic

decision based on the specific requirements of a system. A reliable network is not defined by a single, superior technology, but by the intelligent application of the right tool for the right job. The continued evolution of switching principles and their strategic combination in modern network design is what enables the reliable, scalable, and dynamic communication systems that underpin our interconnected world.

3. Broadcast Techniques

3.1 Introduction

In computer networks, communication between devices can take different forms depending on how data is delivered to its destination. These forms are collectively referred to as broadcast techniques,

since they determine whether a message is sent to one device, a group of devices, or all devices on the network. Understanding these methods is important because they directly affect network efficiency, scalability, and performance.

The three primary communication methods are:

1. Broadcast – sending data to every device in a network segment.
2. Unicast – one-to-one transmission from a single sender to a single receiver.
3. Multicast – one-to-many transmission, targeting a specific group of devices.

This section explores each method, outlining their characteristics, advantages, disadvantages, and practical applications.

3.2 Unicast Communication

3.2.1 Definition and Core Concepts

Figure 6: Illustration of unicast transmission from one sender to a single receiver.

Unicast communication is the fundamental one-to-one model in networking: data packets are sent from one source address to one destination address. Each unicast packet carries a unique destination

identifier, so exactly one host (and only one) processes it. In unicast, the sending host specifies a single recipient IP (and corresponding MAC), and network devices forward the packet along a dedicated path. This contrasts with broadcast (one-to-all) or multicast (one-to-many) modes. Unicast

is the default transmission method for most Internet traffic, meaning each request-response pair (like

a web page load or email) involves a unique sender-receiver connection.

Key concepts in unicast include:

1. Sender and Receiver: One device originates the packet (sender) and one targeted device (receiver) processes it. This direct pairing resembles a point-to-point connection, much like a phone call between two endpoints.
2. Unique IP/MAC Addresses: Every device has a unique IP address (logical layer-3 address) and a unique MAC address (layer-2 address). Unicast packets carry one IP destination. On

an Ethernet LAN, the packet's MAC address has its most-significant bit set to 0, indicating a unicast frame.

3. Point-to-Point Communication: The network establishes a single path from source to destination. Routers examine the packet's destination IP and consult routing tables (using protocols like OSPF or BGP) to forward it hop by hop. Switches use the destination MAC to send the frame only to the correct output port, minimizing unnecessary traffic.

4. Reliability and Order (TCP): Common transport-layer protocols like TCP build on unicast to provide guaranteed delivery and sequencing. Each unicast stream can be managed (retransmissions, congestion control) independently.

5. Application Support: Virtually all Internet applications (HTTP, FTP, SSH, etc.) use unicast. For example, when a browser fetches a web page, it opens a unicast TCP connection to the server's IP; similarly, file transfers and remote logins are unicast exchanges.

3.2.2 Technical Mechanisms (OSI Layers, Addressing, Routing, Protocols)

Unicast communication spans multiple layers of the network stack. At the network layer (Layer 3), each host is assigned a unique IP address. In IPv4, any address not reserved for broadcast or special

use can serve as a unicast address; IPv6 likewise defines global and link-local unicast prefixes for point-to-point hosts. A unicast packet's header includes the destination IP, which routers use to forward it. On the data link layer (Layer 2), the packet is encapsulated in a frame whose destination MAC address is that of the target host. Ethernet frames with a unicast destination (MSB = 0) are delivered by switches to the single port where the recipient resides.

1. IP Addressing: Unicast uses standard IP addressing. In IPv4, the entire address space (except 0.0.0.0/8 and 224.0.0.0/4) is generally available for unicast. IPv6 unicast addresses include unique-local and global addresses, which are structured hierarchically. Routers rely on these addresses to determine where to send packets. IP headers contain the unicast destination, and routers decrement the Time-To-Live as usual while forwarding.

2. MAC Addressing: On Ethernet, unicast frames are sent to a specific MAC address (e.g. 00:1A:2B:3C:4D:5E). A switch learns MAC-to-port mappings and forwards the frame only to the port for that MAC. Unicast MAC addresses are distinguished by having the least significant bit of the first octet set to 0.

3. Routing Protocols: For unicast, routers use well-known routing protocols to build and update forwarding tables. Interior gateway protocols like OSPF (link-state) or RIPv2 (distance-vector) advertise routes, while BGP (path-vector) handles inter-domain routing. These protocols ensure each router knows the best next hop for any unicast destination. IONOS explains that routers use OSPF and RIPv2 in particular to populate their routing tables for IPv4 unicast traffic. When a host on a different subnet needs to be reached, the source sends the packet to its default gateway; routers then pass it along the unicast route until it reaches the destination's LAN.

4. ARP and NDP: On local networks, unicast IP packets require a MAC-layer address. IPv4 uses the Address Resolution Protocol (ARP) to map the destination IP to a MAC. The ARP request itself is a broadcast, but once the MAC is known, further IP packets are pure unicast.

In IPv6, ARP is replaced by Neighbor Discovery (a set of ICMPv6 messages); one host sends a neighbor solicitation to the multicast address corresponding to the target IP, and the target replies with a unicast neighbor advertisement.

5. Transport Layer: Unicast communication most commonly employs TCP when reliability and order are needed. TCP adds headers for source/destination port, sequence numbers, and acknowledgments. Alternatively, UDP is used for unicast when speed is more important than guaranteed delivery (e.g. DNS queries, real-time voice/video in some cases). In either case, the transport layer multiplexes user data into IP packets that remain one-to-one.

6. Switching vs. Routing: On local area networks (LANs), unicast frames do not get flooded. Ethernet switches forward a unicast frame only on the port learned for the destination MAC; this conserves bandwidth on other ports. By contrast, a broadcast frame would flood all ports. Routers treat unicast IP packets differently: they consult routing tables and forward the packet to the next router or final interface. Because unicast is explicitly routed, it can traverse multiple subnets, unlike broadcast.

3.2.3 Real-World Applications and Use Cases

Unicast forms the backbone of everyday network applications. In practice, every standard client-server or peer-to-peer interaction uses unicast. Key real-world examples include:

1. Web Browsing and Internet Services: Each HTTP(S) request from a browser to a web server uses a unicast TCP connection. When a user navigates to a website, the browser resolves the server's domain name to an IP and sends a unicast request. The server then replies over the same unicast stream. Each image, document, or API response is delivered individually to that user. This holds for all web-based services (REST APIs, AJAX calls, etc.). Similarly, other TCP-based services like FTP file transfers and remote database queries are unicast.
2. Email and Messaging: Email protocols (SMTP for sending, IMAP/POP3 for receiving) operate over unicast connections. When your mail client fetches messages or sends email to an SMTP server, it opens a direct unicast session. Instant messaging and chat services (using proprietary protocols or XMPP) also exchange data one-to-one between client and server (and sometimes peer-to-peer).
3. File Sharing and Cloud Storage: Uploading or downloading files via services like Dropbox, Google Drive, or FTP involves unicast streams for each file transfer. Even if multiple users download the same file, each user receives it via a separate unicast session from the server or nearest cache.
4. Video on Demand and Streaming: On-demand streaming services (Netflix, YouTube, Amazon Prime, etc.) use unicast for each viewer's stream. Each client device establishes its own unicast connection to the streaming server or CDN edge. While the content (e.g. a movie file) might be identical for all viewers, the delivery uses individualized streams so that each user can seek or adapt bitrate independently.
5. Video Conferencing and VoIP: Virtual meeting platforms (Zoom, Microsoft Teams, etc.) typically use unicast connections between the presenter, participants, and the central server/media mixer. Each participant's audio and video are sent over separate unicast streams

(often with end-to-end encryption). As FastPix notes, corporate apps such as Zoom can involve millions of daily unicast video streams. Individual voice calls (VoIP) also rely on unicast RTP/UDP streams.

6. Online Gaming: Many multiplayer games use a client-server model with unicast UDP or TCP streams. Each player's device exchanges data directly with the game server. Even peer-to-peer gaming (or in-game chat) often falls back on unicast data exchanges between each pair of players (or via a relay).

7. Remote Access and Security: Protocols like SSH and RDP for remote terminal access create secure unicast tunnels between client and server. Network management tools often use unicast (e.g. SNMP polling, sFlow data). VPN tunnels (IPsec, OpenVPN) are unicast: encrypted packets travel point-to-point between the remote client and VPN gateway.

8. Virtual Private Networks and Tunneling: Corporate sites and clouds use unicast tunnels to link remote sites. Techniques like GRE or IPsec send packets unicast through intermediary routers, preserving the one-to-one nature of the connection.

9. Internet of Things (IoT) and Home Devices: Many IoT devices communicate unicast with cloud servers or controllers. For example, a smart thermostat may send temperature readings to its manufacturer's cloud via a unicast MQTT connection. (Occasionally IoT protocols use multicast discovery, but actual data reports/commands are usually unicast.)

10. Content Distribution Networks (CDNs): Even CDNs use unicast internally. When a client requests cached content from the nearest edge server, it does so over a unicast link. The CDN ensures availability and reliability through numerous unicast server instances.

3.2.4 Advantages of Unicast Communication

Unicast communication offers several important benefits:

1. Simplicity and Ubiquity: Unicast is straightforward to implement and manage, as it involves only point-to-point delivery. No special group management is needed; standard routing and switching tables handle it by default. All network devices and software natively support unicast. This universality means developers and network engineers have a vast toolset (TCP/IP stack, firewalls, NAT, monitoring tools) optimized for unicast traffic.

2. Targeted Delivery: Data sent via unicast reaches only the intended host. This isolation reduces wasted processing on other machines and avoids interfering with unrelated services. For example, if Host A sends data to Host B, Hosts C, D, E do not even see the unicast frames, so their CPUs or NICs need not filter or drop them. This targeted nature also makes unicast inherently more private; only the target can decrypt or act on the content if end-to-end encryption (such as TLS) is used.

3. Reliability and Control: Unicast communication can leverage reliable transport protocols. TCP ensures packet delivery, in-order arrival, and error correction for each connection. The sender and receiver can acknowledge and retransmit as needed. In contrast, multicast or broadcast often rely on UDP or custom schemes, which lack inherent reliability. With unicast, one can tune connection parameters (window size, timeouts) to optimize performance for each user.

4. Security: Encrypting a unicast stream is straightforward, since only two parties share keys.

Technologies like SSL/TLS work seamlessly on a one-to-one connection. In broadcast or multicast, encryption is more complex (group key management). Because unicast can be easily paired with authentication and encryption, it provides a high degree of confidentiality and integrity by default. An attacker must intercept or impersonate a specific session, which is harder than eavesdropping on a broadcast.

5. Quality of Service (QoS) and Traffic Control: Network operators can apply QoS policies to individual unicast flows (e.g. prioritize a VoIP call). Fine-grained traffic engineering is possible because each stream is separate. Unicast flows can be shaped, metered, or rerouted independently.

6. Customized Delivery: Each unicast connection can be optimized for its user's needs. For instance, in video streaming, different users can receive different bitrates or formats based on their bandwidth or device. In web services, each client can have personalized content. This adaptability is natural with unicast, since each data path is distinct.

7. Low Latency for Individual Streams: A direct unicast path can minimize delay for time-sensitive applications. For one-to-one communications (like a video call between two people), unicast avoids extra hops and coordination overhead that might exist in multicast or broadcast setups.

8. Compatibility and Protocol Interoperability: Unicast works through NATs and firewalls more easily than multicast/broadcast. Almost all middleboxes and network designs assume unicast. This broad compatibility saves engineers from having to upgrade infrastructure.

3.2.5 Disadvantages of Unicast Communication

Despite its many strengths, unicast has notable drawbacks, especially in scenarios involving many recipients or large-scale distribution:

1. Inefficiency for One-to-Many: The chief disadvantage of unicast is that it does not scale well when sending identical data to multiple receivers. Each receiver requires a separate stream from the sender. For example, if one server must stream a video to 100 viewers, it must send the same data 100 times. This multiplies bandwidth usage on the server's network links and multiplies load on the server CPU/network interface. Compared to multicast (one stream) or even broadcast (one stream within a LAN), unicast wastes resources by duplicating traffic.

2. Bandwidth Consumption: Because separate copies of the data traverse each link, unicast can quickly exhaust network links. In high-demand scenarios (e.g. live event streaming to many clients), the aggregate unicast traffic may far exceed available bandwidth. FastPix points out that unicast "becomes costly as the audience grows" and can "completely exhaust" source and network resources. The linear growth of traffic means networks must either overprovision or employ caching/CDNs to handle loads that multicast could inherently reduce.

3. Server and Infrastructure Load: In addition to network links, the sender's hardware is strained. Each outgoing connection uses CPU and memory resources. Managing thousands of concurrent unicast connections may require large server farms or load balancers. This adds operational cost. By contrast, with multicast the server deals with a single distribution stream.

4. Latency and Congestion: As unicast traffic volume increases, network congestion and

queuing delay can rise. In a heavily subscribed scenario, routers and switches must handle many flows, which can introduce jitter or packet loss affecting quality. In large multicast groups, the network overhead would be far smaller, but with unicast each packet must be enqueued and forwarded many times.

5. Cost: From a business perspective, the per-user cost of unicast delivery is higher for large audiences. Network operators or content providers must pay for more bandwidth and servers.

For services with millions of users (like global streaming), unicast delivery can become very expensive, whereas multicast could reduce costs if it were available end-to-end.

6. No Inherent Group Awareness: Unicast lacks mechanisms for one-to-many coordination. Unlike multicast, there is no concept of a “group address” or membership. The sender must keep track of every recipient individually. This complicates applications that want to handle dynamic groups, because they must implement their own group-management at the application level.

3.2.6 Role of Unicast in Modern Networks

Unicast remains the fundamental method of communication in computer networks, forming the basis

for how data is transmitted between two specific endpoints. In both IPv4 and IPv6, hosts are assigned

unique addresses that enable devices to identify and exchange information directly. The operation of switches and routers is optimized around this model: switches forward frames only to the port associated with the destination address, while routers use unicast routing tables to deliver packets across subnets and networks. In practice, this makes unicast communication efficient, predictable, and seamlessly integrated into the existing infrastructure without requiring additional configuration.

The ubiquity of unicast is evident in its application across local, enterprise, and global networks. Everyday traffic such as web browsing, email, and remote access depends on one-to-one delivery, and modern technologies like virtual LANs, cloud platforms, and data-center overlays are also built on unicast forwarding. Even in cases where other techniques are theoretically more efficient, such as video streaming or system updates distributed to many recipients, networks frequently rely on optimized unicast solutions, supported by caching and content delivery networks to reduce duplication. While multicast and broadcast continue to serve important purposes in specific contexts,

unicast persists as the dominant mode of communication because of its simplicity, reliability, and integration with existing protocols. It is the primary means by which the Internet and most private networks operate, and despite known limitations in scalability for one-to-many distribution, it continues to underpin the design and operation of modern network systems.

3.3 Multicast Communication

3.3.1 Definition and Core Concepts

Multicast communication is a group-oriented model: a single sender transmits data to multiple receivers at once. In IP networking, multicast involves addressing a data stream to a “multicast group” rather than to a single host. All hosts that have joined that group receive the packets. In

practical terms, multicast is “point-to-multipoint” communication. For example, an application sends packets with a destination IP in the special multicast range (224.0.0.0–239.255.255.255 for IPv4, or the range in IPv6). Any host that has registered interest in that IP (via IGMP/MLD) will get a copy. Unlike broadcast, which reaches every node, multicast delivers only to those interested parties, and unlike unicast it replicates the data only where necessary.

Key concepts in multicast communication include:

1. **Multicast Group:** A set of hosts that want to receive a common data stream. Each group is identified by a multicast IP address (class D in IPv4). Membership is dynamic: hosts can join or leave the group at any time using protocols like IGMP (IPv4) or MLD (IPv6).
2. **Source:** A single sender (or multiple senders, in some applications) transmits to the group address. Only one packet is sent out from the source; the network takes care of duplicating it for each branch that leads to group members.
3. **Group Addressing:** Packets use a multicast destination address. In IPv4, this is an address from 224.0.0.0/4. (For example, 239.10.10.10 might represent a video channel.) In IPv6, multicast addresses begin with certain prefixes. These IP addresses are distinct from the usual unicast and broadcast ranges.
4. **Selective Delivery:** Only hosts that have explicitly requested the multicast stream will receive it. Routers and switches along the path forward packets only toward network segments containing group members. This differentiates multicast from broadcast (which does not filter receivers).
5. **Layer-2 vs Layer-3:** On Ethernet, a multicast IP maps to a special MAC address (for IPv4, Ethernet MACs starting with 01:00:5E) so that switches know to potentially flood it. Switches use IGMP snooping (if enabled) to flood multicast frames only to ports with interested hosts. At the network layer, multicast relies on special routing algorithms (described below).

3.3.2 Technical Mechanisms (OSI Layers, Addressing, Routing, Protocols)

Multicast operation involves several specialized mechanisms across the network stack:

1. **Addressing:** Multicast uses reserved address ranges. In IPv4, the class D range 224.0.0.0 through 239.255.255.255 is dedicated to multicast groups. For IPv6, the FF00::/8 prefix is used, with further scoping bits to indicate link-local vs global scope. Unlike unicast, which identifies a single interface, a multicast address identifies a group. Hosts join a group by signaling through IGMP (IPv4) or MLD (IPv6). The router tracks which hosts have joined which groups on each interface.
2. **Ethernet MAC Addresses:** When an IP packet is sent to a multicast group, its destination MAC address is a special form that indicates multicast. For IPv4 multicast, switches look at the first few bits of the MAC (set to 01:00:5E). This tells a switch that the frame should be treated as multicast (and potentially forwarded to multiple ports). For IPv6, MAC addresses start with 33:33. Properly configured switches will replicate frames to all ports with group members.
3. **Routing Protocols:** Multicast requires routers to build a distribution tree. Standard unicast routing protocols are not sufficient. Instead, routers use multicast-specific protocols.

Common ones include:

- o Protocol Independent Multicast (PIM): PIM has two main modes. In dense mode (PIM-DM), routers flood multicast packets everywhere and prune back where no members exist. In sparse mode (PIM-SM), routers send join messages toward a “Rendezvous Point” (RP) and form a tree only to branches with members. PIM is “protocol independent” because it relies on the existing unicast routing information (it does not require a separate distance-vector for multicast).
- o Distance Vector Multicast Routing Protocol (DVMRP): An early protocol that creates a multicast routing table by building a tree using reverse-path forwarding (RPF). It was designed for dense networks and uses techniques from RIP.
- o Multicast OSPF (MOSPF): Extends OSPF to include multicast group membership. Routers build a complete topology and then use reverse-path checks to forward multicast along shortest-path trees.

In all cases, the Reverse Path Forwarding (RPF) algorithm is central: when a router receives a multicast packet, it checks if the packet arrived on the interface that leads back to the source according to the unicast routing table. If yes, the router forwards (replicating to other interfaces that have members). Otherwise it discards the packet to prevent loops. Truncated RPF (TRPF) further optimizes by not forwarding to segments without group members.

4. Group Membership Protocols: Hosts use IGMP (Internet Group Management Protocol) for IPv4 (and MLD in IPv6) to communicate with local routers. When an application wants to receive a stream, its host sends an IGMP “join” message for that multicast address. Routers listen for these messages to build the list of groups and local interfaces. They periodically query and hosts respond to show continuing membership. If no hosts on an interface have joined a given group, the router stops forwarding that multicast on that interface. This limits traffic to interested networks.

5. Switching Support: On a LAN, switches can snoop IGMP messages and maintain a list of which ports have members of each group. This way, when a multicast frame arrives, the switch can forward it only to ports with interested hosts rather than flooding all ports. This IGMP snooping is a common feature that preserves bandwidth on the LAN.

6. Transport Layer: Multicast applications often use UDP because it supports one-to-many. There is no built-in reliable multicast in TCP/IP (multicast reliability must be handled at the application level). Protocols like RTP/UDP or proprietary multicast formats are common. Because packets may follow different paths or encounter varying loss, applications may use forward error correction or request-selective retransmissions from neighbors.

7. OSPF/Other Uses: Even some traditional protocols use multicast addresses. For instance, OSPF routers send hello packets to 224.0.0.5 (all OSPF routers) and 224.0.0.6 (all OSPF DRs). Other protocols like VRRP use 224.0.0.18. These are examples of multicast for control-plane group messaging.

3.3.3 Real-World Applications and Use Cases

Multicast is used whenever the same data must be delivered to multiple hosts efficiently. Notable

applications include:

1. **Live Video and IPTV:** Internet Protocol Television (IPTV) services use multicast to broadcast live TV channels or live events to many set-top boxes simultaneously. Instead of streaming individually to each viewer, one multicast stream carries the content to all subscribers at once. This saves massive bandwidth on the distribution network. Similarly, corporate or university networks may use IP multicast to broadcast live video lectures or town-hall meetings to many monitors or endpoints. (As the networkacademy.io tutorial notes, multicast allows one video stream to reach only those hosts that have joined the group.)
2. **Videoconferencing and Webinars:** In large video conferences or webinars, multicast can distribute the presenter's video/audio to all participants. In this way, the host sends one stream, and the network replicates it to each campus or office location that has members. This is more efficient than having the presenter's server send thousands of unicast streams.
3. **Financial Market Feeds:** Stock exchanges and financial data providers often distribute ticker updates via multicast. Traders' workstations subscribe to specific multicast groups for market data. This ensures all subscribers see the same information simultaneously without overwhelming back-end servers.
4. **Service Discovery (mDNS/DNS-SD):** Protocols like Multicast DNS (mDNS) and DNS-Based Service Discovery use multicast to locate services on a LAN. For example, Apple's Bonjour protocol sends DNS queries to the IPv4 multicast address 224.0.0.251 (and IPv6 FF02::FB) to discover printers, file shares, or other services. Hosts responding announce their presence via multicast. This allows zero-configuration networking without a central DNS server.
5. **Routing Protocols:** Internally, many routing protocols use multicast addresses to exchange information. For example, OSPF (Open Shortest Path First) routers send hello and LSDB update packets to multicast addresses (224.0.0.5/224.0.0.6 for IPv4). This is a form of multicast use for control messages, ensuring only routers participate.
6. **Online Gaming and Real-Time Apps:** Some multiplayer games use multicast on local networks for entity updates, or to broadcast game state to nearby players. (However, on the wider Internet, games usually revert to unicast or application-layer solutions since IP multicast is not available end-to-end.)
7. **Software/Database Updates:** In data centers, a server may multicast a software patch or database update to many servers simultaneously. For example, synchronizing configuration to clustered appliances can be done via multicast. This reduces the time needed compared to sending updates individually.
8. **Industrial Control and IoT:** In some industrial and IoT networks (especially those based on Ethernet or Wi-Fi), multicast addresses are used to command groups of sensors or actuators. For instance, a controller might send a firmware upgrade packet to all devices on a subnet via multicast.
9. **Wireless Broadcast Services:** Cellular networks include multicast/broadcast services (e.g. 5G's MBMS) to deliver the same content (like emergency alerts or live video) to all phones in a cell. These rely on multicast at lower layers.

3.3.4 Advantages of Multicast Communication

Multicast communication yields significant efficiency gains and scalability when delivering identical data to multiple receivers:

1. **Bandwidth Efficiency:** A single multicast stream can serve all group members. Network links carry only one copy of the data per link regardless of the number of receivers downstream. This is far more efficient than sending N separate unicast streams. Multicast cuts down the number of transmissions because one message can be received by several destinations. For example, an IPTV channel sent via multicast will use the same bandwidth on the network whether 10 people or 10,000 people tune in.
2. **Scalability:** Because multicast does not multiply traffic with more receivers, it scales gracefully. Adding more subscribers has minimal impact on bandwidth usage. This makes multicast suitable for very large audiences. An enterprise can support streaming to thousands of desktops without requiring thousands of times more network capacity. In contrast, unicast's cost grows linearly with audience size. Sources highlights multicast's scalability for large user groups.
3. **Server/Source Load:** The sender only generates one copy of the data. This dramatically reduces CPU and network load on the source. In unicast, the server must encode and send each stream separately; with multicast, encoding happens once and is replicated by the network. In effect, the distribution burden shifts from the sender to the network, which is better optimized for replication.
4. **Cost-Effectiveness:** For large-scale distribution, multicast cuts infrastructure costs. Network administrators do not need extra bandwidth for multiple copies, and fewer server resources are needed. For example, a TV network using IP multicast to distribute channels over its backbone saves on transmission costs.
5. **Traffic Containment:** Only the network segments with interested receivers carry the traffic. Routers forward multicast packets only on links leading to group members (after pruning others). This means other parts of the network remain unaffected, making efficient use of resources.
6. **Dynamic Group Management:** IGMP/MLD allow membership to be dynamic. Users can join or leave streams on the fly without requiring the sender to know the list of receivers in advance. The network automatically adjusts which routers and switches forward traffic. This flexibility is advantageous in live scenarios (e.g. viewers tuning in or dropping out of a broadcast).
7. **Support for Many-to-Many:** While core multicast is one-to-many, some protocols support many-to-many (multiple senders to multiple receivers) by using distinct source-group addresses. This is useful for group conferencing or sensor networks, although such use cases still rely on the basic one-to-many efficiency.

3.3.5 Disadvantages of Multicast Communication

Despite its efficiency, multicast has several drawbacks that limit its use:

1. **Complex Implementation:** Enabling multicast requires special network configuration.

Routers must run multicast routing protocols (e.g. PIM) and maintain group tables. Switches may need IGMP snooping enabled. This adds administrative overhead. Setting up and managing multicast is “a little bit more complicated” than unicast, since proper protocols and group management are needed.

2. Limited Support: Multicast must be supported end-to-end. Many network devices and ISPs do not forward multicast by default. Lack of global support means multicast is usually confined to controlled environments. Users on the public Internet generally cannot subscribe to arbitrary multicast groups without specific ISP cooperation. This limits multicast’s practical deployment to private networks or specific multicast-enabled domains.

3. Reliability Challenges: There is no built-in reliable delivery in IP multicast. It normally uses UDP-like transport. Lost packets are not automatically retransmitted, so applications must implement their own error recovery if needed. This is a disadvantage relative to unicast/TCP. If network conditions cause loss or varying delays, ensuring all members receive an identical, complete stream requires complex coordination (e.g. ARQ or FEC). This makes multicast less straightforward to use for data that must be perfectly synchronized.

4. Security Risks: Multicast streams are inherently less secure. Data sent to a multicast address can be received by any host that joins the group. Controlling access is difficult. If encryption is used, distributing keys to all group members is a challenge. Multicast is “less secure” compared to unicast. An attacker who somehow learns a group address could join it and eavesdrop on the traffic. This potential for unauthorized reception is a serious concern for sensitive data.

5. Incompatibility with Firewalls/NAT: Most firewalls and NATs do not handle multicast. Because of this, many corporate or home networks simply block multicast traffic. This creates barriers for multicast crossing network boundaries, again restricting it to specific segments.

6. Group Management Overhead: In networks with many multicast groups and members, routers must track each (group, source) pair on each interface. This can consume CPU and memory. In highly dynamic environments with frequent joins/leaves, the control traffic and recalculations can be significant. Misconfigurations can also cause traffic to flood unexpectedly if groups are not pruned correctly.

7. Potential for Redundant Traffic: In dense mode or misconfigured networks, multicast can inadvertently flood segments with data even if no members exist (until pruned). This can waste bandwidth. Without strict pruning, a multicast packet might traverse parts of the network where it is not needed.

3.3.6 Role of Multicast in Modern Networks

Multicast provides a structured method for one-to-many communication, where a single stream from

the source is replicated by the network only at points where multiple receivers exist. Unlike broadcast, multicast does not reach all devices indiscriminately, and unlike unicast, it avoids the inefficiency of sending identical data separately to each recipient. Its use is concentrated in controlled environments such as enterprise backbones, service provider networks, and specialized

industries where scalability and bandwidth efficiency are critical.

In practice, multicast is enabled through group addressing and supported by standard protocols such as IGMP and MLD for host membership, and PIM for routing across networks. These mechanisms allow devices to dynamically join or leave multicast groups, ensuring that data flows only where needed. On local networks, switches employ features such as IGMP snooping to forward traffic efficiently, while in larger infrastructures multicast trees are established to optimize delivery paths. This design makes multicast a technically sound solution for distributing identical content to many receivers with minimal overhead.

However, despite its efficiency, multicast has not become a universal default. Its deployment requires deliberate configuration, careful management of routing and group policies, and support across the underlying infrastructure. As a result, it remains common in contexts such as IPTV, financial trading networks, and media distribution within data centers, but limited on the open Internet where simpler unicast-based solutions dominate. Looking ahead, multicast is expected to retain its relevance in areas such as 5G broadcasting, IoT group messaging, and large-scale media services, though its complexity ensures that it will continue to be applied selectively rather than universally.

3.4 Broadcast Communication

3.4.1 Definition and Core Concepts

Broadcast communication means sending a message from one sender to all nodes on a network segment. In other words, one source addresses data to every possible recipient. At the link layer, this is typically done by using the all-ones destination MAC address. At the network layer (IPv4), broadcasts use a special address: the limited broadcast 255.255.255.255 (reaching all hosts in the sender's IP subnet) or a directed broadcast (e.g. 192.168.1.255 for a /24 network). When a host sends a broadcast packet, every other host in the same broadcast domain receives and processes it.

In practice, broadcasting is characterized by:

1. **One-to-All Delivery:** By definition, a broadcast is delivered to every device on the LAN. No filtering by destination is done at the sender; the packet is simply sent to the broadcast address. For example, ARP requests use a broadcast so that all local hosts see the query for the MAC corresponding to a given IP.
2. **No Prior Address Knowledge:** Because broadcasts are sent to all, the sender does not need to know individual addresses. The special broadcast address implicitly replaces individual recipient addresses.
3. **Broadcast Domain:** Broadcasts are limited to a single network (or VLAN). Routers, by default, do not forward broadcast packets to other networks. This confines the scope of a broadcast to the local segment.
4. **Network Layer Definition:** Some resources define broadcast as a type of multi-point connection. The IONOS digital guide states that "a data packet is transmitted from one point

to all participants in a network” using the broadcast address. This highlights that broadcast is a fundamental communication mode alongside unicast and multicast, with its own reserved addresses.

Figure 7: Illustration of broadcast from one sender to all receivers in the network.

A core concept is that of the broadcast address. In IPv4, the address composed of all 1 bits in the host portion (given the netmask) is the directed broadcast address for that subnet. The limited broadcast 255.255.255.255 reaches every node on the current network segment. In summary, broadcasting is an “all-nodes” communication: any packet sent to the broadcast address is delivered

to all hosts. This distinguishes broadcast from unicast (one-to-one) and multicast (one-to-many to a subset): broadcast indiscriminately targets the entire LAN.

3.4.2 Technical Mechanisms (OSI Layers, Addressing, Routing, Protocols)

Broadcasting operates mainly at the data-link and network layers:

1. Addressing: All Ethernet devices examine such frames. At Layer 3 (IP), IPv4 uses special addresses for broadcast. The most common is 255.255.255.255 (limited broadcast), which all hosts on the local IP subnet receive. Alternatively, the directed broadcast address of the subnet (e.g. 192.168.1.255 for mask 255.255.255.0) also reaches all hosts in that subnet.

(IPv6 has no broadcast; instead it uses multicast, broadcast replaces recipient addresses with a special address, “mainly used when the addresses of the individual network participants are unknown”).

2. Switching Behaviour: On a local Ethernet, a switch will flood a broadcast frame out all ports (except the ingress port). This ensures all hosts see the frame. Unlike multicast, there is no selective port forwarding; every segment must receive the broadcast. In effect, the switch acts like a network-wide repeater for that frame.

3. IP Forwarding: Routers by default do not forward broadcast packets between networks. The network is divided into broadcast domains by routers. A packet sent to 255.255.255.255 on one LAN is dropped by any router it reaches (though directed broadcast behavior can be enabled on some routers for legacy reasons, it is usually disabled for security). Thus broadcasts are confined to one LAN. This is why broadcasts are also called “local broadcasts.”

4. Layer 2 vs Layer 3: Some consider broadcast primarily a Layer-2 concept. Indeed, a limited broadcast at 255.255.255.255 has no layer-3 destination within a subnet—it is immediately delivered to the link layer. In practical terms, most broadcasts are triggered by an IP stack but handled by the MAC layer.

5. No Delivery Guarantees: Like multicast, IP broadcasts typically use a connectionless protocol. A broadcast packet is sent once; hosts may listen and act on it or ignore it. There are no acknowledgments or retransmissions inherent in a broadcast. If a host’s NIC or software drops the broadcast, the sender will not know.

6. Common Protocol Use: Essential network protocols leverage broadcast addressing. For example, ARP (Address Resolution Protocol) sends a broadcast to map an IP address to a MAC. DHCP uses broadcasts extensively during the IP lease process. These protocols are

fundamental to IP networking. Beyond them, some older routing protocols (like RIP v1) broadcast updates, though modern protocols (RIP v2, OSPF) use multicast addresses instead.

3.4.3 Real-World Applications and Use Cases

Broadcasts are used primarily for network control and discovery tasks on LANs. Common real-world uses include:

1. **Address Resolution (ARP):** Whenever a host needs to determine the MAC address for an IP (on the same subnet), it sends an ARP request to the broadcast address. All hosts receive the request, but only the one with the matching IP replies. This is perhaps the single most common use of broadcast on IPv4 networks.
2. **Dynamic Host Configuration (DHCP):** A client that joins a network with no IP configuration sends DHCPDISCOVER as a broadcast. The DHCP server(s) on the LAN receive the broadcast and respond with offers. The client then broadcasts again to choose an offer. Broadcasting is essential here because the client initially does not know the server's address. (TechTarget notes that a DHCP client sends a broadcast request on the LAN to find a DHCP server.)
3. **Wake-on-LAN:** Magic packets to wake sleeping machines are sent as broadcasts (so that the target host can receive it without a dedicated address).
4. **Vendor and Service Discovery:** Some discovery protocols use broadcast. For instance, older service discovery or network scanning tools will broadcast pings or queries to find active hosts. Network administrators sometimes use broadcast pings to sweep a LAN.
5. **Routing Updates (Legacy):** Some legacy routing protocols, like RIPv1 or older proprietary protocols, broadcast route updates to all routers in the broadcast domain. Modern routing protocols have largely moved to multicast for this purpose.
6. **Local Gaming and Peer Discovery:** In a home or office LAN, certain multiplayer games or peer-to-peer apps broadcast to locate game hosts or peers. For example, a gaming console may broadcast a packet to find a game server on the LAN.
7. **Spanning Tree Protocol (STP):** Although STP BPDUs are technically multicast at MAC layer (01:80:C2:00:00:00) to the "STP all bridges" address, for practical purposes they function as broadcasts to network switches.
8. **Other LAN Utilities:** Some messaging systems and network services on a LAN (like Windows NetBIOS broadcasts in the old days) used broadcasts for announcements. While much of this has moved to DNS or multicast, broadcast remains used by some legacy protocols.

3.4.4 Advantages of Broadcast Communication

Broadcast offers several advantages for local network communication:

1. **Ease of Communication:** Sending a broadcast requires no per-recipient addressing. A single packet can reach every node without establishing individual connections. This simplicity saves the sender from needing to know all IP addresses on the network. For tasks like address resolution (ARP) or auto-configuration, this is extremely convenient.
2. **No Prior Coordination Needed:** Since the sender doesn't have to maintain a list of recipients, new hosts can simply listen for broadcasts and join the network without additional setup.

This property enables plug-and-play behavior. For example, a brand-new device can announce itself or request configuration via a broadcast immediately upon joining a LAN.

3. Rapid Distribution to All: Broadcasting is the fastest way to push a message to every connected host simultaneously. If a network administrator needs to send an emergency alert or if a service (like DHCP) needs to respond to an unknown client, using broadcast ensures no node is missed. It's essentially a built-in "mail-to-all" function.

4. Diagnostic Utility: Broadcast can help identify or troubleshoot network issues. Sending a broadcast can help identify corrupt resources or security threats on a network. For instance, a broadcast ping can quickly reveal which hosts are alive on a subnet.

5. No Additional Infrastructure Required: Broadcast does not require special multicast trees or group management. It uses standard Ethernet delivery. that Broadcasting "doesn't require any additional network infrastructure and can simultaneously send data to all nodes in a network". This makes it reliable in small networks without extra setup.

6. Saves Bandwidth for All-receivers Case: If truly every host needs the data, broadcast sends it only once over each link. Compared to sending N unicasts to N hosts, a single broadcast packet saves $N-1$ copies. In the specific case where all hosts must receive a message (e.g. an ARP announcement), broadcast is optimal.

These advantages mean that for certain functions, broadcast is the simplest and most direct solution.

For example, address resolution via ARP relies entirely on broadcast because it efficiently discovers the target without knowing which host holds the address. Likewise, DHCP broadcast discovery works without requiring prior network knowledge.

However, note that the advantage of "reducing traffic" is conditional: it reduces transmission overhead only compared to sending separate unicast frames to each host. It does create traffic on all

devices in the LAN, so it is beneficial primarily when the alternative is sending identical data N times to N hosts. In small or fully interested networks, this can indeed save bandwidth. But in any case, broadcast's simplicity of reaching all nodes with one frame is its chief benefit.

3.4.5 Disadvantages of Broadcast Communication

Broadcast communication also has significant drawbacks:

1. Broadcast Storms: If many devices respond to a broadcast simultaneously, or if multiple broadcasts occur, the network can be overwhelmed. Every host on the LAN must process each broadcast packet, consuming CPU and link bandwidth. Sources warns that if many devices broadcast at once, it "would lead to higher network traffic and ultimately result in network failure". This self-amplifying problem can bring down networks.

2. Security and Privacy Issues: A broadcast is received by every node, including malicious ones. Sensitive information sent as a broadcast can be intercepted by any host. There is no confidentiality; all data is visible to everyone in the domain. Broadcast "doesn't support large amounts of devices" and lacks security – "limited control i.e. the data is sent to everyone". This means you must trust all local hosts, which is often not acceptable.

3. Excessive Traffic for Some Receivers: In many cases, broadcasts are received by hosts that

don't need the data. This wastes processing: each host must examine every broadcast, even if it ultimately discards it. On a busy network, this overhead can degrade performance.

4. Limited Scope: Broadcast packets do not traverse routers, so they only reach the local broadcast domain. This means they cannot be used for communication across subnets. This confinement is intentional (routers drop broadcasts) but also limits usefulness to a single LAN. Any attempt to broadcast beyond a single segment requires complex workarounds (and is usually disabled to prevent amplification).

5. Not Scalable: In large networks with many hosts, broadcasts become increasingly problematic. If dozens or hundreds of devices generate periodic broadcasts (e.g. ARP, DHCP renewals, device discovery), the sheer volume can saturate the network (the so-called broadcast storm). Also, large networks often have too many hosts for a single broadcast domain, so network architects split them into VLANs to contain broadcasts.

6. Lack of Delivery Guarantees: Broadcast is inherently connectionless. If a broadcast frame collides or is lost, the sender has no way to know. Important network setup can thus fail silently if packets are dropped.

7. Network Clutter: Every broadcast appears as traffic on the local medium (wired or wireless). Excessive broadcast traffic contributes to congestion for other legitimate unicast traffic. For example, on Wi-Fi, broadcast and multicast frames are sent at the lowest basic rate, slowing down the entire WLAN.

3.4.6 Role of Broadcast in Modern Networks

Broadcast communication allows a single packet to reach all hosts within a local network segment, making it useful for discovery and configuration tasks where device addresses are not yet known. In

IPv4, broadcasts are still employed for essential functions such as ARP and DHCP, but their use is carefully constrained. Routers define broadcast domains by limiting traffic to the local subnet, and administrators segment networks with VLANs to keep broadcast scope small. Switches forward broadcasts to all ports by default, but modern equipment includes controls to limit their impact, reflecting the need to prevent network congestion and security risks. Because broadcasts can easily

generate unnecessary overhead, they are viewed as a mechanism to be minimized rather than expanded.

Modern practices increasingly replace broadcast with more efficient alternatives. IPv6 eliminates the concept entirely, relying instead on targeted multicast groups for tasks such as neighbor discovery and router advertisement. Legacy services that once depended on broadcast, such as NetBIOS, have largely shifted to DNS or multicast-based solutions. Even in wireless environments, broadcast frames are inefficient, often requiring conversion into repeated unicasts for reliability. These developments show a clear trend: broadcast remains supported, but networks are designed to

reduce its presence whenever possible.

From an evaluative perspective, broadcast is simple and indispensable for a limited set of functions,

yet unsuitable for large-scale data distribution. Its strengths lie in enabling initial communication without prior addressing knowledge, but its weaknesses include inefficiency, lack of security, and the potential for broadcast storms. Consequently, engineers restrict its role to control-plane operations while relying on unicast or multicast for broader data exchange. Broadcast therefore persists as a necessary but tightly managed mechanism, one whose continued decline is evident in the architectural choices of modern IPv6-based networks.

3.5 Comparative Analysis

3.5.1 Efficiency and Scalability

When comparing unicast, multicast, and broadcast, their differences become clear in how they use network resources as the number of recipients grows.

Bandwidth usage:

- Unicast sends a separate stream to each recipient. Thus, total bandwidth grows linearly with the number of receivers. If one server sends the same data to 100 clients, it must transmit 100 copies. This can quickly saturate links.
- Multicast sends a single stream per network branch, regardless of recipients. Even if 100 clients subscribe, the network carries essentially one copy along each link of the distribution tree. This conserves bandwidth. For example, a video stream in a campus with 100 viewers uses one stream on each hallway link instead of 100.
- Broadcast sends one copy to all nodes, using minimal link bandwidth if everyone needs it. If all devices must receive the data, a single broadcast is more efficient than 100 unicasts. However, if only some nodes need it, broadcast wastes bandwidth on uninterested hosts.

Processing and forwarding:

- Unicast packets are routed normally and forwarded by switches only to the destination port. There is no duplication in the network beyond what routers do per hop.
- Multicast packets are duplicated by routers at branch points. Switches flood multicast frames only to ports with group members (if IGMP snooping is enabled), reducing unnecessary processing on hosts. The routers handle the complexity of replication.
- Broadcast packets are flooded to all ports of every switch, so every host and router interface on the LAN must process or at least examine the frame. This can burden devices with unneeded work. Broadcast requires minimal decision-making by routers (they simply do not forward it), but at the cost of heavy local flooding.

Scalability with group size:

- Unicast scales poorly as the group grows, because each new member adds a full stream. This also means the server and routers must handle more sessions.
- Multicast scales gracefully: adding more members barely changes network load, since it still carries only one stream per link. Router CPU may rise slightly to track extra group membership, but data-plane cost is almost constant.
- Broadcast does not scale across multiple networks (limited to one segment) and burdens every device equally no matter how many hosts actually need the data. It is only efficient if truly everyone needs the same data.

Example: Imagine sending a live video to 1,000 users. With unicast, the server must send 1,000 streams (1000× bandwidth). With multicast, the server sends 1 stream; network equipment replicates as needed (roughly 1× bandwidth except at branch points). With broadcast (assuming a single LAN with 1000 hosts), the sender does one broadcast and every host processes it – bandwidth per link is 1×, but 1000 hosts each handle the packet. Multicast and broadcast are far more efficient here; unicast would require vastly more resources.

In summary, multicast offers the best efficiency and scalability for large recipient counts, because it minimizes duplicate transmissions. Unicast is efficient only for small groups or singular communication; it becomes inefficient at scale. Broadcast can quickly overwhelm the network if not all nodes need the data, so it is only efficient in the special case where every host must receive the message.

3.5.2 Security and Reliability

Security:

- Unicast connections can be secured end-to-end. Protocols like TLS/SSL are designed for unicast streams, allowing encryption and authentication between exactly two endpoints. An attacker must target that specific unicast address to intercept data. This makes unicast inherently more secure against eavesdropping.
- Multicast data is visible to the entire group. Securing multicast requires distributing encryption keys to all group members, which is complex. Unauthorized hosts that join the group can receive the traffic. There is no straightforward way to hide a multicast stream from someone who has joined the group address. Thus, multicast is considered less secure by default.
- Broadcast is the least secure: any host on the local network receives the traffic. Confidential data cannot be protected when broadcast to all. Even network information (like ARP requests) is exposed to every LAN device, including malicious ones. Networks often trust that LANs are controlled, but in open environments this poses a risk.

Reliability:

- Unicast typically uses reliable transport (TCP) or can incorporate acknowledgments. Lost packets can be retransmitted by the sender, ensuring data integrity and order. Flow control prevents overwhelming the receiver. This reliability is a major advantage for unicast.
- Multicast usually uses UDP without built-in retransmission. Ensuring reliability requires higher-level protocols or application logic (FEC, selective NACKs, etc.). This makes multicast inherently less reliable for guaranteed delivery. If a packet is lost on one branch of the multicast tree, there is no automatic recovery.
- Broadcast similarly uses connectionless delivery. No acknowledgments or retries are provided. If an important broadcast (like a DHCP renewal) is lost, the protocol must handle it (e.g. DHCP times out and retries). Generally, broadcast is used for control messages where occasional loss triggers a retry mechanism, but it is not reliable in a single attempt.

Control and Isolation:

□ Unicast ensures that only the designated recipient processes the data. This isolation helps contain problems: a misbehaving unicast stream affects only one destination's resources. Firewalls and access controls can be applied easily to unicast addresses.

□ Multicast can be confined to authorized group members via network policies, but any member can potentially disrupt the stream (e.g. by sending unauthorized data if allowed). Controlling which hosts may join a multicast group requires extra mechanisms.

□ Broadcast offers no isolation at all. A faulty or malicious host can send a broadcast that goes to all devices, potentially causing denial-of-service. For example, a broadcast storm or a malicious DHCP reply could disrupt the entire network.

In evaluation, unicast is the safest and most reliable of the three. It works with mature security and transport protocols. Broadcast and multicast trade off some of that safety and reliability for their scalability benefits. Multicast sits in the middle: more targeted than broadcast (only group members

get data) but still inherently less private and reliable than unicast. Broadcast is the simplest but least

secure and least resilient. Network architects recognize these trade-offs: they often prefer unicast for

sensitive data and accept broadcast/multicast only when the efficiency gains justify the extra complexity in security and reliability.

3.5.3 Use Cases and Domain Relevance

Each communication mode finds its niche based on domain requirements:

□ Unicast: Universally used on the global Internet and most LANs. It is relevant wherever individualized communication or flexible routing is needed. Typical domains include web services, cloud computing, corporate networks (client-server apps), IoT device management, and general data exchange. Any scenario where you must deliver content uniquely to each requester (on-demand video, interactive applications, file downloads) relies on unicast. Because of its compatibility, unicast is dominant in virtually all networks.

□ Multicast: Best suited to environments where efficient distribution to many receivers is required. Service providers (ISPs, cable operators) use multicast for IPTV and live streaming. Enterprises might use it for large group communications (e.g. corporate town halls or large multicasts in private networks). Specialized domains like financial trading employ multicast for market data feeds. Wireless and cellular networks are exploring multicast/broadcast for group content (emergency alerts, software pushes). Multicast is also used in certain industrial and IoT scenarios for group commands (e.g. smart grid). In summary, multicast's relevance is highest where many recipients share the same content simultaneously, and where the network can be controlled (e.g. a managed campus or ISP network).

□ Broadcast: Relevant primarily within a local network segment. It underpins LAN technologies. Broadcast is ubiquitous in small networks, used for ARP, DHCP, network discovery, and initial bootstrapping in almost every IPv4 network. In domain terms, every Ethernet LAN or VLAN uses broadcast for these tasks. However, broadcast's role outside

local subnets is minimal: routers break networks into smaller broadcast domains to keep broadcast traffic manageable. In home and office networks, broadcast is common but kept to network-layer protocols. In data centers and cloud environments, broadcast is usually suppressed or tunneled, with VLANs isolating each tenant's broadcasts. Essentially, broadcast remains relevant at layer 2 for fundamental protocols, but its use beyond that is limited.

An overview of domain relevance:

- Enterprise Networks: All three modes appear: unicast for apps and services, multicast for distribution within campuses, broadcast for DHCP/ARP. Enterprises carefully segment broadcasts with VLANs and use multicast for specific needs (e.g. video distribution).
- Service Provider Networks: Unicast for general Internet traffic, multicast for IPTV or VoIP conferencing, broadcast is typically local-only (e.g. within a subscriber's home network).
- Wireless/Wi-Fi: Unicast is the norm. Broadcast (and multicast) traffic is sometimes optimized (converted to multiple unicasts) to ensure delivery to all clients. Wi-Fi uses broadcast for management (beacons, ARP).
- Cloud and Data Centers: Largely unicast. Broadcast domains are tightly controlled (per tenant VLANs). Multicast may be used within clusters (e.g. HPC broadcast) but is not common in public cloud services.
- Mobile/5G: Unicast for standard Internet use; broadcast/multicast are used for specialized services (e.g. MBMS for video, multicast for group messaging in IoT).
- IoT/Smart Home: Unicast to cloud services, multicast to discover devices or propagate group commands (e.g. a smart bulb group command). Broadcast is often avoided, but used for legacy IPv4 device discovery.

Thus, the choice depends on the domain. If you are streaming content on a managed network (IPTV or corporate video), multicast is most relevant. If you are browsing the web or using client-server apps, unicast is the tool. If you are configuring devices or establishing initial connections on a LAN, broadcast is necessary. Each mode complements the others: real networks use a mixture, choosing unicast for one-to-one, multicast for efficient one-to-many, and broadcast for fundamental link-layer tasks.

3.5.4 Emerging Trends and Future Directions

Network communication is evolving, but unicast, multicast, and broadcast continue to play roles:

- Software-Defined Networking (SDN) and Network Virtualization: SDN allows dynamic control of forwarding rules. In the future, SDN controllers could set up multicast trees on demand or convert broadcasts into selective floods. Network virtualization overlays (like VXLAN with EVPN) are improving how broadcast/multicast traffic is handled across data centers. For instance, EVPN provides efficient replication of broadcast, unknown unicast, and multicast across fabric. These trends aim to make group communication more manageable and efficient in virtualized environments.
- 5G and Beyond: Next-generation mobile standards incorporate group broadcast modes. For example, 5G specifies "Multicast Broadcast Service" (MBS) to distribute the same content

(news, alerts, video) to many users in a cell. Future wireless networks may generalize multicast to support massive IoT, where a command might be multicast to all devices of a certain type. This could revive broader use of multicast in service-provider domains.

□ Internet of Things (IoT): Many IoT protocols leverage group communication. Constrained Application Protocol (CoAP) supports IP multicast for resource discovery. Protocols like Zigbee or Thread use mesh broadcasts or multicast for device discovery and group commands. As IoT deployments grow, efficient one-to-many messaging becomes crucial, potentially boosting multicast adoption (or its application-layer equivalents) at the edge.

□ Content Distribution and Edge Computing: Content providers continue to explore ways to mimic multicast benefits. Content Delivery Networks (CDNs) and peer-to-peer overlays effectively create application-layer multicast, distributing content in a staged fashion. Edge computing may introduce localized “mini-CDNs” where data is distributed via multicast within edge clusters before unicast to each edge node. While not IP multicast, it reflects the desire for efficient distribution.

□ Multicast over the Internet: There is ongoing research into enabling multicast on the broader Internet. Technologies like BIER (Bit Indexed Explicit Replication) aim to allow multicast routing without per-group state in routers. While not yet mainstream, such advances may someday make multicast viable beyond controlled domains.

□ Security Enhancements: There is work on secure multicast (e.g. group key management protocols, encrypted group membership) to address multicast’s security issues. If these mature, multicast might handle sensitive data more safely. Similarly, broadcast suppression techniques and secure VLAN overlays continue to evolve to mitigate broadcast storm risks.

□ IPv6 Adoption: As IPv6 becomes dominant, all IPv4 broadcasts (like ARP) are naturally replaced by IPv6’s multicast-based mechanisms. This effectively eliminates one form of broadcast. Emerging applications on IPv6 will inherently use multicast for what used to be broadcast services, making network behavior more uniform.

□ Time-Sensitive and Deterministic Networks: In certain industrial or vehicular networks, time-synchronization and low-latency delivery to multiple nodes are important. New Ethernet standards and TSN (Time-Sensitive Networking) are exploring deterministic group messaging. These may incorporate concepts akin to multicast or broadcast at the data link layer with strict schedules.

3.6 Tabular Comparison between Unicast, Multicast, and Broadcast

Feature	Unicast	Multicast	Broadcast
Communication	One-to-one (single sender, single receiver)	One-to-many (single sender, multiple subscribers)	One-to-all (single sender, all hosts in LAN)
Model	sender, receiver	single sender, multiple subscribers	single sender, all hosts in LAN
Layer	Addressing	Unique IP per host	Class D IP Broadcast
IP	IPv4/IPv6 unicast (224.0.0.0/4 in IPv4; FF00::/8 in IPv6)	(255.255.255.255 or FF00::/8 in directed)	addresses)

Link Layer Unicast MAC (MSB Multicast MAC Broadcast MAC

Addressing = 0) (e.g., 01:00:5E for (FF:FF:FF:FF:FF:FF)

IPv4; 33:33 for

IPv6)

OSI Layers Operates at all layers Primarily Network Data Link (broadcast

(3 & 4 prominent) layer (L3) plus domain) and Network

UDP; app layer for (IPv4)

control

Routing/Forwarding Standard routing (IP Special multicast No routing (confined to

lookup per packet) routing LAN); switches flood all

(IGMP/MLD +

PIM/others)

Network Scope Global (across Domain-defined Local only (does not

subnets/Internet) (can span subnets cross router unless

with routers allowed)

enabled)

Efficiency Low for multiple High for multiple Only efficient if all hosts

receivers (bandwidth receivers need data ($1 \times$ link)

$\propto$ #receivers) (bandwidth $\approx$ single

stream)

Scalability Poor for large groups Good for large Not scalable (limited

(linear growth of groups (near- domain, can flood

load) constant load) network)

Reliability High (TCP or Often lower Low (UDP, no ACKs,

controlled UDP) (usually UDP, no same as multicast)

native ACKs)

Security Easy to secure (per- Harder (group key Hardest (all nodes receive

connection distribution needed, data, no privacy)

TLS/IPsec) open to group)

Usage Examples HTTP web browsing, IPTV, video ARP, DHCP, local

file transfer, VoIP conferencing (one- announcements (one-to-

(one-to-one) to-many), routing all)

updates

Resource High with many Low per additional High on all hosts; every

Consumption receivers (server and receiver; network host processes it

network burden) handles distribution

Key Advantage Direct and reliable Efficient Simple and wide-

delivery to one distribution to many reaching; no address

endpoint needed

with single

transmission

Key Disadvantage Inefficient if many Complex to set up; Can overwhelm network;
recipients less reliable per se reaches unwanted hosts

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Level 400 • Second Semester • Handout

COMPUTER NETWORKS

COMMUNICATION

Lecture Notes

Course Code – CSC 423

DEPARTMENT OF COMPUTER SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

Internet overview

The Internet is a giant network of networks.

- A network may include PCs, and other devices like servers or printers.□
- A network is connected through a communication channel.□
- Early research was performed by the US Department of Defense in 1962. This research group established ARPAnet (Advanced Research Project Agency) in order to connect the US Defense Department network.□

What did the Internet come from?

- Original aim was to create a network that would allow users of a research computer at one university to be able to ‘talk to’ research computers at other universities.□
- A side benefit of ARPAnet’s design was that, because messages could be routed or rerouted in more than one direction, the network could continue to function even if parts of it were destroyed in the event of a military attack or other disaster.□
- The users of the Internet took a direction of their own.□

History of the Internet

- The first long distance communication took place in 1965 between a computer in MIT and California. In 1969, four computer clients were connected together via ARPAnet.□

How old is the Internet?

- Leonard Kleinrock is accredited with the idea of packet switching, which describes how data can be sent across a network. The Ethernet was developed by Xerox during this period. This was inspired by Robert Metcalfe’s PhD on ‘packet networks’.□
- An Ethernet is a protocol for describing how computers can be connected in a LAN (Local Area network).□
- Through the use of Ethernet and ARPAnet the US were able to develop a working network.□
- In the late 1970s and early 1980s other networks were developed, e.g. CSNET, USNET and BITNET. In 1973 Vint Cerf and Bob Kahn created the TCP/IP communication protocols.□
- TCP/IP: Transfer Control Protocol/Internet Protocol is a set of rules that describe how computers can communicate over a network.□

□ To send information over the Internet, a computer packs data into Internet Protocol (IP) packets and labels them with the correct address. They are then sent across a packet switched interconnected network.□

Introduction to Data Communication

The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Computer Network

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Software modules in one system are used to communicate with one or more software modules in the distance System. Such interfaces across a distance are termed as “peer-to-peer” interfaces; and the local interfaces are termed as “service” interfaces. The modules on each end are organized as a sequence of functions called “layers”. The set of modules organized as layers is also commonly called a “protocol stack”.

Over the years, some layered models have been standardized. The ISO Open Systems Interconnection (ISO/OSI) layered model has seven layers and was developed by a set of committees under the auspices of International Standards Organization (ISO).

Classification of Computer Networks

1. Based on Transmission Mode

Transmission mode defines the direction of signal flow between two linked devices. There are three types of transmission modes.

□ Simplex

In simplex mode, the communication is unidirectional. Among the stations only one can transmit and the other can only receive.

□ Half-Duplex

In half-Duplex mode, the communication is bidirectional. In this both station can sent and receive but not at the same time.

□ Full-Duplex

In Full-Duplex mode, both stations can transmit and receive simultaneously.

2. Based on Time Transmission Type

□ Synchronous Transmission

In synchronous Transmission both the sender and the receiver use the same time cycle for the transmission. We send bits one after another without start/stop bits or gaps. It is the responsibility of the receiver to group the bits. Bit stream is delivered with a fixed delay and given error rate. Each bit reaches the destination with the same time delay after leaving the source.

□ Asynchronous Transmission

In Asynchronous Transmission we send one start bit at the beginning and one stop bit at the end of each byte. There may be a gap between each byte. Bit stream is divided into packets. Packets are received with varying delays, so packets can arrive out of order. Some packets are not received correctly.

3. Based on Authentication

□ Peer to Peer Connection□

In peer-to-peer networks, there are no dedicated servers. All the computers are equal and, therefore, are termed as peers. Normally, each computer functions as both a client and a server. No one can control the other computers.

□ Server Based Connection□

Most networks have a dedicated server. A dedicated server is a computer on a network which functions as a server, and cannot be used as a client or a workstation. A dedicated server is optimized to service requests from network clients. A server can control the clients for its services.

4. Based on Geographical location

□ Local Area Networks (LAN)

LAN is a small high speed network. In LAN few numbers of systems are interconnected with networking device to create network. As the distance increases between the nodes or system it speed decreases. So it is limited to few meters only. Networks which cover close geographical area. LAN used to link the devices in a single office, building or campus. It provides high speeds over short distance. Systems are connecting directly to Network. The LAN is owned by private people.

□ Wide Area Network (WAN)

WAN is collection of network (or LAN). This network speed is less than the LAN network speed. WAN network connect systems indirectly. WAN spread over the world may be spread over more than one city country or continent. Systems in this network are connected indirectly. Generally WAN network are slower speed than LAN's. The WAN network are owned or operated by network providers. If it is owned by a single owner then it is called Enterprise network. Often these types have combination of more than one topology.

□ MAN (Metropolitan Area Network)

Metropolitan area network is an extension of local area network to spread over the city. It may be a single network or a network in which more than one local area network can share their resources.

5. Based on Reliability

Reliability is maintained by authentication.

□ Connection-oriented

This type of communication establishes a session connection before data can be sent. This method is often called a "reliable" network service. It can guarantee that data

will arrive in the same order.

□ Connection less

This type of communication does not require a session connection between sender and receiver for data transfer. The sender simply starts sending packets to the destination. A connectionless network provides minimal services.

Topology

The term topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus and ring (see figure 1).

Figure 1 Categories of topology

1. Mesh: In a mesh topology, every device has a dedicated point to point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. (see figure 2)

Figure 2 A fully connected mesh topology (five devices)

A mesh offers several advantages over other network topologies. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices. Second, a mesh topology is robust. If one link becomes unavailable it does not incapacitate the entire system. Third, there is the advantage of privacy or security. Whenever message travels along a dedicated line, only the intended recipient sees it. Finally, point to point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

The main disadvantages of a mesh are related to the amount of cabling and that of I/O ports required. First, because every device must be connected to every other device, installation and reconnection are difficult. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings or floors) can accommodate. Finally, the hardware required to connect each link (I/O ports and cables) can be prohibitively expensive. For these reasons, a mesh topology is usually implemented in a limited fashion, for example, as a backbone connecting the main computers of a hybrid network that can include several other topologies.

One practical example of a mesh topology is the connection of a telephone regional office in which each regional office needs to be connected to every other regional office.

2. Star Topology: In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. These devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices. The controller acts as an exchange. If one device wants to send data to another it sends the data to the controller, which then relay the data to the other connected devices (see figure 3)

Figure 3: A star topology connecting four stations

A Star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect to any number of others. This factor also makes it easy to install and reconfigure. Far less cabling needs to be housed, and additions, moves and deletions involve only one connection between that device and the hub.

Other advantages include robustness. If one link fails, only that link is affected. All other remain active. This factor also lends itself to easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

3. Bus Topology: A bus topology is multipoint. One long cable acts as a backbone to link all the devices in the network (see figure 4)

Figure 4: A bus topology connecting three stations

Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason, there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path and then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.

Disadvantages include difficult reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices. Signal reflection at the taps can cause degradation of quality.

In addition, a fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Bus topology was the one of the first topologies used in the design of early local area networks

4. Ring Topology: In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along (see figure 5).

Figure 5 A ring topology connecting six stations

A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two

connections. The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified. Generally, a signal is circulating at all times in a ring. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Today, the need for higher speed LANS has made this topology less popular

5. Hybrid Topology: A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology as shown in Figure 6.

Figure 6 A hybrid topology: a star backbone with three bus networks

All computers, regardless of topology, communicate by addressing data to one or more computers and transmitting it across cable as electronic signals. Data is broken into packets and sent as electronic signals that travel on the cable. Only the computer to which the data is addressed accepts it.

Protocol

Protocols mean set of rules. It is a formal description of message formats and the rules two or more machines has to follow to exchange messages. The key elements of a protocol are syntax, semantics and timing.

□ Syntax

Syntax refers to the structure or format of the data, meaning the order in which they are presented.

□ Semantics

Semantics refers to the meaning of each section of bits.

□ Timing

Timing refers to when data should be sent and how fast it can be sent.

Internetworking Technologies

Internetworking Technologies tell how the Internet accommodating multiple underlying hardware technologies and how they are interconnected and formed the network, and set of communication standard which the network used to inter-operate.

The lowercase internet means multiple networks connected together, using a common protocol suite. The uppercase Internet refers to the collection of hosts around the world that can communicate with each other using TCP/IP. While the Internet is an internet, the reverse is not true.

Network Infrastructure or Transmission Infrastructure:

Network infrastructure is divided into two parts.

1. Access Networks

An access network is the part of a telecommunications network which connects end system to the first router or subscribers to their immediate service provider as shown in figure 7.

Figure 7 Network Infrastructure

It is different from core network which connects all the routers to each other and ISP (Internet service provider). An access network may be a so-called local area network within a company or university, a dial telephone line with a modem, or a high-speed cable-based or phone-based access network.

Access networks can be loosely divided into three categories:

- Residential access networks, connecting a home end system into the network.
- Institutional access networks, connecting an end system in a business or educational institution into the network.
- Mobile access networks, connecting a mobile end system into the network

Core Networks:

Core network connects all the routers to each other and ISP (Internet service provider). It is a main back bone for internet. Core network uses circuit switching and packet switching for data transmission.

ISP (Internet Service Provider):

In internet bottom-to-top the hierarchy consists of end systems (PCs, workstations, etc.) connected to local Internet Service Providers (ISPs). The local ISPs are in turn connected to regional ISPs, which are in turn connected to national and international ISPs. The national and international ISPs are connected together at the highest tier in the hierarchy.

Let's begin at the top of the hierarchy and work our way down. Residing at the very top of the hierarchy are the national ISPs, which are called National Backbone Provider (NBPs). The NBPs form independent backbone networks that span North America (and typically abroad as well). Just as there are multiple long-distance telephone companies in the USA, there are multiple NBPs that compete with each other for traffic and customers. The existing NBPs include internetMCI, SprintLink, PSINet, UUNet Technologies, and AGIS. The NBPs typically have high-bandwidth transmission links, with bandwidths ranging from 1.5 Mbps to 622 Mbps and higher. Each NBP also has numerous hubs which interconnect its links and at which regional ISPs can tap into the NBP.

The NBPs themselves must be interconnected to each other. To see this, suppose one regional ISP, say MidWestnet, is connected to the MCI NBP and another regional ISP, say EastCoastnet, is connected to Sprint's NBP. How can traffic be sent from MidWestnet to EastCoastnet? The solution is to introduce switching centers, called Network Access Points (NAPs), which interconnect the NBPs, thereby allowing each regional ISP to pass traffic to any other regional ISP. To keep us all confused, some of the NAPs are not referred to as NAPs but instead as MAEs (Metropolitan Area Exchanges).

Component of Internet:

A network (or internet) is formed using Hardware (or network device) and network software or Application and protocols.

Hardware or Network device:

1. Hub:

- It is used to connect systems or nodes or networks.
- It has direct connection to a node (point to point connection).
- It suffers from high collision of data, results to data loss.
- A hub takes data from input port and retransmits the input data on output port.

2. Repeater:

- A repeater is a device which regenerates or amplifies the data or signal so that it can travel to the other segment of cable.
- It is used to connect two networks that use same technology and protocol.
- It does not filter or translate any data.
- Works in physical layer.

3. Bridge:

- It is used to connect two networks.
- It divides the collision domain based on number of ports or interface present in a bridge.
- It uses the packet switches that forward and filter the frames using LAN destination address.
- Bridge examines the destination address of frame and forwards it to the interface or port which leads to the destination.
- It uses the routing table for routing frame from one node to other using MAC address.
- It works in Data Link Layer.

4. Switch:

- It is similar to bridge. It has a greater number of interfaces as compared to bridge.
- It allows direct communication between the nodes.
- It works in Data Link Layer.
- It uses MAC address for data transmission and communication.

5. Router:

- It is used to connect different types of network (types- architecture/ Protocol).
- It works similar to bridge but it uses IP address for routing data.
- Router can't be used for connecting systems.
- It works in Network Layer.

6. Gateways:

Gateways make communication possible between systems that use different communication protocols, data formatting structures, languages and architectures.

Gateways repackage data going from one system to another. Gateways are usually dedicated servers on a network and are task-specific.

System, Software and Protocols:

Basically, two types of system are used in Internet

- Client system: User which access data from internet.
- Server System: Host data for users using HTML files.

Software or Applications and protocols:

- Chat- IRC (Internet Relay Chat) is used for live discussions on the Internet.□
- Ecommerce - Taking orders for products and services on the Internet.□

- E-mail - Exchanging electronic letters, messages, and small files.□
- FTP - File Transfer Protocol is the most common method of transferring files between computers via the Internet.□
- Hosting - Making information available to others on the Internet.□
- Search Engines - These tools are really a part of the World Wide Web and are often used when looking for information because the Web has grown so large and is without any inherent organizational structure.□
- Telnet - Creation of a dumb terminal session to a host computer in order to run software applications on the host system.□
- World Wide Web - This is largest, fastest growing, part of the Internet, the part for which Internet browsers like Netscape's Navigator and Microsoft's Explorer were designed. Business is the leading factor fueling the rapid growth of the Web making information, advertising, and product ordering readily available to everyone with Web access.□
- TCP/IP□
- Browser□

WAN Protocols

- Frame Relay□

Frame relay is used to connect large number of sites in the network because it is relatively inexpensive to do so. The service provider gives you a frame relay circuit and is charged for the amount of data and the bandwidth you use as oppose to T1 circuit that charges with a flat monthly rate whether you use partial bandwidth or the full bandwidth regardless. Frame relay is a high-performance WAN protocol that operates at the Data Link layer and the Physical layer of the OSI model.

- Integrated Services Digital Network (ISDN)□

Integrated Services Digital Network (ISDN) is designed to run over existing telephone networks. It can deliver end to end digital service carrying voice and data. ISDN operates at OSI model, physical layer, data link layer and network layer. It can carry multimedia and graphics with all other voice, data services. ISDN supports all upper layer protocols and you can choose PPP, HDLC or LAPD as your encapsulation protocol. It has two offerings, Primary rate which is 23B+D channels. 23, 64 kbps and one 64kbps mainly used for signaling. The other is the Basic Rate which has 2B+D channels two 64kbps and one 16kbps. At data link layer ISDN supports two protocols; LAPB and LAPD. LAPB is used to mainly transfer data from upper layers and has three types of frames. I-Frames carry upper layer information and carries out sequencing, flow control, error detection and recovery. S- Frames carry control information for the I-frame. LAPD provides an additional multiplexing function to the upper layers enabling number of network entities to operate over a single physical access. Each individual link procedure acts independently of others. The multiplex procedure combines and distributes the data link channels according to the address information of the frame.

Each link is associated with a specific Service Access Point (SAP), which is identified in the part of the address field.

□ High Level Data Link Control (HDLC)□

High Level Data Link Control (HDLC) is a bit oriented data link layer frame protocol that has many versions similar to LAP, LAPB, and LAPD. CISCO routers default encapsulation is HDLC, but it is proprietary to CISCO.

OSI model

OSI (Open System Interconnection), developed by the International Organization for Standardization (ISO), was the solution designed to promote interoperability between vendors. It defines architecture for communications that support distributed processing. The OSI model describes the functions that allow systems to communicate successfully over a network. Using what is called a layered approach, communications functions are broken down into seven distinct layers.

Figure 8 Interaction between layers in OSI model.

The seven layers, beginning with the bottom layer of the OSI model, are shown in figure 8. Routers are used as intermediate node to create a link between A and B end system.

OSI model layers are dependent on each other. Each layer serves the upper layer and also depends upon the services from the lower layer.

OSI model also provide the layer abstraction. Layers are dependent on each other for services but in terms of protocol they are independent.

In each layer information is added into original data as header but in data link layer trailer is added into the data as shown in figure 9.

Figure 9 Exchange of data using OSI model

OSI Model Layer

Layer 1: Physical Layer

- It defines the transmission of data across the communications medium and translation of binary data into signals.
- Mode of transmission over the link i.e Simplex or Half Duplex or Full Duplex
- It defines the transmission rate of bits per second.

Layer 2: Data Link Layer

- It divides the data into number of frames.
- It uses the MAC address for sending frames from one node to other.
- It provides flow control, error control and access control.

Layer 3: Network Layer

- It divides data into number of packets.
- It uses IP address for routing packets to their destination.
- It provides end to end connection.

Layer 4: Transport Layer

- It divides message into segments and also reassemble the segments to

create original message.

- It can be either connection-oriented or connectionless.
- It uses service-point address or port address for process-to-process communication.
- Flow control and error control also provided by transport layer.

Layer 5: Session Layer

- Session Layer establishes, maintains and synchronizes the interaction among communicating systems.

Layer 6: Presentation Layer

- It is concerned with the syntax and semantics of the information exchanged between two systems.
- It translates information from text/numeric into bit stream.
- It also encrypts the information for security purpose and compress the information to reduce the number of bits in the information.

Layer 7: Application Layer

- It provides the interface to the end user and supports for services such as Email, file transfer and distributed information service.

OSI Model and Protocol stack

Layer Protocol

Application HTTP, FTP, SMTP,

TELNET

Presentation JPG, GIF, MPEG,

Session TCP 3-way Handshaking

Transport TCP, UDP

Network IP, IPX

Data Link Ethernet, Token Ring, HDLC

Physical X.21, RS-232, DS, DS3

TCP/IP model

- TCP/IP protocol suite was developed before the OSI model.
- TCP/IP is a set of protocols developed to allow cooperating computers to share resources across a network.
- In 1969 the Defense Advanced research projects Agency (DARPA) funded a research and development project to create an experimental packet switching network. This network is called ARPANET.
- In 1975 the ARPANET was converted from an experimental network to an operational network, and the responsibility for administering the network was given to the Defense Communication Agency (DCA).
- The TCP/IP protocols were adopted as Military Standards (MIL STD) in 1983, and all hosts connected to the network were required to convert to the new protocols.
- DARPA funded to implement TCP/IP in Berkely Unix.
- In 1983, the old ARPANET was divided into MILNET and smaller ARPANET. The Internet was

used to refer to the entire network; MILNET and ARPANET.

Advantages of TCP/IP

Open protocol standards, freely available and developed independently from any specific computer hardware or operating system. A common addressing scheme which is enable to connect the most widely used networks. It may use any protocols. It connects dissimilar systems. It provides client/server framework. It provides access to the Internet

Differences of the OSI and TCP/IP models

TCP/IP combines the presentation and session layer into its application layer. TCP/IP combines the OSI data link and physical layers into one layer. TCP/IP appears simpler because it has fewer layers. TCP/IP transport layer using UDP does not always guarantee reliable delivery of packets as the transport layer in the OSI model does.

Packet Switching fundamentals

A switched network consists of a series of interlined device called switches (shown in figure 10). It is a device which can create a temporary connection between two or more system linked to the switch. In switched network some of the nodes are systems and other are used for routing.

The end systems (communicating devices) are labeled A, B, C, D, and so on, and the switches are labeled I, II, III, IV, and V. Each switch is connected to multiple links.

Figure 10 Switched Network

There are three methods of switching

1. Circuit Switched Networks
2. Packet Switched Networks
 - A. Datagram Networks
 - B. Virtual- circuit Networks
3. Message Switched Networks

1. Circuit Switched Network:

- In circuit-switched networks, a dedicated path is needed for communication between the end systems are reserved for the duration of the session.
- Each connection uses only one dedicated channel on each link.
- Each link is divided into n channels by using FDM (frequency division Multiplexing) or TDM (TimeDivision multiplexing).

Figure 11 trivial circuit-switched network

In the above figure one link is divided into n channel (here n=3). A circuit switched network requires the following three phase during the session.

1. Setup Phase: First of all two system needs to create dedicated circuit or path for communication. For example, in figure 11 when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I. Switch I finds a channel between itself and switch II that can be dedicated for this purpose. Switch I then send the request to switch II, which finds a dedicated channel between itself and switch III. Switch III informs system M of about system A.

To establish a path system M must send an acknowledgement for the request of A. Only after system A receives this acknowledgement the connection is established. Only end to end addressing is required for establishing connection between two end systems.

2. Data Transfer Phase

After the establishment of the dedicated path (channels), the two systems can transfer data.

3. Teardown Phase

When one of the systems needs to disconnect, a signal is sent to each switch to release the resources.

2. Packet Switched Networks

a) Datagram Networks

□ In packet switched network message is divided into number of packets. Each packet is of fixed size defined by network or protocol.□

□ Datagram switched network is also known as Connectionless packet switching□

□ There is no dedicated link between source and destination.□

□ No dedicated Resources are allocated for packet. Resources are allocated on demand and it follows first come first basis. When a switch receives a packet, irrespective of the source or destination, the packet must wait if the other packets being processed.□

□ A single message is divided into number of packets. During the transfer of packets from source to destination, each packet is treated independently. Destination can receive unordered packets and later packet can be ordered and combine the packets to extract the message.□

□ Packets are referred as datagrams in this type of switching. Datagram switching is normally done at the network layer.□

□ The datagram networks are referred to as connectionless networks. Connectionless means switches have no connection state information.□

□ There is no setup and teardown phase. So, a routing table is required in every switch to route packet from source to destination. A Routing table is based on the destination address. The routing table updated periodically. The destination addresses and the corresponding forwarding output ports are recorded in the tables. This is different from the table of a circuit switched network in which each entry is created when the setup phase is completed and deleted when the teardown phase is over. Figure 4` shows the routing table for a switch.□

Destination Output Port

address

1234 1

4444 2

6666 3

.....

2222 3

Figure 12 Routing table for a switch

Destination Address

Every packet in a datagram network carries a header that contains information of the destination address of the packet. When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded. This address, unlike the address in a virtual-circuit-switched network, remains the same during the entire journey of the packet.

Efficiency:

The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred. If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources can be reallocated during these minutes for other packets from other sources.

Delay

There may be greater delay in a datagram network than in a virtual-circuit network. Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded. In addition, since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.

Switching in the Internet is done by using the datagram approach to packet switching at the network layer.

b) Virtual -Circuit Networks:

A virtual-circuit network uses the characteristics of both the circuit switched network and the datagram network. A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer. Virtual-circuit network is also known as Connection-oriented packet switching

Addressing

Two types of addressing is used in virtual-circuit network

- Global Address: It is an address which can uniquely identify the systems (source or destination) in a network or internet. This address is used to create virtual circuit identifier only.
- Virtual Circuit Identifier: The identifier that is actually used for data transfer is known as virtual circuit identifier (VCI). It is a number which is used in a frame between two switches. This VCI changes from one switch to another. Every switch uses a fixed range of values for VCI.

Three phases of Virtual -Circuit Networks:

1. Data Transfer Phase

- To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit.
- The table, in its simplest form, has four columns.
- This means that the switch holds four pieces of information for each virtual circuit that is already setup.

□ Figure 12 shows such a switch and its corresponding table. Figure 13 shows a frame arriving at port 1 with a VCI of 14. When the frame arrives, the switch looks in its table to find port 1 and VCI of 14. When it is found, the switch knows to change the VCI to 22 and send out the frame

from port 3.

□ The data transfer phase is active until the source sends all its frames to the destination.

□ The procedure at the switch is the same for each frame of a message.

□ The process creates a virtual circuit, not a real circuit, between the source and destination.

Figure 13 Switch and tables in a virtual-circuit network

2. Setup Phase

In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the setup request and the acknowledgment.

Figure 14 Switch and tables in a virtual-circuit network

- Setup Request:

A setup request frame is sent from the source to the destination. Figure 6 shows the process.

- a. Source A sends a setup frame to switch 1.
- b. Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. The switch, in the setup phase, acts as a packet switch; it has a routing table which is different from the switching table. For the moment, assume that it knows the output port. The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.
- c. Switch 2 receives the setup request frames. The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).
- d. Switch 3 receives the setup request frame. Again, three columns are completed: Incoming port (2), incoming VCI (22), and outgoing port (3).
- e. Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and no other sources.

- Acknowledgment:

A special frame, called the acknowledgment frame, completes the entries in the switching tables. Figure 15 shows the process.

- a. The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed. The frame also carries VCI 77, chosen by the destination as the

incoming VCI for frames from A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.

- b. Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.
- c. Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.
- d. Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.
- e. The source uses this as the outgoing VCI for the data frames to be sent to destination B.

Figure 15 Setup acknowledgment in a virtual-circuit network

1. Teardown Phase

In this phase, source A, after sending all frames to B, sends a special frame called a teardown request. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables.

Note: In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Efficiency of Virtual-Circuit Networks:

Virtual-Circuit Networks uses the resources efficiently and it reduces the waiting time of data frame.

Delay in Virtual-Circuit Networks:

In a virtual-circuit network, there is a delay for setup and for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets. Figure 8 shows the delay for a packet traveling through two switches in a virtual-circuit network.

Figure 16 Delay in a virtual-circuit network

The packet is traveling through two switches (routers). There are three transmission times (3T), three propagation times (3T), data transfer depicted by the sloping lines, a setup delay (which includes transmission and propagation in two directions), and a teardown delay (which includes transmission and propagation in one direction).

Packet Switching versus Circuit Switching

Packet Switching Circuit Switching

1. Packet switched network is implemented in the physical layer. 1. Circuit-switched network is implemented in the physical layer.
2. Message is divided into number of packets or frames. 2. Complete message is transfer from source to destination.
3. Resources are allocated on demand if available (or free). 3. Resources are reserved during the data transfer.
4. Less efficient and more costly. 4. More efficient and less costly. 5. As the number of users increases, the
5. A link can be used by any number of user. bandwidth for each user decreases. But most of the

6. It has unpredictable delay due to waiting of time fixed number of user can use a link because packets/frames at switch, if resources are not number of channel is fixed. available.
7. Suitable for most of the Internet applications. 6. It has less delay in data transmission.
7. Suitable for real time applications.

Internet Standards:

Internet Engineering Task Force (IETF):

The IETF is an open international community concerned with the development and operation of the Internet and its architecture. The IETF was formally established by the Internet Architecture Board (IAB) in 1986. The IETF meets three times a year; much of its ongoing work is conducted via mailing lists by working groups. Typically, based upon previous IETF proceedings, working groups will convene at meetings of the IETF to discuss the work of the IETF working groups. The IETF is administered by the Internet Society, whose WWW site contains lots of high-quality, Internet-related material.

The IETF Internet Engineering Task Force) is the body that defines standard Internet operating protocols such as TCP/IP. The IETF is supervised by the Internet Society Internet Architecture Board (IAB). IETF members are drawn from the Internet Society's individual and organization membership. Standards are expressed in the form of Requests for Comments (RFCs). The Internet Engineering Task Force (IETF) is an organized activity of the Internet Society (ISOC). It is an open standard organisations, with no formal membership or membership requirements. All participants and managers are volunteers, though their work is usually funded by their employers or sponsors.

The IETF started out as an activity supported by the US federal government, but since 1993 it has operated as a standards development function under the auspices of the Internet society, an international membership-based non-profit organization. The mission of the IETF is to make the Internet work better by producing high quality, relevant technical documents that influence the way people design, use, and manage the Internet.

Institute of Electrical and Electronics Engineers (IEEE) :

The Institute of Electrical and Electronics Engineers (IEEE) is a professional association with its corporate office in New York City and its operations center in Piscataway, New Jersey. It was formed in 1963 from the amalgamation of the American Institute of Electrical Engineers and the Institute of Radio Engineers. Today it is the world's largest association of technical professionals with more than 400,000 members in chapters around the world. Its objectives are the educational and technical advancement of electrical and electronic engineering, telecommunications, computer engineering and allied disciplines.

The IEEE is best known for developing standards for the computer and electronics industry. In particular, the IEEE 802 standards for LANs are widely followed. IEEE manages the Ethernet address space and assigns addresses as needed.

IEEE is one of the leading standards-making organizations in the world. IEEE performs its

standards making and maintaining functions through the IEEE Standards Association(IEEE-SA). IEEE standards affect a wide range of industries including: power and energy, biomedical and healthcare, Information Technology (IT), telecommunications, transportation, nanotechnology, information assurance, and many more. In 2013, IEEE had over 900 active standards, with over 500 standards under development. One of the more notable IEEE standards is the IEEE 802 LAN/MAN group of standards which includes the IEEE 802.3 Ethernet standard and the IEEE 802.11 Wireless Networking standard.

Asynchronous Transfer Mode (ATM):

Asynchronous Transfer Mode, a network technology based on transferring data in cells or packets of a fixed size. The cell used with ATM is relatively small compared to units used with older technologies. The small, constant cell size allows ATM equipment to transmit video, audio, and computer data over the same network, and assure that no single type of data hogs the line. Some people think that ATM holds the answer to the Internet band width problem, but others are skeptical. ATM creates a fixed channel, or route, between two points whenever data transfer begins. This differs from TCP/IP, in which messages are divided into packets and each packet can take a different route from source to destination. This difference makes it easier to track and bill data usage across an ATM network, but it makes it less adaptable to sudden surges in network traffic.

ATM (asynchronous transfer mode) is a dedicated-connection switching technology that organizes digital data into 53-byte cell units and transmits them over a physical medium using digital signal technology. Individually, a cell is processed asynchronously relative to other related cells and is queued before being multiplexed over the transmission path.

Asynchronous transfer mode was designed with cells in mind. This is because voice data is converted to packets and is forced to share a network with burst data (large packet data) passing through the same medium. So, no matter how small the voice packets are, they always encounter full- sized data packets, and could experience maximum queuing delays. This is why all data packets should be of the same size. The fixed cell structure of ATM means it can be easily switched by hardware without the delays introduced by routed frames and software switching. This is why some people believe that ATM is the key to the Internet bandwidth problem. ATM creates fixed routes between two points before data transfer begins, which differs from TCP/IP, where data is divided into packets, each of which takes a different route to get to its destination. This makes it easier to bill data usage. However, an ATM network is less adaptable to a sudden network traffic surge.

The ATM provides data link layer services that run on the OSI's Layer 1 physical links. It functions much like small-packet switched and circuit-switched networks, which makes it ideal for real-time, low-latency data such as VoIP and video, as well as for high-throughput data traffic like file transfers. A virtual circuit or connection must be established before the two end points can actually exchange data.

ATM services generally have four different bit rate choices:

□ Available Bit Rate: Provides a guaranteed minimum capacity but data can be bursted to higher capacities when network traffic is minimal.

□ Constant Bit Rate: Specifies a fixed bit rate so that data is sent in a steady stream. This is analogous to a leased line.

□ Unspecified Bit Rate: Doesn't guarantee any throughput level and is used for applications such as file transfers that can tolerate delays.

□ Variable Bit Rate (VBR): Provides a specified throughput, but data is not sent evenly. This makes it a even popular choice for voice and videoconferencing.

ATM Service includes:

- Voice and video
- Packetized voice and video
- Systems Network Architecture (SNA)
- WAN/VPN connectivity
- Web hosting
- E-commerce
- Client-server (terminal-host) data
- LAN interconnection
- LAN emulation
- Remote access
- File transfer
- Internet/intranet/extranet access
 - E-mail messaging
 - Text imaging
- Forms processing

Use in internet: ATM is normally utilized by Internet service providers on their private long-distance networks. ATM operates at the data link layer (Layer 2 in the OSI model) over either fiber or twisted- pair cable

International Telecommunication Union (ITU)

The ITU coordinates the shared global use of the radio spectrum, promotes international cooperation in assigning satellite orbits, works to improve telecommunication infrastructure in the developing world, and assists in the development and coordination of worldwide technical standards. The ITU is active in areas including broadband Internet, latest-generation wireless technologies, aeronautical and maritime navigation, radio astronomy, satellite-based meteorology, convergence in fixed-mobile phone, Internet access, data, voice, TV broadcasting, and next-generation networks.

ITU also organizes worldwide and regional exhibitions and forums, such as ITU TELECOM WORLD, bringing together representatives of government and the telecommunications and ICT industry to exchange ideas, knowledge and technology.

ITU, based in Geneva, Switzerland, is a member of the United Nations Development Group

ITU has been an intergovernmental public-private partnership organization since its inception. Its membership includes 193 Member States and around 700 public and private sector companies as

well as international and regional telecommunication entities, known as Sector Members and Associates, which undertake most of the work of each Sector.

- An organization based on public-private partnership since its inception, ITU currently has a membership of 193 countries and over 700 private-sector entities and academic institutions. ITU is headquartered in Geneva, Switzerland, and has twelve regional and area offices around the world.
- ITU membership represents a cross-section of the global ICT sector, from the world's largest manufacturers and carriers to small, innovative players working with new and emerging technologies, along with leading R&D institutions and academia.
- Founded on the principle of international cooperation between governments (Member States) and the private sector (Sector Members, Associates and Academia), ITU is the premier global forum through which parties work towards consensus on a wide range of issues affecting the future direction of the ICT industry.

Internet Protocol (IP)

- The Internet Protocol (IP) is a network-layer (Layer 3) protocol that contains addressing information and some control information that enables packets to be routed.
- IP is documented in RFC 791 and is the primary network-layer protocol in the Internet protocol suite. Along with the Transmission Control Protocol (TCP), IP represents the heart of the Internet protocols.
- IP has two primary responsibilities: providing connectionless, best-effort delivery of datagrams through an internetwork; and providing fragmentation and reassembly of datagrams to support data links with different maximum-transmission unit (MTU) sizes.

IP Packet Format

Packets in the IP layer are called datagrams. A datagram divided into two parts : Header and Data. Header can be from 20 to 60 bytes and contains information for routing and delivery of data.

IP packet fields Details:

- Version: Indicates the version of IP currently used.
- IP Header Length (IHL): Indicates the datagram header length in 32-bit words.
- Type-of-Service: Specifies how an upper-layer protocol would like a current datagram to be handled, and assigns datagrams various levels of importance.
- Total Length: Specifies the length, in bytes, of the entire IP packet, including the data and header.
- Identification: Contains an integer that identifies the current datagram. This field is used to help piece together datagram fragments.
- Flags: Consists of a 3-bit field of which the two low-order (least-significant) bits control fragmentation. The low-order bit specifies whether the packet can be fragmented. The middle bit specifies whether the packet is the last fragment in a series of fragmented packets. The third or high-order bit is not used.
- Fragment Offset: Indicates the position of the fragment's data relative to the beginning

of the data in the original datagram, which allows the destination IP process to properly reconstruct the original datagram.

□ Time-to-Live: Maintains a counter that gradually decrements down to zero, at which point the datagram is discarded. This keeps packets from looping endlessly.

□ Protocol: Indicates which upper-layer protocol receives incoming packets after IP processing is complete.

□ Header Checksum: Helps ensure IP header integrity.

□ Source Address: Specifies the sending node.

□ Destination Address: Specifies the receiving node.

IP Addresses

- TCP/IP version 4 or IPv4 uses 32-bit for logical address and IPv6 uses 128-bit for logical address.
- An IP address represented in dotted decimal notation. Example- 123.22.33.44
- IP address is divided into net id or network id and host id.
- IP Addresses are divided into five classes: Class A, Class B, Class c, Class C, Class D, Class E.

IP Starting First Last Address No. of No. of Host

Address Binary Address Network

Class Value

Class A 0 1.0.0.0 126.255.255.254 27-1 224- 2

Class B 10 128.0.0.0 191.255.255.254 214 216 -2

Class C 110 192.0.0.0 223.255.255.254 221 28-2

Class D 1110 224.0.0.0 239.255.255.254 Multicast

Class E 1111 240.0.0.0 254.255.255.254 Undefined

Class A:

- It uses first octet for network address to uniquely identify the network and rest three octet for host address to uniquely identify the host on that network.
- An important rule is that network address cannot have all 8 bits 0 (zero).
- First bit is set to zero for class A, so following 7 bits in the first octet use to distinguish the network from other network.
- It means $2^7 - 1 = 127$ network i.e. 0 to 126
- Similar to the rule that the network portion of the address cannot be all 0s, the host portion of the address cannot be all 0s and it cannot be all 1s.
- A host portion with all 1s refers to an IP broadcast address.
- And the host portion with all 0s is a reference to the network.
- Class A network is: $2^{24} - 2 = 16,777,214$ number of hosts.
- You subtract 2 because addresses with all 0s and all 1s are invalid.

Class B:

□ It uses first two octets for network address to uniquely identify the network and rest two octet for host address to uniquely identify the host on that network.

□ 10 in the first 2 bits, the following 6 bits in the first octet and all 8 bits in the second octet for total 14 bits are used to distinguish this network from all other networks.

□ Hence $2^{14} = 16,384$ number of Class B networks.

□ And $2^{16} - 2 = 65534$ number of hosts on class B network.

Class C:

□ It uses first three octets for network address to uniquely identify the network and last octet for host address to uniquely identify the host on that network.

□ 110 in the first 3 bits, the following 5 bits in the first octet, all 8 bits in the second octet and all 8 bits in the third octet for total 21 bits are used to distinguish this network from all other networks.

□ Hence $2^{21} = 2,097,152$ number of Class C networks.

□ And $2^8 - 2 = 254$ number of hosts on class C network.

Class D:

□ In the first octet, the first 4 bits are 1110.

□ Class D addresses are called Multicast Address which cannot be used for host.

□ The purpose of a multicast address is to enable a server somewhere to send data to a Class D address that no one host has so that several hosts can listen to that address at the same time. When you are watching TV on the Internet or listening to the radio on the Internet, your computer is listening to a Class D address. No server is sending data directly to your workstation; instead, a server is sending data to the multicast address. Any host can use software to listen for data at that address, and many hosts can be listening at once.

Class E:

□ In the first octet, the first 4 bits are 1110.

□ Class E addresses are reserved addresses and are invalid host addresses. They are used for experimental purposes by the IETF.

Special Address:

□ Address use for Private use

Class A: 10.0.0.0 to 10.255.255.255

Class B: 172.16.0.0 to 172.31.255.255

Class C: 192.168.0.0 to 192.168.255.255

□ Loop Back Address

127.0.0.0 to 127.255.255- For testing the TCP/IP connection. It cannot be used for

host addressing.

FIREWALL

Many organizations have confidential or proprietary information, such as trade secrets, product development plans, marketing strategies, etc., which should be protected from unauthorized access

and modification. One possible approach is to use suitable encryption/decryption technique for transfer of data between two secure sites, as we have discussed in the previous lesson. Although these techniques can be used to protect data in transit, it does not protect data from digital pests and

hackers. To accomplish this, it is necessary to perform user authentication and access control to protect the networks from unauthorized traffic. This is known as firewalls.

A firewall system is an electronic security guard and electronic barrier at the same time. It protects and controls the interface between a private network and an insecure public network as shown in the

simplified diagram of Fig. 17. It is responsible for partitioning a designated area such that any damage on one side cannot spread to the other side. It prevents bad things from happening, i.e. loss of

information, without preventing good things from happening, that is controlled exchange of information with the outside world. It essentially enforces an access control policy between two networks. The manner in which this is implemented varies widely, but in principle, the firewall can be considered as a pair of mechanisms: one that is used to block traffic, and the other that is used to

permit traffic. Some firewalls place more emphasis on blocking traffic, while others emphasize on permitting traffic. Probably the most important issue to understand of a firewall is the access control

policy it implements. If a firewall administrator has no idea about what or whom he is protecting his network, what should be allowed and what should be prohibited, a firewall really won't help his organization. As firewall is a mechanism for enforcing policy, which affects all the persons behind it, it imposes heavy responsibility on the administrator of the firewall. In this lesson various issues related to Firewalls are discussed.

Figure 17 Schematic diagram of a firewall

Why a Firewall is needed?

There is no need for a firewall if each and every host of a private network is properly secured. Unfortunately, in practice the situation is different. A private network may consist of different platforms with diverse OS and applications running on them. Many of the applications were designed and developed for an ideal environment, without considering the possibility of the existence of bad guys. Moreover, most of the corporate networks are not designed for security. Therefore, it is essential to deploy a firewall to protect the vulnerable infrastructure of an enterprise.

Access Control Policies

Access control policies play an important role in the operation of a firewall. The policies can be

broadly categorized in to the following four types:

Service Control:

- Determines the types of internet services to be accessed.
- Filters traffic based on IP addresses and TCP port numbers.
- Provides Proxy servers that receives and interprets service requests before it is passed on

Direction Control:

- Determines the direction in which a particular service request may be initiated and allowed to flow through the firewall

User Control:

- Controls access to a service according to which user is attempting to access it.
- Typically applied to the users inside the firewall perimeter.
- Can be applied to the external users too by using secure authentication technique

Behavioral Control:

- Controls how a particular service is used
- For example, a firewall may filter email to eliminate spam
- Firewall may allow only a portion of the information on a local web server to an external user

Firewall Capabilities

Important capabilities of a firewall system are listed below:

- defines a single choke point to keep unauthorized users out of protected network
- It prohibits potentially vulnerable services from entering or leaving the network
- It provides protection from various kinds of IP spoofing
- It provides a location for monitoring security-related events
- Audits and alarms can be implemented on the firewall systems
- A firewall is a convenient platform for several internet functions that are not security related
- A firewall can serve as the platform for IPSec using the tunnel mode capability and can be used to implement VPNs

Limitations of a Firewall

Main limitations of a firewall system are given below:

- Firewall cannot protect against any attacks that bypass the firewall. Many organizations buy expensive firewalls but neglect numerous other back-doors into their network.
- A firewall does not protect against the internal threats from traitors. An attacker may be able to break into network by completely bypassing the firewall, if he can find a ``helpful'' insider who can be fooled into giving access to a modem pool
- Firewalls can't protect against tunneling over most application protocols. For example, firewall cannot protect against the transfer of virus-infected programs or files

Types of Firewalls

The firewalls can be broadly categorized into the following three types:

Packet Filters: Packet filtering router applies a set of rules to each incoming IP packet and then forwards or discards it. Packet filter is typically set up as a list of rules based on matches of fields in the IP or TCP header. An example table of telnet filter rules is given in Fig. 8.3.2. The packet filter

operates with positive filter rules. It is necessary to specify what should be permitted, and everything

that is explicitly not permitted is automatically forbidden.

Application-level Gateway: Application-level gateway, also called a Proxy Server acts as a relay of application-level traffic. Users contact gateways using an application and the request is successful after authentication. The application gateway is service specific such as FTP, TELNET, SMTP or HTTP.

Circuit Level Gateway: Circuit-level gateway can be a standalone or a specialized system. It does not allow end-to-end TCP connection; the gateway sets up two TCP connections. Once the TCP connections are established, the gateway relays TCP segments from one connection to the other without examining the contents. The security function determines which connections will be allowed

and which are to be disallowed.

Network Address Translation

NAT works by using one set of addresses for communications on the internet and a separate set of addresses for communication on the private network. IANA set aside three ranges of IP addresses given below for communication on the internal network.

Class A addresses: 10.0.0.0 – 10.255.255.255

Class B addresses: 172.16.0.0 – 172.31. 255.255

Class C addresses: 192.168.0.0 – 192.168.255.255

As these addresses are reserved for internal network addressing, these are not routable. The Firewall

performs translation of an internal address to an external IP address and vice versa to facilitate communication between the private and the public network, as shown in Fig. 18. However, the NAT affords a substantial degree of security by preventing direct communication. Moreover, NAT allows the use of same IP addresses in different private networks. This prolongs the life expectancy of IPv4 on the internet. Without NAT the supply of IP addresses would have exhausted long back.

Figure 18 Function of a Network Address Translator

Paper 10 of 10 — CSC 423

Level 400 • Second Semester • Handout

COMPUTER NETWORKS / COMMUNICATION

AOC

- The Internet, Computer Networks and Data Communication
- Classifications of Computer Networks
- Network Topologies
- Elements of a Protocol
- Networking Devices
- Application Protocols
- Functions of The OSI and TCP/IP Model (And Their Differences)
- Packet Switching Fundamentals
- Packet & Circuit Switching
- International Internet Standards
- IP [Addresses, Classifications, Packet Formats, Field Details] And NAT
- Network Security Policies
- TCP/IP Vulnerabilities and Solutions
- Network Security Protocols.

COMPUTER NETWORKS

COMMUNICATION

Lecture Notes

Course Code – CSC 423

DEPARTMENT OF COMPUTER SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

Internet overview

The Internet is a giant network of networks.

- A network may include PCs, and other devices like servers or printers.□
- A network is connected through a communication channel.□
- Early research was performed by the US Department of Defense in 1962. This research group established ARPAnet (Advanced Research Project Agency) in order to connect the US Defense Department network.□

What did the Internet come from?

- Original aim was to create a network that would allow users of a research computer at one university to be able to ‘talk to’ research computers at other universities.□
- A side benefit of ARPAnet’s design was that, because messages could be routed or rerouted in more than one direction, the network could continue to function even if parts

of it were destroyed in the event of a military attack or other disaster.□

□ The users of the Internet took a direction of their own.□

History of the Internet

□ The first long distance communication took place in 1965 between a computer in MIT and California. In 1969, four computer clients were connected together via ARPAnet.□

How old is the Internet?

□ Leonard Kleinrock is accredited with the idea of packet switching, which describes how data can be sent across a network. The Ethernet was developed by Xerox during this period. This was inspired by Robert Metcalfe's PhD on 'packet networks'.□

□ An Ethernet is a protocol for describing how computers can be connected in a LAN (Local Area network).□

□ Through the use of Ethernet and ARPAnet the US were able to develop a working network.□

□ In the late 1970s and early 1980s other networks were developed, e.g. CSNET, USNET and BITNET. In 1973 Vint Cerf and Bob Kahn created the TCP/IP communication protocols.□

□ TCP/IP: Transfer Control Protocol/Internet Protocol is a set of rules that describe how computers can communicate over a network.□

□ To send information over the Internet, a computer packs data into Internet Protocol (IP) packets and labels them with the correct address. They are then sent across a packet switched interconnected network.□

Introduction to Data Communication

The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Computer Network

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Software modules in one system are used to communicate with one or more software modules in the distance System. Such interfaces across a distance are termed as "peer-to-peer" interfaces; and the local interfaces are termed as "service" interfaces. The modules on each end are organized as a sequence of functions called "layers". The set of modules organized as layers is also commonly called a "protocol stack".

Over the years, some layered models have been standardized. The ISO Open Systems Interconnection (ISO/OSI) layered model has seven layers and was developed by a set of committees under the auspices of International Standards Organization (ISO).

Classification of Computer Networks

1. Based on Transmission Mode

Transmission mode defines the direction of signal flow between two linked devices. There are three types of transmission modes.

□ Simplex

In simplex mode, the communication is unidirectional. Among the stations only one can transmit and the other can only receive.

□ Half-Duplex

In half-Duplex mode, the communication is bidirectional. In this both station can sent and receive but not at the same time.

□ Full-Duplex

In Full-Duplex mode, both stations can transmit and receive simultaneously.

2. Based on Time Transmission Type

□ Synchronous Transmission

In synchronous Transmission both the sender and the receiver use the same time cycle for the transmission. We send bits one after another without start/stop bits or gaps. It is the responsibility of the receiver to group the bits. Bit stream is delivered with a fixed delay and given error rate. Each bit reaches the destination with the same time delay after leaving the source.

□ Asynchronous Transmission

In Asynchronous Transmission we send one start bit at the beginning and one stop bit at the end of each byte. There may be a gap between each byte. Bit stream is divided into packets. Packets are received with varying delays, so packets can arrive out of order. Some packets are not received correctly.

3. Based on Authentication

□ Peer to Peer Connection□

In peer-to-peer networks, there are no dedicated servers. All the computers are equal and, therefore, are termed as peers. Normally, each computer functions as both a client and a server. No one can control the other computers.

□ Server Based Connection□

Most networks have a dedicated server. A dedicated server is a computer on a network which functions as a server, and cannot be used as a client or a workstation. A dedicated server is optimized to service requests from network clients. A server can control the clients for its services.

4. Based on Geographical location

□ Local Area Networks (LAN)

LAN is a small high speed network. In LAN few numbers of systems are interconnected with networking device to create network. As the distance increases between the nodes or system it speed decreases. So it is limed to few meters only. Networks which cover close geographical area. LAN used to link the devices in a single office, building or campus. It provides high speeds over short distance. Systems are connecting directly to Network.

The LAN is owned by private people.

□ Wide Area Network (WAN)

WAN is collection of network (or LAN). This network speed is less than the LAN network speed. WAN network connect systems indirectly. WAN spread over the world may be spread over more than one city country or continent. Systems in this network are connected indirectly. Generally WAN network are slower speed than LAN's. The WAN network are owned or operated by network providers. If it is owned by a single owner then it is called Enterprise network. Often these types have combination of more than one topology.

□ MAN (Metropolitan Area Network)

Metropolitan area network is an extension of local area network to spread over the city. It may be a single network or a network in which more than one local area network can share their resources.

5. Based on Reliability

Reliability is maintained by authentication.

□ Connection-oriented

This type of communication establishes a session connection before data can be sent. This method is often called a "reliable" network service. It can guarantee that data will arrive in the same order.

□ Connection less

This type of communication does not require a session connection between sender and receiver for data transfer. The sender simply starts sending packets to the destination. A connectionless network provides minimal services.

Topology

The term topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus and ring (see figure 1).

Figure 1 Categories of topology

1. Mesh: In a mesh topology, every device has a dedicated point to point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. (see figure 2)

Figure 2 A fully connected mesh topology (five devices)

A mesh offers several advantages over other network topologies. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices. Second, a mesh topology is robust. If one link becomes unavailable it does not incapacitate the entire system. Third, there is the advantage of privacy or security. Whenever message travels along a dedicated line, only the intended recipient sees it. Finally, point to point links make fault identification and fault isolation

easy. Traffic can be routed to avoid links with suspected problems. This enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

The main disadvantages of a mesh are related to the amount of cabling and that of I/O ports required. First, because every device must be connected to every other device, installation and reconnection are difficult. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings or floors) can accommodate. Finally, the hardware required to connect each link (I/O ports and cables) can be prohibitively expensive. For these reasons, a mesh topology is usually implemented in a limited fashion, for example, as a backbone connecting the main computers of a hybrid network that can include several other topologies.

One practical example of a mesh topology is the connection of a telephone regional office in which each regional office needs to be connected to every other regional office.

2. Star Topology: In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. These devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices. The controller acts as an exchange. If one device wants to send data to another it sends the data to the controller, which then relay the data to the other connected devices (see figure 3)

Figure 3: A star topology connecting four stations

A Star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect to any number of others. This factor also makes it easy to install and reconfigure. Far less cabling needs to be housed, and additions, moves and deletions involve only one connection between that device and the hub.

Other advantages include robustness. If one link fails, only that link is affected. All other remain active. This factor also lends itself to easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is registered in a star than in some other topologies (such as ring or bus).

3. Bus Topology: A bus topology is multipoint. One long cable acts as a backbone to link all the devices in the network (see figure 4)

Figure 4: A bus topology connecting three stations

Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason, there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages of a bus topology include ease of installation. Backbone cable can be laid along the

most efficient path and then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.

Disadvantages include difficult reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices. Signal reflection at the taps can cause degradation of quality.

In addition, a fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Bus topology was the one of the first topologies used in the design of early local area networks

4. Ring Topology: In a ring topology, each device has a dedicated point to point connection with only the two devices on either side of it. A signal is passed along the ring in one direction from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along (see figure 5).

Figure 5 A ring topology connecting six stations

A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified. Generally, a signal is circulating at all times in a ring. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Today, the need for higher speed LANS has made this topology less popular

5. Hybrid Topology: A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology as shown in Figure 6.

Figure 6 A hybrid topology: a star backbone with three bus networks

All computers, regardless of topology, communicate by addressing data to one or more computers and transmitting it across cable as electronic signals. Data is broken into packets and sent as electronic signals that travel on the cable. Only the computer to which the data is addressed accepts it.

Protocol

Protocols mean set of rules. It is a formal description of message formats and the rules two or more machines has to follow to exchange messages. The key elements of a protocol are syntax, semantics and timing.

□ Syntax

Syntax refers to the structure or format of the data, meaning the order in which they are presented.

□ Semantics

Semantics refers to the meaning of each section of bits.

□ Timing

Timing refers to when data should be sent and how fast it can be sent.

Internetworking Technologies

Internetworking Technologies tell how the Internet accommodating multiple underlying hardware technologies and how they are interconnected and formed the network, and set of communication standard which the network used to inter-operate.

The lowercase internet means multiple networks connected together, using a common protocol suite. The uppercase Internet refers to the collection of hosts around the world that can communicate with each other using TCP/IP. While the Internet is an internet, the reverse is not true.

Network Infrastructure or Transmission Infrastructure:

Network infrastructure is divided into two parts.

1. Access Networks

An access network is the part of a telecommunications network which connects end system to the first router or subscribers to their immediate service provider as shown in figure 7.

Figure 7 Network Infrastructure

It is different from core network which connects all the routers to each other and ISP (Internet service provider). An access network may be a so-called local area network within a company or university, a dial telephone line with a modem, or a high-speed cable-based or phone-based access network.

Access networks can be loosely divided into three categories:

- Residential access networks, connecting a home end system into the network.
- Institutional access networks, connecting an end system in a business or educational institution into the network.
- Mobile access networks, connecting a mobile end system into the network

Core Networks:

Core network connects all the routers to each other and ISP (Internet service provider). It is a main back bone for internet. Core network uses circuit switching and packet switching for data transmission.

ISP (Internet Service Provider):

In internet bottom-to-top the hierarchy consists of end systems (PCs, workstations, etc.) connected to local Internet Service Providers (ISPs). The local ISPs are in turn connected to regional ISPs, which are in turn connected to national and international ISPs. The national and international ISPs are connected together at the highest tier in the hierarchy.

Let's begin at the top of the hierarchy and work our way down. Residing at the very top of the hierarchy are the national ISPs, which are called National Backbone Provider (NBPs). The NBPs form independent backbone networks that span North America (and typically abroad as

well). Just as there are multiple long-distance telephone companies in the USA, there are multiple NBPs that compete with each other for traffic and customers. The existing NBPs include internetMCI, SprintLink, PSINet, UUNet Technologies, and AGIS. The NBPs typically have high-bandwidth transmission links, with bandwidths ranging from 1.5 Mbps to 622 Mbps and higher. Each NBP also has numerous hubs which interconnect its links and at which regional ISPs can tap into the NBP.

The NBPs themselves must be interconnected to each other. To see this, suppose one regional ISP, say MidWestnet, is connected to the MCI NBP and another regional ISP, say EastCoastnet, is connected to Sprint's NBP. How can traffic be sent from MidWestnet to EastCoastnet? The solution is to introduce switching centers, called Network Access Points (NAPs), which interconnect the NBPs, thereby allowing each regional ISP to pass traffic to any other regional ISP. To keep us all confused, some of the NAPs are not referred to as NAPs but instead as MAEs (Metropolitan Area Exchanges).

Component of Internet:

A network (or internet) is formed using Hardware (or network device) and network software or Application and protocols.

Hardware or Network device:

1. Hub:

- ☐ It is used to connect systems or nodes or networks.
- ☐ It has direct connection to a node (point to point connection).
- ☐ It suffers from high collision of data, results to data loss.
- ☐ A hub takes data from input port and retransmits the input data on output port.

2. Repeater:

- ☐ A repeater is a device which regenerates or amplifies the data or signal so that it can be travel to the other segment of cable.
- ☐ It is used to connect two networks that use same technology and protocol.
- ☐ It does not filter or translate any data.
- ☐ Work in physical layer.

3. Bridge:

- ☐ It is used to connect two networks.
- ☐ It divides the collision domain based on number of ports or interface present in a bridge.
- ☐ It uses the packet switches that forward and filter the frames using LAN destination address.
- ☐ Bridge examines the destination address of frame and forwards it to the interface or port which leads to the destination.
- ☐ It uses the routing table for routing frame from one node to other using MAC address.
- ☐ It works in Data Link Layer.

4. Switch:

- ☐ It is similar to bridge. It has a greater number of interfaces as compared to bridge.
- ☐ It allows direct communication between the nodes.
- ☐ It works in Data Link Layer.

□ It uses MAC address for data transmission and communication.

5. Router:

□ It is used to connect different types of network (types- architecture/ Protocol).

□ It work similar to bridge but it uses IP address for routing data.

□ Router can't be used for connecting Systems.

□ It works in Network Layer.

6. Gateways:

Gateways make communication possible between systems that use different communication protocols, data formatting structures, languages and architectures.

Gateways repackage data going from one system to another. Gateways are usually dedicated servers on a network and are task-specific.

System, Software and Protocols:

Basically, two types of system are used in Internet

□ Client system: User which access data from internet.

□ Server System: Host data for users using HTML files.

Software or Applications and protocols:

□ Chat- IRC (Internet Relay Chat) is used for live discussions on the Internet.□

□ Ecommerce - Taking orders for products and services on the Internet.□

□ E-mail - Exchanging electronic letters, messages, and small files.□

□ FTP - File Transfer Protocol is the most common method of transferring files between computers via the Internet.□

□ Hosting - Making information available to others on the Internet.□

□ Search Engines - These tools are really a part of the World Wide Web and are often used when looking for information because the Web has grown so large and is without any inherent organizational structure.□

□ Telnet - Creation of a dumb terminal session to a host computer in order to run software applications on the host system.□

□ World Wide Web - This is largest, fastest growing, part of the Internet, the part for which Internet browsers like Netscape's Navigator and Microsoft's Explorer were designed. Business is the leading factor fueling the rapid growth of the Web making information, advertising, and product ordering readily available to everyone with Web access.□

□ TCP/IP□

□ Browser□

WAN Protocols

□ Frame Relay□

Frame relay is used to connect large number of sites in the network because it is relatively inexpensive to do so. The service provider gives you a frame relay circuit and is charged for the amount of data and the bandwidth you use as oppose to T1 circuit that

charges with a flat monthly rate whether you use partial bandwidth or the full bandwidth regardless. Frame relay is a high-performance WAN protocol that operates at the Data Link layer and the Physical layer of the OSI model.

□ Integrated Services Digital Network (ISDN)□

Integrated Services Digital Network (ISDN) is designed to run over existing telephone networks. It can deliver end to end digital service carrying voice and data. ISDN operates at OSI model, physical layer, data link layer and network layer. It can carry multimedia and graphics with all other voice, data services. ISDN supports all upper layer protocols and you can choose PPP, HDLC or LAPD as your encapsulation protocol. It has two offerings, Primary rate which is 23B+D channels. 23, 64 kbps and one 64kbps mainly used for signaling. The other is the Basic Rate which has 2B+D channels two 64kbps and one 16kbps. At data link layer ISDN supports two protocols; LAPB and LAPD. LAPB is used to mainly transfer data from upper layers and has three types of frames. I-Frames carry upper layer information and carries out sequencing, flow control, error detection and recovery. S- Frames carry control information for the I-frame. LAPD provides an additional multiplexing function to the upper layers enabling number of network entities to operate over a single physical access. Each individual link procedure acts independently of others. The multiplex procedure combines and distributes the data link channels according to the address information of the frame. Each link is associated with a specific Service Access Point (SAP), which is identified in the part of the address field.

□ High Level Data Link Control (HDLC)□

High Level Data Link Control (HDLC) is a bit oriented data link layer frame protocol that has many versions similar to LAP, LAPB, and LAPD. CISCO routers default encapsulation is HDLC, but it is proprietary to CISCO.

OSI model

OSI (Open System Interconnection), developed by the International Organization for Standardization (ISO), was the solution designed to promote interoperability between vendors. It defines architecture for communications that support distributed processing. The OSI model describes the functions that allow systems to communicate successfully over a network. Using what is called a layered approach, communications functions are broken down into seven distinct layers.

Figure 8 Interaction between layers in OSI model.

The seven layers, beginning with the bottom layer of the OSI model, are shown in figure 8. Routers are used as intermediate node to create a link between A and B end system.

OSI model layers are dependent on each other. Each layer serves the upper layer and also depends upon the services from the lower layer.

OSI model also provide the layer abstraction. Layers are dependent on each other for services but in terms of protocol they are independent.

In each layer information is added into original data as header but in data link layer trailer is added into the data as shown in figure 9.

Figure 9 Exchange of data using OSI model

OSI Model Layer

Layer 1: Physical Layer

- It defines the transmission of data across the communications medium and translation of binary data into signals.
- Mode of transmission over the link i.e Simplex or Half Duplex or Full Duplex
- It defines the transmission rate of bits per second.

Layer 2: Data Link Layer

- It divides the data into number of frames.
- It uses the MAC address for sending frames from one node to other.
- It provides flow control, error control and access control.

Layer 3: Network Layer

- It divides data into number of packets.
- It uses IP address for routing packets to their destination.
- It provides end to end connection.

Layer 4: Transport Layer

- It divides message into segments and also reassemble the segments to create original message.
- It can be either connection-oriented or connectionless.
- It uses service-point address or port address for process-to-process communication.
- Flow control and error control also provided by transport layer.

Layer 5: Session Layer

- Session Layer establishes, maintains and synchronizes the interaction among communicating systems.

Layer 6: Presentation Layer

- It is concerned with the syntax and semantics of the information exchanged between two systems.
- It translates information from text/numeric into bit stream.
- It also encrypts the information for security purpose and compress the information to reduce the number of bits in the information.

Layer 7: Application Layer

- It provides the interface to the end user and supports for services such as Email, file transfer and distributed information service.

OSI Model and Protocol stack

Layer Protocol

Application HTTP, FTP, SMTP,
TELNET

Presentation JPG, GIF, MPEG,

Session TCP 3-way Handshaking

Transport TCP, UDP

Network IP, IPX

Data Link Ethernet, Token Ring, HDLC

Physical X.21, RS-232, DS, DS3

TCP/IP model

- TCP/IP protocol suite was developed before the OSI model.
- TCP/IP is a set of protocols developed to allow cooperating computers to share resources across a network.
- In 1969 the Defense Advanced research projects Agency (DARPA) funded a research and development project to create an experimental packet switching network. This network is called ARPANET.
- In 1975 the ARPANET was converted from an experimental network to an operational network, and the responsibility for administering the network was given to the Defense Communication Agency (DCA).
- The TCP/IP protocols were adopted as Military Standards (MIL STD) in 1983, and all hosts connected to the network were required to convert to the new protocols.
- DARPA funded to implement TCP/IP in Berkely Unix.
- In 1983, the old ARPANET was divided into MILNET and smaller ARPANET. The Internet was used to refer to the entire network; MILNET and ARPANET.

Advantages of TCP/IP

Open protocol standards, freely available and developed independently from any specific computer hardware or operating system. A common addressing scheme which is enable to connect the most widely used networks. It may use any protocols. It connects dissimilar systems. It provides client/server framework. It provides access to the Internet

Differences of the OSI and TCP/IP models

TCP/IP combines the presentation and session layer into its application layer. TCP/IP combines the OSI data link and physical layers into one layer. TCP/IP appears simpler because it has fewer layers. TCP/IP transport layer using UDP does not always guarantee reliable delivery of packets as the transport layer in the OSI model does.

Packet Switching fundamentals

A switched network consists of a series of interlined device called switches (shown in figure 10). It is a device which can create a temporary connection between two or more system linked to the switch. In switched network some of the nodes are systems and other are used for routing.

The end systems (communicating devices) are labeled A, B, C, D, and so on, and the switches are labeled I, II, III, IV, and V. Each switch is connected to multiple links.

Figure 10 Switched Network

There are three methods of switching

1. Circuit Switched Networks

2. Packet Switched Networks

A. Datagram Networks

B. Virtual- circuit Networks

3. Message Switched Networks

1. Circuit Switched Network:

□ In circuit-switched networks, a dedicated path is needed for communication between the end systems are reserved for the duration of the session.

□ Each connection uses only one dedicated channel on each link.

□ Each link is divided into n channels by using FDM (frequency division Multiplexing) or TDM (TimeDivision multiplexing).

Figure 11 trivial circuit-switched network

In the above figure one link is divided into n channel (here $n=3$). A circuit switched network requires the following three phase during the session.

1. Setup Phase: First of all two system needs to create dedicated circuit or path for communication. For example, in figure 11 when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I. Switch I finds a channel between itself and switch II that can be dedicated for this purpose. Switch I then send the request to switch II, which finds a dedicated channel between itself and switch III. Switch III informs system M of about system A.

To establish a path system M must send an acknowledgement for the request of A. Only after system A receives this acknowledgement the connection is established. Only end to end addressing is required for establishing connection between two end systems.

2. Data Transfer Phase

After the establishment of the dedicated path (channels), the two systems can transfer data.

3. Teardown Phase

When one of the systems needs to disconnect, a signal is sent to each switch to release the resources.

2. Packet Switched Networks

a) Datagram Networks

□ In packet switched network message is divided into number of packets. Each packet is of fixed size defined by network or protocol.□

□ Datagram switched network is also known as Connectionless packet switching□

□ There is no dedicated link between source and destination.□

□ No dedicated Resources are allocated for packet. Resources are allocated on demand and it follows first come first basis. When a switch receives a packet, irrespective of the source or destination, the packet must wait if the other packets being processed.□

□ A single message is divided into number of packets. During the transfer of packets from source to destination, each packet is treated independently. Destination can receive unordered packets and later packet can be ordered and combine the packets to extract the message.□

- Packets are referred as datagrams in this type of switching. Datagram switching is normally done at the network layer.□
- The datagram networks are referred to as connectionless networks. Connectionless means switches have no connection state information.□
- There is no setup and teardown phase. So, a routing table is required in every switch to route packet from source to destination. A Routing table is based on the destination address. The routing table updated periodically. The destination addresses and the corresponding forwarding output ports are recorded in the tables. This is different from the table of a circuit switched network in which each entry is created when the setup phase is completed and deleted when the teardown phase is over. Figure 4` shows the routing table for a switch.□

Destination Output Port

address

1234 1

4444 2

6666 3

..... .

2222 3

Figure 12 Routing table for a switch

Destination Address

Every packet in a datagram network carries a header that contains information of the destination address of the packet. When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded. This address, unlike the address in a virtual-circuit-switched network, remains the same during the entire journey of the packet.

Efficiency:

The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred. If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources can be reallocated during these minutes for other packets from other sources.

Delay

There may be greater delay in a datagram network than in a virtual-circuit network. Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded. In addition, since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.

Switching in the Internet is done by using the datagram approach to packet switching at the network layer.

b) Virtual –Circuit Networks:

A virtual-circuit network uses the characteristics of both the circuit switched network and the datagram network. A virtual-circuit network is normally implemented in the data link layer,

while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer. Virtual-circuit network is also known as Connection-oriented packet switching

Addressing

Two types of addressing is used in virtual-circuit network

- Global Address: It is an address which can uniquely identify the systems (source or destination) in a network or internet. This address is used to create virtual circuit identifier only.
- Virtual Circuit Identifier: The identifier that is actually used for data transfer is known as virtual circuit identifier (VCI). It is a number which is used in a frame between two switches. This VCI changes from one switch to another. Every switch uses a fixed range of values for VCI.

Three phases of Virtual -Circuit Networks:

1. Data Transfer Phase

- To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit.
- The table, in its simplest form, has four columns.
- This means that the switch holds four pieces of information for each virtual circuit that is already setup.
- Figure 12 shows such a switch and its corresponding table. Figure 13 shows a frame arriving at port 1 with a VCI of 14. When the frame arrives, the switch looks in its table to find port 1 and VCI of 14. When it is found, the switch knows to change the VCI to 22 and send out the frame from port 3.
- The data transfer phase is active until the source sends all its frames to the destination.
- The procedure at the switch is the same for each frame of a message.
- The process creates a virtual circuit, not a real circuit, between the source and destination.

Figure 13 Switch and tables in a virtual-circuit network

2. Setup Phase

In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the setup request and the acknowledgment.

Figure 14 Switch and tables in a virtual-circuit network

- Setup Request:

A setup request frame is sent from the source to the destination. Figure 6 shows the process.

- a. Source A sends a setup frame to switch 1.
- b. Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. The switch, in the setup phase, acts as a packet switch; it has a

routing table which is different from the switching table. For the moment, assume that it knows the output port. The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.

c. Switch 2 receives the setup request frames. The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).

d. Switch 3 receives the setup request frame. Again, three columns are completed: Incoming port (2), incoming VCI (22), and outgoing port (3).

e. Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and no other sources.

- Acknowledgment:

A special frame, called the acknowledgment frame, completes the entries in the switching tables. Figure 15 shows the process.

a. The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed. The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.

b. Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.

c. Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.

d. Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.

e. The source uses this as the outgoing VCI for the data frames to be sent to destination B.

Figure 15 Setup acknowledgment in a virtual-circuit network

1. Teardown Phase

In this phase, source A, after sending all frames to B, sends a special frame called a teardown request. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables.

Note: In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Efficiency of Virtual-Circuit Networks:

Virtual-Circuit Networks uses the resources efficiently and it reduces the waiting time of data frame.

Delay in Virtual-Circuit Networks:

In a virtual-circuit network, there is a delay for setup and for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets. Figure 8 shows the delay for a packet traveling through two switches in a virtual-circuit network.

Figure 16 Delay in a virtual-circuit network

The packet is traveling through two switches (routers). There are three transmission times (3T), three propagation times (3T), data transfer depicted by the sloping lines, a setup delay (which includes transmission and propagation in two directions), and a teardown delay (which includes transmission and propagation in one direction).

Packet Switching versus Circuit Switching

Packet Switching Circuit Switching

1. Packet switched network is implemented in the physical layer.
2. Message is divided into number of packets or frames.
3. Resources are allocated on demand if available (or free).
4. Less efficient and more costly.
5. As the number of users increases, the bandwidth for each user decreases. But most of the link can be used by any number of user.
6. It has unpredictable delay due to waiting of time fixed number of user can use a link because packets/frames at switch, if resources are not available.
7. Suitable for most of the Internet applications.
8. It has less delay in data transmission.
9. Suitable for real time applications.

Internet Standards:

Internet Engineering Task Force (IETF):

The IETF is an open international community concerned with the development and operation of the Internet and its architecture. The IETF was formally established by the Internet Architecture Board (IAB) in 1986. The IETF meets three times a year; much of its ongoing work is conducted via mailing lists by working groups. Typically, based upon previous IETF proceedings, working groups will convene at meetings of the IETF to discuss the work of the IETF working groups. The IETF is administered by the Internet Society, whose WWW site contains lots of high-quality, Internet-related material.

The IETF Internet Engineering Task Force) is the body that defines standard Internet operating protocols such as TCP/IP. The IETF is supervised by the Internet Society Internet Architecture Board (IAB). IETF members are drawn from the Internet Society's individual and organization membership. Standards are expressed in the form of Requests for Comments

(RFCs). The Internet Engineering Task Force (IETF) is an organized activity of the Internet Society (ISOC). It is an open standard organisations, with no formal membership or membership requirements. All participants and managers are volunteers, though their work is usually funded by their employers or sponsors.

The IETF started out as an activity supported by the US federal government, but since 1993 it has operated as a standards development function under the auspices of the Internet society, an international membership-based non-profit organization. The mission of the IETF is to make the Internet work better by producing high quality, relevant technical documents that influence the way people design, use, and manage the Internet.

Institute of Electrical and Electronics Engineers (IEEE) :

The Institute of Electrical and Electronics Engineers (IEEE) is a professional association with its corporate office in New York City and its operations center in Piscataway, New Jersey. It was formed in 1963 from the amalgamation of the American Institute of Electrical Engineers and the Institute of Radio Engineers. Today it is the world's largest association of technical professionals with more than 400,000 members in chapters around the world. Its objectives are the educational and technical advancement of electrical and electronic engineering, telecommunications, computer engineering and allied disciplines.

The IEEE is best known for developing standards for the computer and electronics industry. In particular, the IEEE 802 standards for LANs are widely followed. IEEE manages the Ethernet address space and assigns addresses as needed.

IEEE is one of the leading standards-making organizations in the world. IEEE performs its standards making and maintaining functions through the IEEE Standards Association (IEEE-SA). IEEE standards affect a wide range of industries including: power and energy, biomedical and healthcare, Information Technology (IT), telecommunications, transportation, nanotechnology, information assurance, and many more. In 2013, IEEE had over 900 active standards, with over 500 standards under development. One of the more notable IEEE standards is the IEEE 802 LAN/MAN group of standards which includes the IEEE 802.3 Ethernet standard and the IEEE 802.11 Wireless Networking standard.

Asynchronous Transfer Mode (ATM):

Asynchronous Transfer Mode, a network technology based on transferring data in cells or packets of a fixed size. The cell used with ATM is relatively small compared to units used with older technologies. The small, constant cell size allows ATM equipment to transmit video, audio, and computer data over the same network, and assure that no single type of data hogs the line. Some people think that ATM holds the answer to the Internet band width problem, but others are skeptical. ATM creates a fixed channel, or route, between two points whenever data transfer begins. This differs from TCP/IP, in which messages are divided into packets and each packet can take a different route from source to destination. This difference makes it easier to track and bill data usage across an ATM network, but it makes it less adaptable to sudden surges in network traffic.

ATM (asynchronous transfer mode) is a dedicated-connection switching technology that organizes digital data into 53-byte cell units and transmits them over a physical medium using digital signal technology. Individually, a cell is processed asynchronously relative to other related cells and is queued before being multiplexed over the transmission path.

Asynchronous transfer mode was designed with cells in mind. This is because voice data is converted to packets and is forced to share a network with burst data (large packet data) passing through the same medium. So, no matter how small the voice packets are, they always encounter full-sized data packets, and could experience maximum queuing delays. This is why all data packets should be of the same size. The fixed cell structure of ATM means it can be easily switched by hardware without the delays introduced by routed frames and software switching. This is why some people believe that ATM is the key to the Internet bandwidth problem. ATM creates fixed routes between two points before data transfer begins, which differs from TCP/IP, where data is divided into packets, each of which takes a different route to get to its destination. This makes it easier to bill data usage. However, an ATM network is less adaptable to a sudden network traffic surge.

The ATM provides data link layer services that run on the OSI's Layer 1 physical links. It functions much like small-packet switched and circuit-switched networks, which makes it ideal for real-time, low-latency data such as VoIP and video, as well as for high-throughput data traffic like file transfers. A virtual circuit or connection must be established before the two end points can actually exchange data.

ATM services generally have four different bit rate choices:

- Available Bit Rate: Provides a guaranteed minimum capacity but data can be bursted to higher capacities when network traffic is minimal.
- Constant Bit Rate: Specifies a fixed bit rate so that data is sent in a steady stream. This is analogous to a leased line.
- Unspecified Bit Rate: Doesn't guarantee any throughput level and is used for applications such as file transfers that can tolerate delays.
- Variable Bit Rate (VBR): Provides a specified throughput, but data is not sent evenly. This makes it a very popular choice for voice and videoconferencing.

ATM Service includes:

- Voice and video
- Packetized voice and video
- Systems Network Architecture (SNA)
- WAN/VPN connectivity
- Web hosting
- E-commerce
- Client-server (terminal-host) data
- LAN interconnection
- LAN emulation
- Remote access

- File transfer
- Internet/intranet/extranet access
 - E-mail messaging
 - Text imaging
- Forms processing

Use in internet: ATM is normally utilized by Internet service providers on their private long-distance networks. ATM operates at the data link layer (Layer 2 in the OSI model) over either fiber or twisted- pair cable

International Telecommunication Union (ITU)

The ITU coordinates the shared global use of the radio spectrum, promotes international cooperation in assigning satellite orbits, works to improve telecommunication infrastructure in the developing world, and assists in the development and coordination of worldwide technical standards. The ITU is active in areas including broadband Internet, latest-generation wireless technologies, aeronautical and maritime navigation, radio astronomy, satellite-based meteorology, convergence in fixed-mobile phone, Internet access, data, voice, TV broadcasting, and next-generation networks.

ITU also organizes worldwide and regional exhibitions and forums, such as ITU TELECOM WORLD, bringing together representatives of government and the telecommunications and ICT industry to exchange ideas, knowledge and technology.

ITU, based in Geneva, Switzerland, is a member of the United Nations Development Group. ITU has been an intergovernmental public-private partnership organization since its inception. Its membership includes 193 Member States and around 700 public and private sector companies as well as international and regional telecommunication entities, known as Sector Members and Associates, which undertake most of the work of each Sector.

□ An organization based on public-private partnership since its inception, ITU currently has a membership of 193 countries and over 700 private-sector entities and academic institutions. ITU is headquartered in Geneva, Switzerland, and has twelve regional and area offices around the world.

□ ITU membership represents a cross-section of the global ICT sector, from the world's largest manufacturers and carriers to small, innovative players working with new and emerging technologies, along with leading R&D institutions and academia.

□ Founded on the principle of international cooperation between governments (Member States) and the private sector (Sector Members, Associates and Academia), ITU is the premier global forum through which parties work towards consensus on a wide range of issues affecting the future direction of the ICT industry.

Internet Protocol (IP)

□ The Internet Protocol (IP) is a network-layer (Layer 3) protocol that contains addressing information and some control information that enables packets to be routed.

□ IP is documented in RFC 791 and is the primary network-layer protocol in the Internet protocol suite. Along with the Transmission Control Protocol (TCP), IP represents the heart of the Internet protocols.

□ IP has two primary responsibilities: providing connectionless, best-effort delivery of datagrams through an internetwork; and providing fragmentation and reassembly of datagrams to support data links with different maximum-transmission unit (MTU) sizes.

IP Packet Format

Packets in the IP layer are called datagrams. A datagram divided into two parts : Header and Data. Header can be from 20 to 60 bytes and contains information for routing and delivery of data.

IP packet fields Details:

- Version: Indicates the version of IP currently used.
- IP Header Length (IHL): Indicates the datagram header length in 32-bit words.
- Type-of-Service: Specifies how an upper-layer protocol would like a current datagram to be handled, and assigns datagrams various levels of importance.
- Total Length: Specifies the length, in bytes, of the entire IP packet, including the data and header.
- Identification: Contains an integer that identifies the current datagram. This field is used to help piece together datagram fragments.
- Flags: Consists of a 3-bit field of which the two low-order (least-significant) bits control fragmentation. The low-order bit specifies whether the packet can be fragmented. The middle bit specifies whether the packet is the last fragment in a series of fragmented packets. The third or high-order bit is not used.
- Fragment Offset: Indicates the position of the fragment's data relative to the beginning of the data in the original datagram, which allows the destination IP process to properly reconstruct the original datagram.
- Time-to-Live: Maintains a counter that gradually decrements down to zero, at which point the datagram is discarded. This keeps packets from looping endlessly.
- Protocol: Indicates which upper-layer protocol receives incoming packets after IP processing is complete.
- Header Checksum: Helps ensure IP header integrity.
- Source Address: Specifies the sending node.
- Destination Address: Specifies the receiving node.

IP Addresses

- TCP/IP version 4 or IPv4 uses 32-bit for logical address and IPv6 uses 128-bit for logical address.
- An IP address represented in dotted decimal notation. Example- 123.22.33.44
- IP address is divided into net id or network id and host id.
- IP Addresses are divided into five classes: Class A, Class B, Class c, Class C, Class D, Class E.

IP Starting First Last Address No. of No. of Host

Address Binary Address Network

Class Value

Class A 0 1.0.0.0 126.255.255.254 27-1 224- 2

Class B 10 128.0.0.0 191.255.255.254 214 216 -2

Class C 110 192.0.0.0 223.255.255.254 221 28-2

Class D 1110 224.0.0.0 239.255.255.254 Multicast

Class E 1111 240.0.0.0 254.255.255.254 Undefined

Class A:

- It uses first octet for network address to uniquely identify the network and rest three octet for host address to uniquely identify the host on that network.
- An important rule is that network address cannot have all 8 bits 0 (zero).
- First bit is set to zero for class A, so following 7 bits in the first octet use to distinguish the network from other network.
- It means $2^7 - 1 = 127$ network i.e. 0 to 126
- Similar to the rule that the network portion of the address cannot be all 0s, the host portion of the address cannot be all 0s and it cannot be all 1s.
- A host portion with all 1s refers to an IP broadcast address.
- And the host portion with all 0s is a reference to the network.
- Class A network is: $2^{24} - 2 = 16,777,214$ number of hosts.
- You subtract 2 because addresses with all 0s and all 1s are invalid.

Class B:

- It uses first two octets for network address to uniquely identify the network and rest two octet for host address to uniquely identify the host on that network.
- 10 in the first 2 bits, the following 6 bits in the first octet and all 8 bits in the second octet for total 14 bits are used to distinguish this network from all other networks.
- Hence $2^{14} = 16,384$ number of Class B networks.
- And $2^{16} - 2 = 65,534$ number of hosts on class B network.

Class C:

- It uses first three octets for network address to uniquely identify the network and last octet for host address to uniquely identify the host on that network.
- 110 in the first 3 bits, the following 5 bits in the first octet, all 8 bits in the second octet and all 8 bits in the third octet for total 21 bits are used to distinguish this network from all other networks.
- Hence $2^{21} = 2,097,152$ number of Class C networks.
- And $2^8 - 2 = 254$ number of hosts on class C network.

Class D:

- In the first octet, the first 4 bits are 1110.

- Class D addresses are called Multicast Address which cannot be used for host.
- The purpose of a multicast address is to enable a server somewhere to send data to a Class D address that no one host has so that several hosts can listen to that address at the same time. When you are watching TV on the Internet or listening to the radio on the Internet, your computer is listening to a Class D address. No server is sending data directly to your workstation; instead, a server is sending data to the multicast address. Any host can use software to listen for data at that address, and many hosts can be listening at once.

Class E:

- In the first octet, the first 4 bits are 1110.
- Class E addresses are reserved addresses and are invalid host addresses. They are used for experimental purposes by the IETF.

Special Address:

- Address use for Private use

Class A: 10.0.0.0 to 10.255.255.255

Class B: 172.16.0.0 to 172.31.255.255

Class C: 192.168.0.0 to 192.168.255.255

- Loop Back Address

127.0.0.0 to 127.255.255- For testing the TCP/IP connection. It cannot be used for host addressing.

FIREWALL

Many organizations have confidential or proprietary information, such as trade secrets, product development plans, marketing strategies, etc., which should be protected from unauthorized access

and modification. One possible approach is to use suitable encryption/decryption technique for transfer of data between two secure sites, as we have discussed in the previous lesson. Although these techniques can be used to protect data in transit, it does not protect data from digital pests and

hackers. To accomplish this, it is necessary to perform user authentication and access control to protect the networks from unauthorized traffic. This is known as firewalls.

A firewall system is an electronic security guard and electronic barrier at the same time. It protects and controls the interface between a private network and an insecure public network as shown in the

simplified diagram of Fig. 17. It is responsible for partitioning a designated area such that any damage on one side cannot spread to the other side. It prevents bad things from happening, i.e. loss of

information, without preventing good things from happening, that is controlled exchange of information with the outside world. It essentially enforces an access control policy between two networks. The manner in which this is implemented varies widely, but in principle, the firewall can

be considered as a pair of mechanisms: one that is used to block traffic, and the other that is used to permit traffic. Some firewalls place more emphasis on blocking traffic, while others emphasize on permitting traffic. Probably the most important issue to understand of a firewall is the access control policy it implements. If a firewall administrator has no idea about what or whom he is protecting his network, what should be allowed and what should be prohibited, a firewall really won't help his organization. As firewall is a mechanism for enforcing policy, which affects all the persons behind it, it imposes heavy responsibility on the administrator of the firewall. In this lesson various issues related to Firewalls are discussed.

Figure 17 Schematic diagram of a firewall

Why a Firewall is needed?

There is no need for a firewall if each and every host of a private network is properly secured. Unfortunately, in practice the situation is different. A private network may consist of different platforms with diverse OS and applications running on them. Many of the applications were designed and developed for an ideal environment, without considering the possibility of the existence of bad guys. Moreover, most of the corporate networks are not designed for security. Therefore, it is essential to deploy a firewall to protect the vulnerable infrastructure of an enterprise.

Access Control Policies

Access control policies play an important role in the operation of a firewall. The policies can be broadly categorized into the following four types:

Service Control:

- Determines the types of internet services to be accessed.
- Filters traffic based on IP addresses and TCP port numbers.
- Provides Proxy servers that receive and interpret service requests before it is passed on

Direction Control:

- Determines the direction in which a particular service request may be initiated and allowed to flow through the firewall

User Control:

- Controls access to a service according to which user is attempting to access it.
- Typically applied to the users inside the firewall perimeter.
- Can be applied to the external users too by using secure authentication technique

Behavioral Control:

- Controls how a particular service is used
- For example, a firewall may filter email to eliminate spam
- Firewall may allow only a portion of the information on a local web server to an external user

Firewall Capabilities

Important capabilities of a firewall system are listed below:

- defines a single choke point to keep unauthorized users out of protected network

- It prohibits potentially vulnerable services from entering or leaving the network
- It provides protection from various kinds of IP spoofing
- It provides a location for monitoring security-related events
- Audits and alarms can be implemented on the firewall systems
- A firewall is a convenient platform for several internet functions that are not security related
- A firewall can serve as the platform for IPSec using the tunnel mode capability and can be used to implement VPNs

Limitations of a Firewall

Main limitations of a firewall system are given below:

- Firewall cannot protect against any attacks that bypass the firewall. Many organizations buy expensive firewalls but neglect numerous other back-doors into their network.
- A firewall does not protect against the internal threats from traitors. An attacker may be able to break into network by completely bypassing the firewall, if he can find a ``helpful" insider who can be fooled into giving access to a modem pool
- Firewalls can't protect against tunneling over most application protocols. For example, firewall cannot protect against the transfer of virus-infected programs or files

Types of Firewalls

The firewalls can be broadly categorized into the following three types:

Packet Filters: Packet filtering router applies a set of rules to each incoming IP packet and then forwards or discards it. Packet filter is typically set up as a list of rules based on matches of fields in the IP or TCP header. An example table of telnet filter rules is given in Fig. 8.3.2. The packet filter operates with positive filter rules. It is necessary to specify what should be permitted, and everything that is explicitly not permitted is automatically forbidden.

Application-level Gateway: Application-level gateway, also called a Proxy Server acts as a relay of application-level traffic. Users contact gateways using an application and the request is successful after authentication. The application gateway is service specific such as FTP, TELNET, SMTP or HTTP.

Circuit Level Gateway: Circuit-level gateway can be a standalone or a specialized system. It does not allow end-to-end TCP connection; the gateway sets up two TCP connections. Once the TCP connections are established, the gateway relays TCP segments from one connection to the other without examining the contents. The security function determines which connections will be allowed and which are to be disallowed.

Network Address Translation

NAT works by using one set of addresses for communications on the internet and a separate set of addresses for communication on the private network. IANA set aside three ranges of IP addresses given below for communication on the internal network.

Class A addresses: 10.0.0.0 – 10.255.255.255

Class B addresses: 172.16.0.0 – 172.31. 255.255

Class C addresses: 192.168.0.0 – 192.168.255.255

As these addresses are reserved for internal network addressing, these are not routable. The Firewall

performs translation of an internal address to an external IP address and vice versa to facilitate communication between the private and the public network, as shown in Fig. 18. However, the NAT affords a substantial degree of security by preventing direct communication. Moreover, NAT allows the use of same IP addresses in different private networks. This prolongs the life expectancy of IPv4 on the internet. Without NAT the supply of IP addresses would have exhausted long back.

Figure 18 Function of a Network Address Translator